

Gasoline Profiles and Impact of Ethanol Blending in Asia

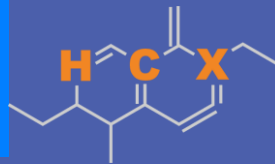
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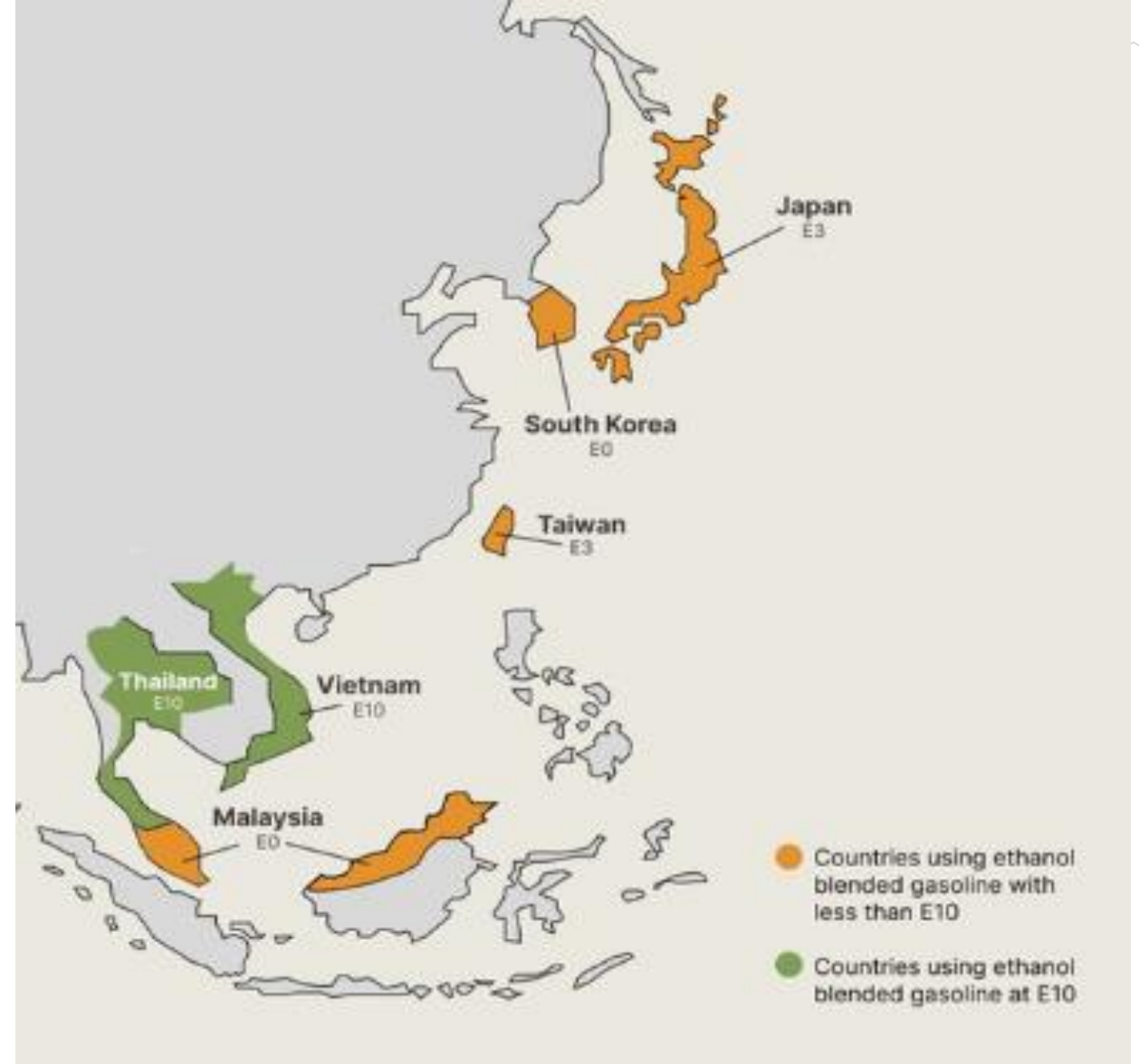


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Ethanol Blending in Asia

There are important fuel quality and environmental impact of vehicle emission challenges in these regions.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Asia to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

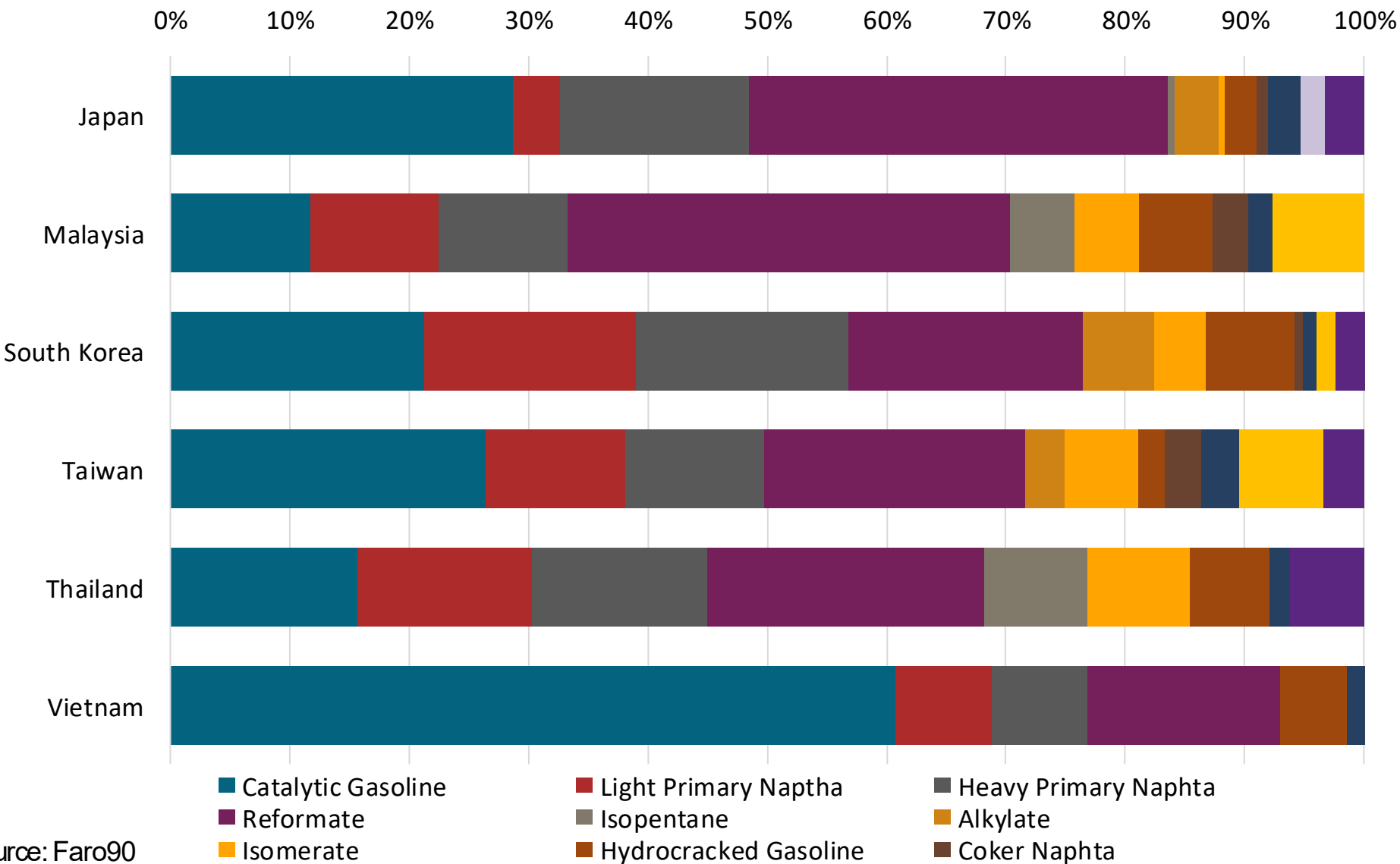


Seven countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Gasoline Component Blending in Asia

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Asia are:

- Catalytic Gasoline
- Light Primary Naptha
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics



Source: Faro90

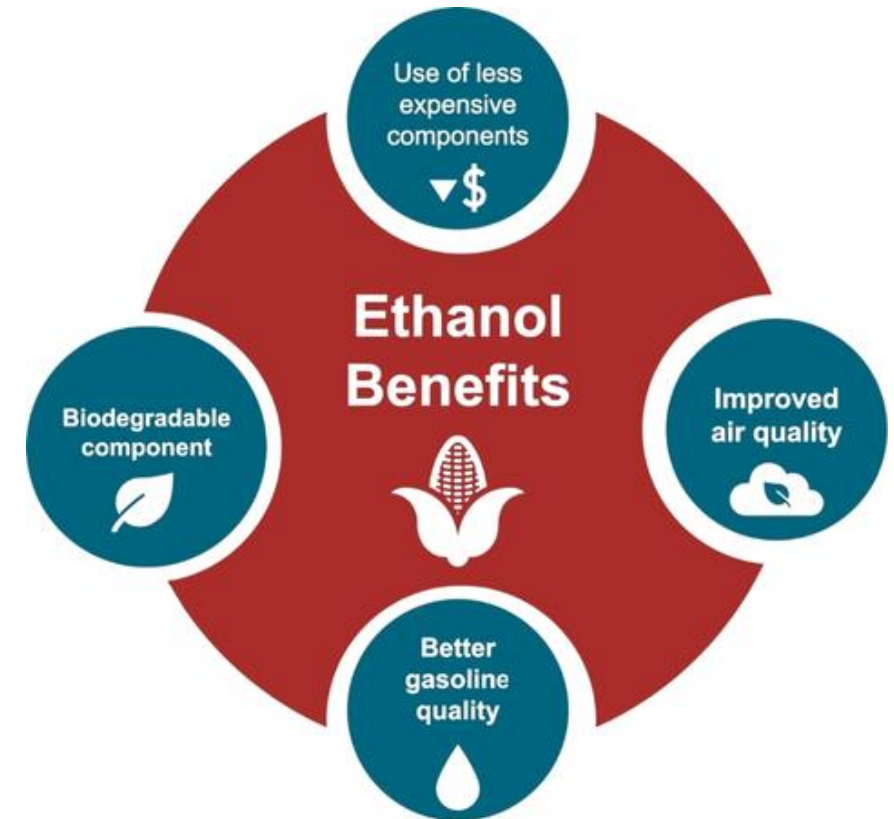
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

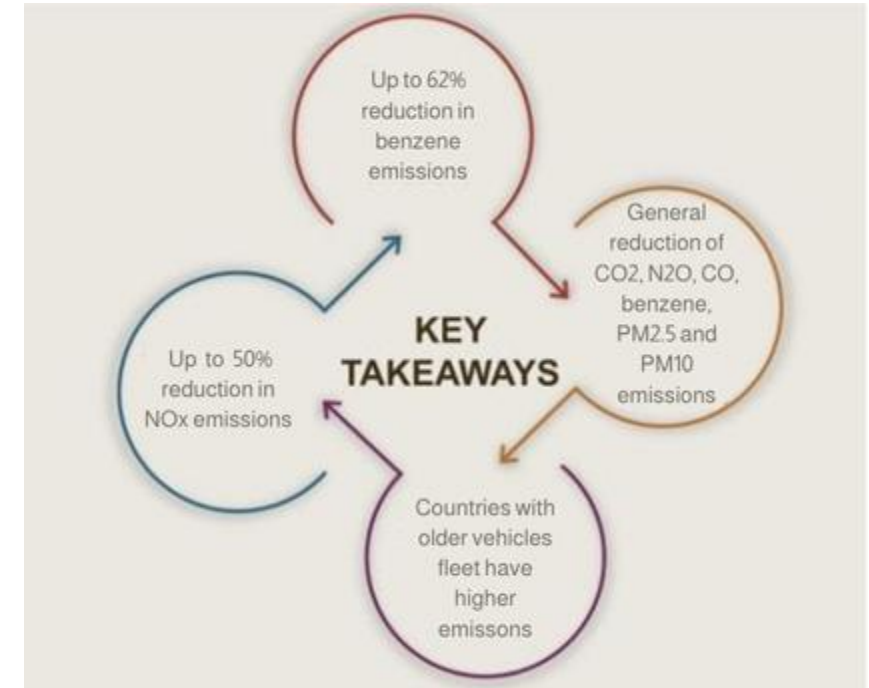
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



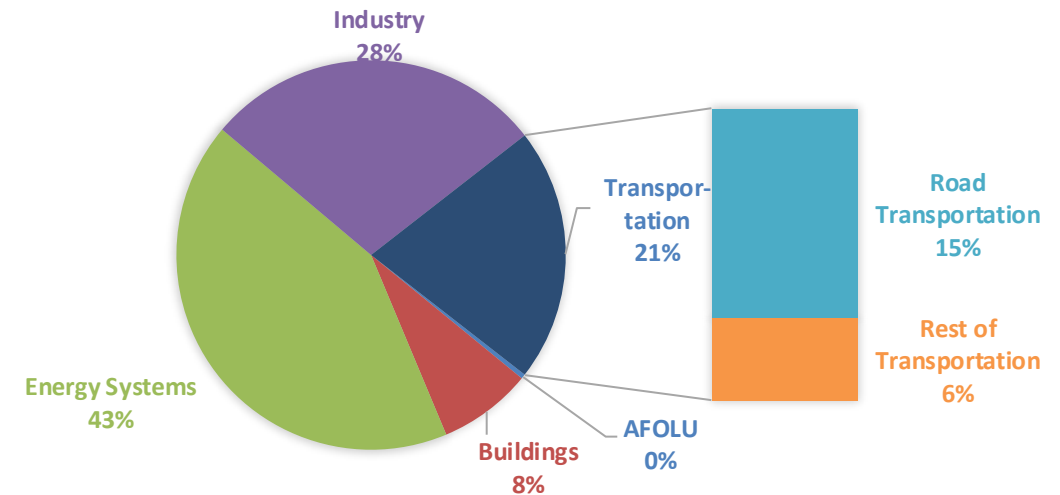
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

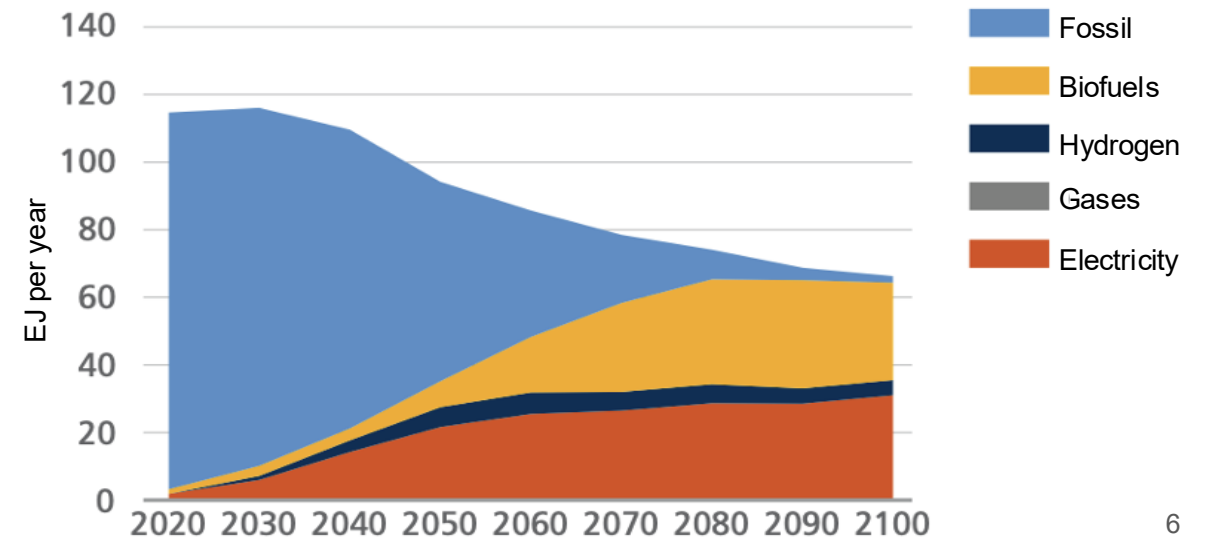
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

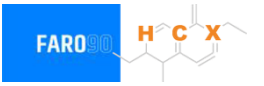
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Case Studies



1. Japan
2. Malaysia
3. South Korea
4. Taiwan
5. Thailand
6. Vietnam

Ethanol Gasoline Blending in Japan



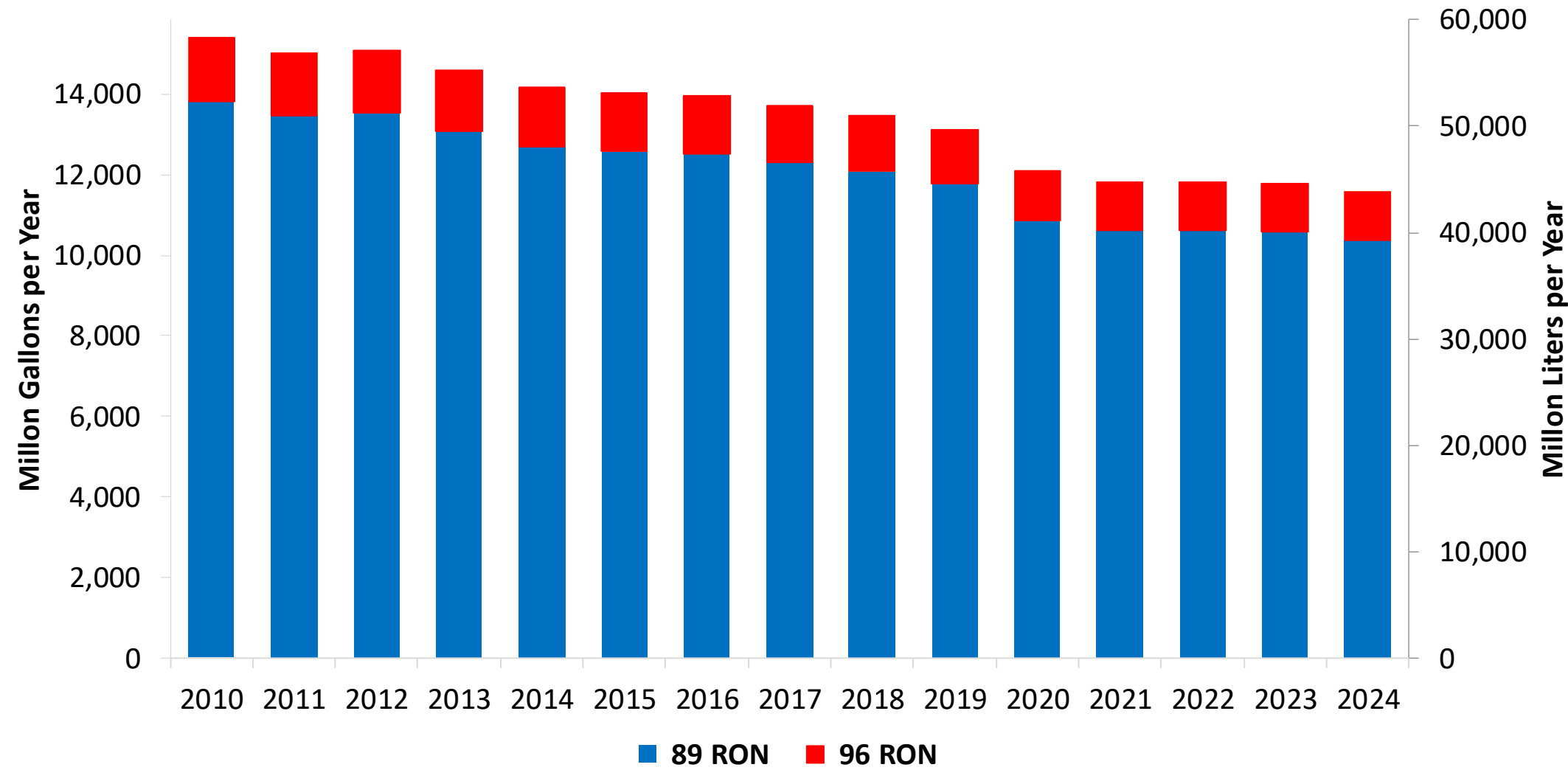
In 2024, gasoline consumption in Japan was 43,826 million liters (11,578 million gallons) and growth rate was -2.5%. Gasoline production and net imports were 42,543 and 1,283 million liters (11,239 and 339 million gallons) correspondingly. Japan complies with the JIS standards and offers two main grades: RON 89 and RON 96) with an average octane of 89.7 RON.

Japan currently allows E3. Last year's consumption was 830 million liters (219 million gallons) equivalent to 1.9%, primarily as the component ethyl tert-butyl ether (ETBE). There is no local production. Ethanol is imported as a blendstock for ETBE production and as ETBE. The Ministry of Economy, Trade and Industry (METI) aiming for E10 (10% ethanol) by 2030 and E20 (20% ethanol) by 2040. This initiative requires infrastructure improvements, as most existing gas stations are only equipped for the E3 blend.

Source: METI, USDA

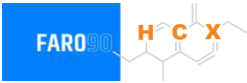
Gasoline Demand in Japan

Gasoline Demand by Grade in Japan

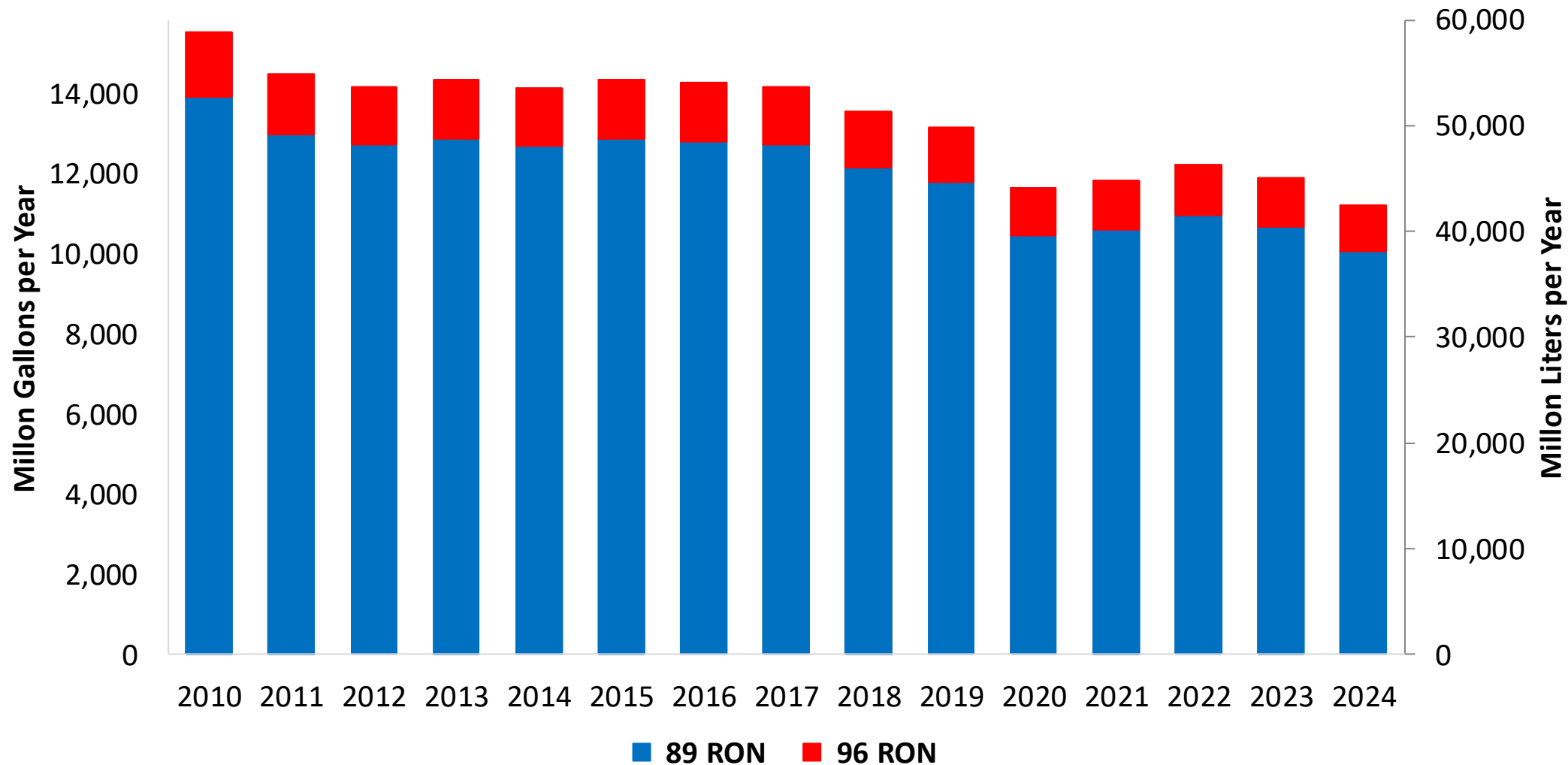


Source: METI

Gasoline Production in Japan



Gasoline Production by Grade in Japan



Source: METI

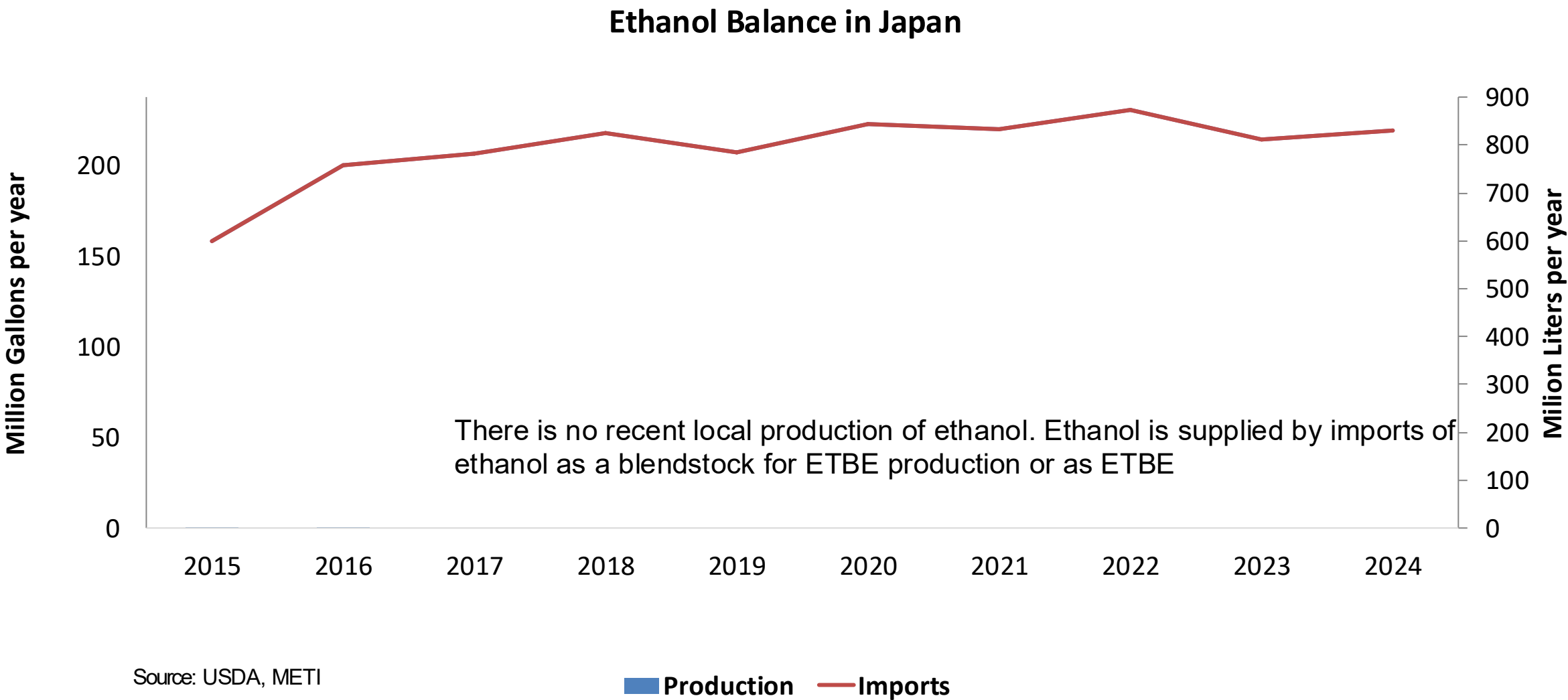
The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexadiene. The structure features a six-membered ring with two double bonds. Substituents include a hydrogen atom (H), a carbon atom (C), and a group labeled 'X' (which is a vinyl group, -CH=CH₂). There are also methyl groups attached to the ring.

The chart displays the annual consumption of 89 RON and 96 RON gasoline from 2010 to 2024. The 89 RON consumption (blue) shows a general upward trend, peaking in 2024 at approximately 1,150 million gallons. The 96 RON consumption (red) also shows an upward trend, peaking in 2024 at approximately 150 million gallons. The total consumption in 2024 reached approximately 1,300 million gallons.

Year	89 RON (Million Gallons)	96 RON (Million Gallons)	Total (Million Gallons)
2010	260	30	290
2011	600	70	670
2012	660	80	740
2013	450	50	500
2014	380	40	420
2015	300	40	340
2016	200	20	220
2017	230	30	260
2018	500	60	560
2019	550	60	610
2020	740	90	830
2021	690	90	780
2022	550	70	620
2023	760	90	850
2024	1150	150	1300

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Ethanol Balance in Japan



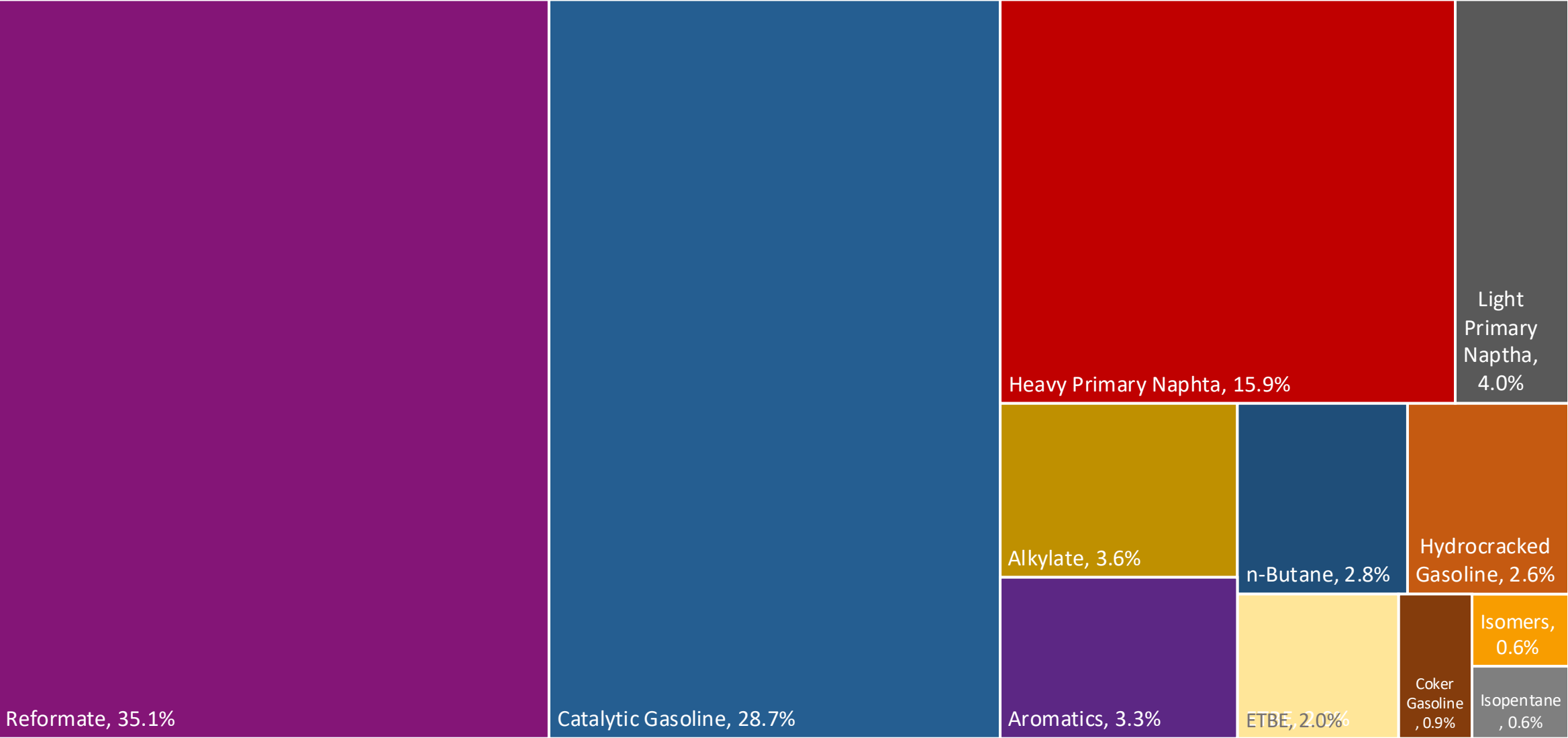
Gasoline Quality in Japan

Name	JIS K 2202: 2023 - Law on the Quality Control of Gasoline and Other Fuels							
Implementation Date	2023							
Applicability	Japan							
Grade	Regular		Premium		E10 Regular *		E10 Premium *	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	89.0		95.0		90.0		96.0	
MON								
AKI								
Benzene (%vol)		1.0		1.0		1.0		1.0
Aromatics (% vol)		-		-		-		-
Olefins (%vol)		-		-		-		-
Lead (g/L)		-		-		-		-
Manganese (mg/L)		-		-		-		-
Sulfur (ppm)		10		10		10		10
Oxygen (%w)		1.3		1.3		3.5		3.5
Ethanol (%vol)		3.0		3.0	3	10.0	3.0	10.0
MTBE (%vol)		7.0		7.0				
PVR 37.8°C (psi)	6.4	11.3	6.4	11.3	6.4	11.3	6.4	11.3
Carbon Intensity (gCOe/MJ)	0	88.74	-	88.74	0	88.74	-	88.74
Kerosene (%vol)	-	4.0	-	4.0	-	4.0	-	4.0

*Not Mandatory. A 55% Carbon Intensity Reduction target has been established and will be revised once LCA has been completed

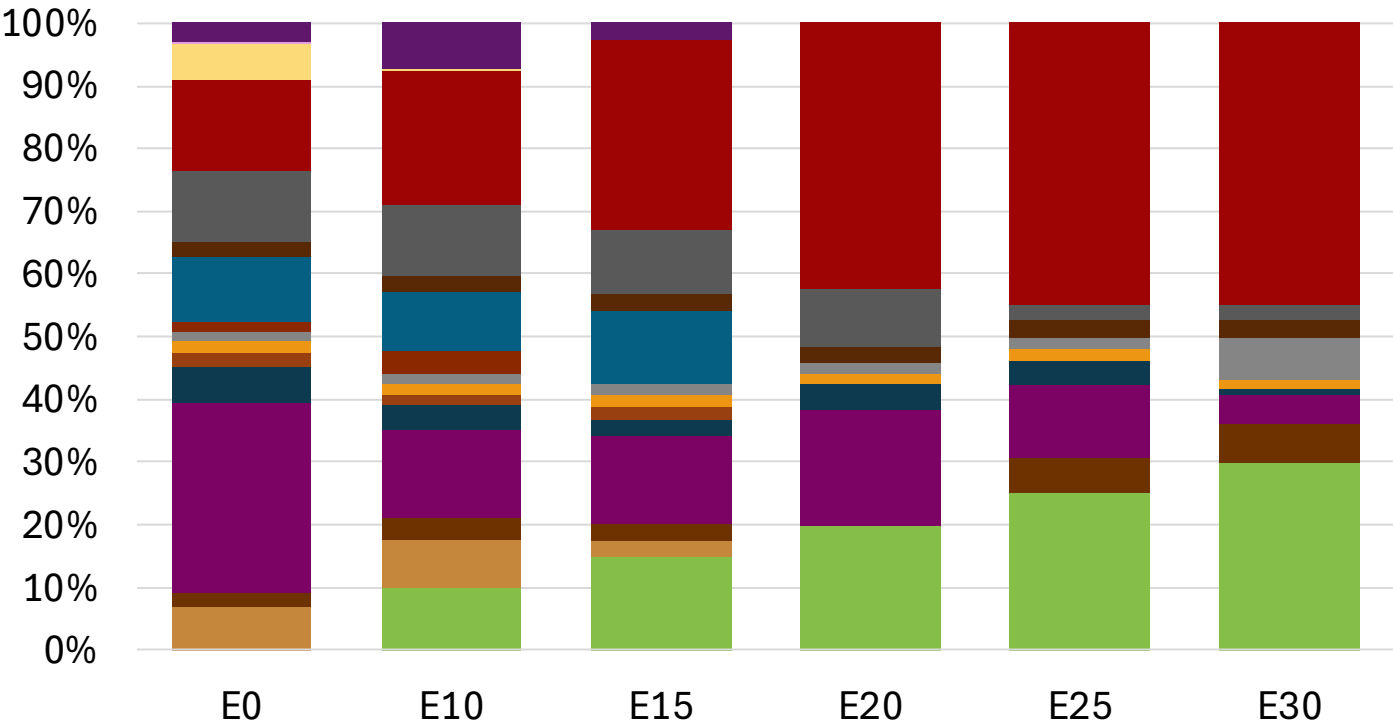
Source: METI

Japan Available Blending Components



Source: Faro90

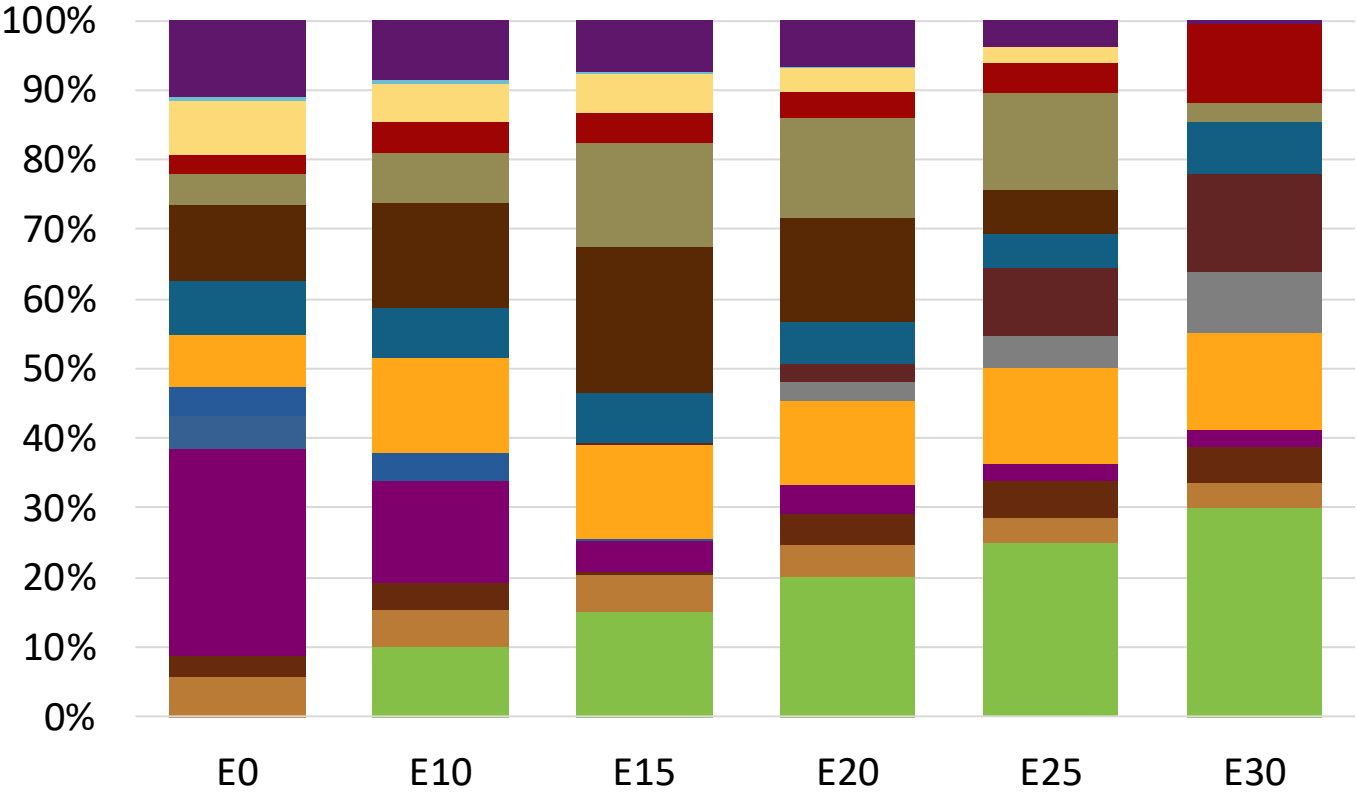
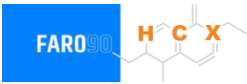
Japan – Gasoline 1 – Constant Octane



Average CIF 2024
Prices

Source: Faro90

Japan - Gasoline 2 - Constant Octane



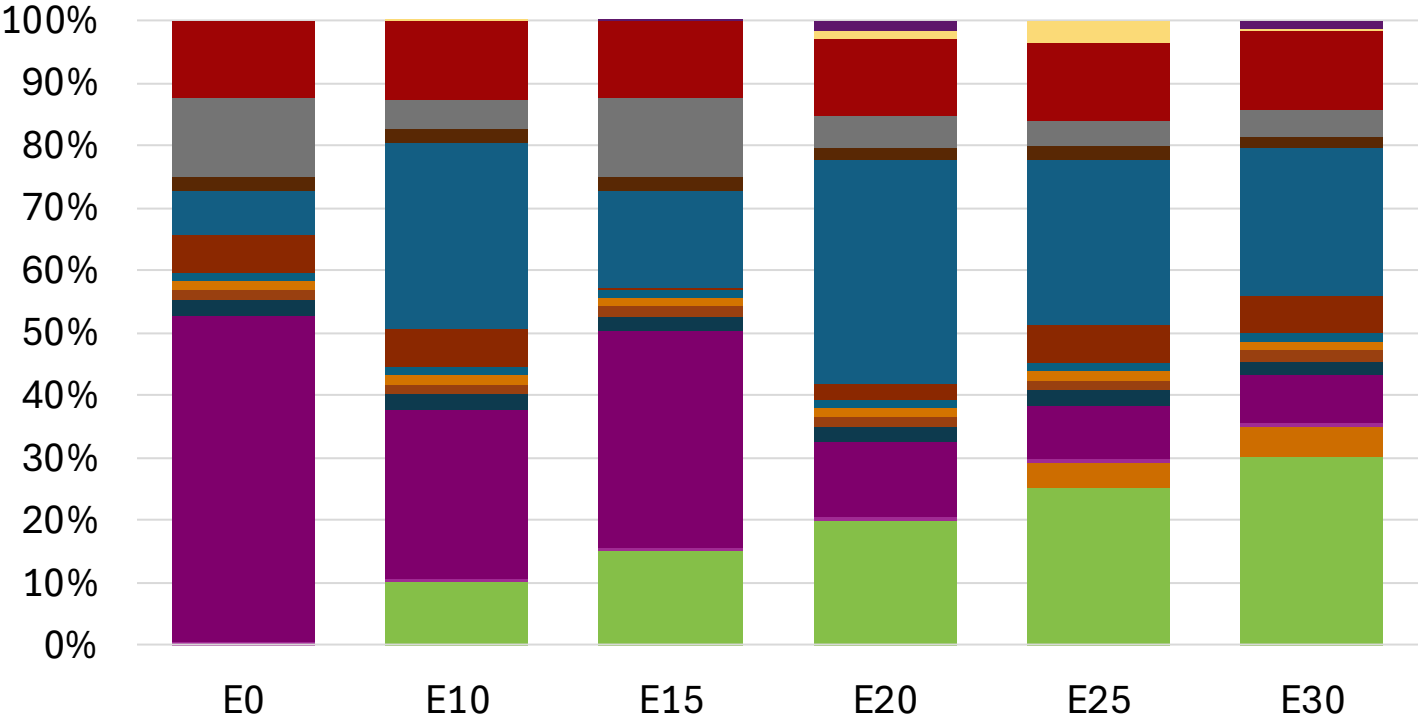
Octane (RON)	96.0					
Price (US\$/gal)	\$ 2.421	\$ 2.344	\$ 2.265	\$ 2.180	\$ 2.090	\$ 1.984

- Ethanol
- Alkylate
- Raffinate
- Reformate
- n-Butane
- Isomerate
- Isopentane
- Hydrocracked Gasoline
- Catalytic Gasoline
- Coker Naphtha
- Light Primary Naphtha
- Heavy Primary Naphtha
- MTBE
- ETBE
- TAME
- Aromatics

Average CIF 2024 Prices

Source: Faro90

Japan – Gasoline 1 – Increased Octane

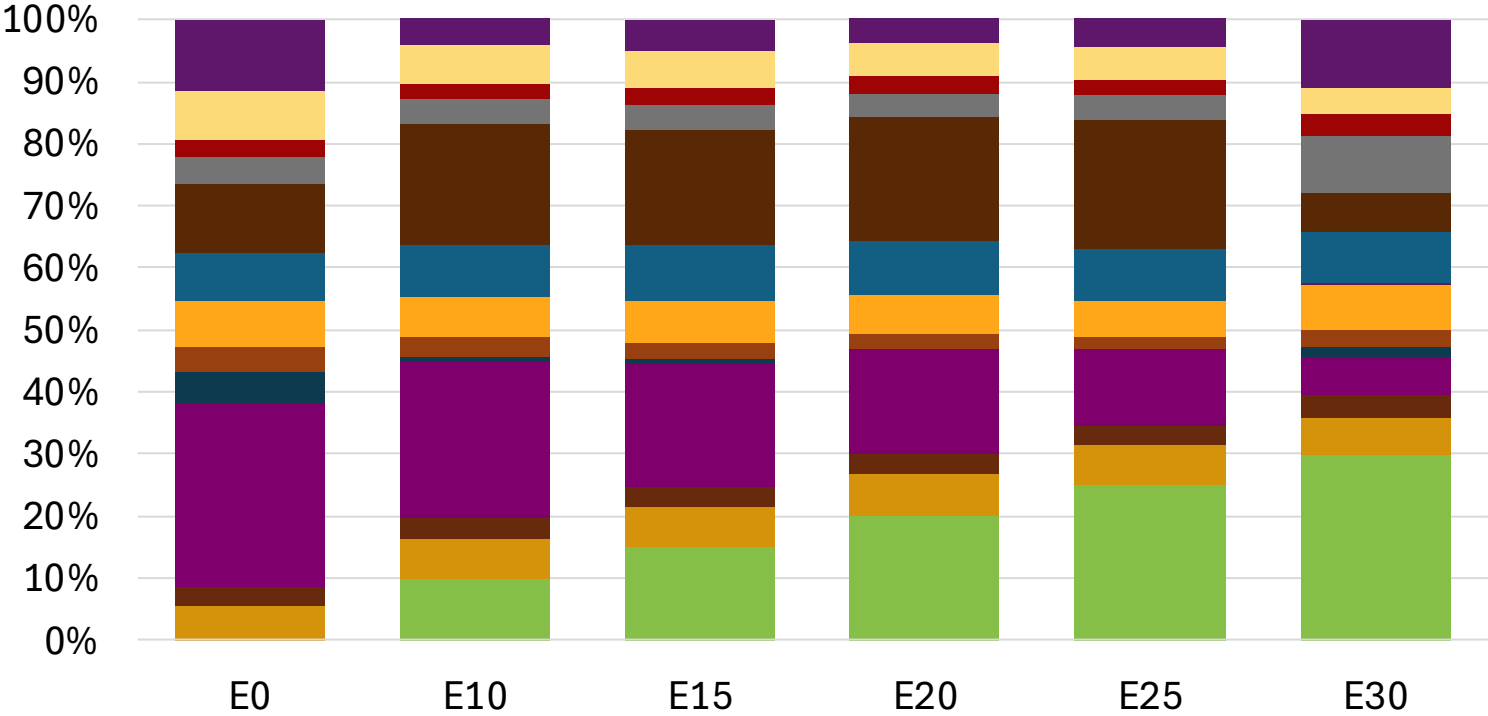


Average CIF 2024
Prices

Octane (RON)	89.2	93.3	95.2	97.1	98.8	100.0
Price (US\$/gal)	\$ 2.223	\$ 2.223	\$ 2.223	\$ 2.223	\$ 2.223	\$ 2.223

Source: Faro90

Japan – Gasoline 2 – Increased Octane

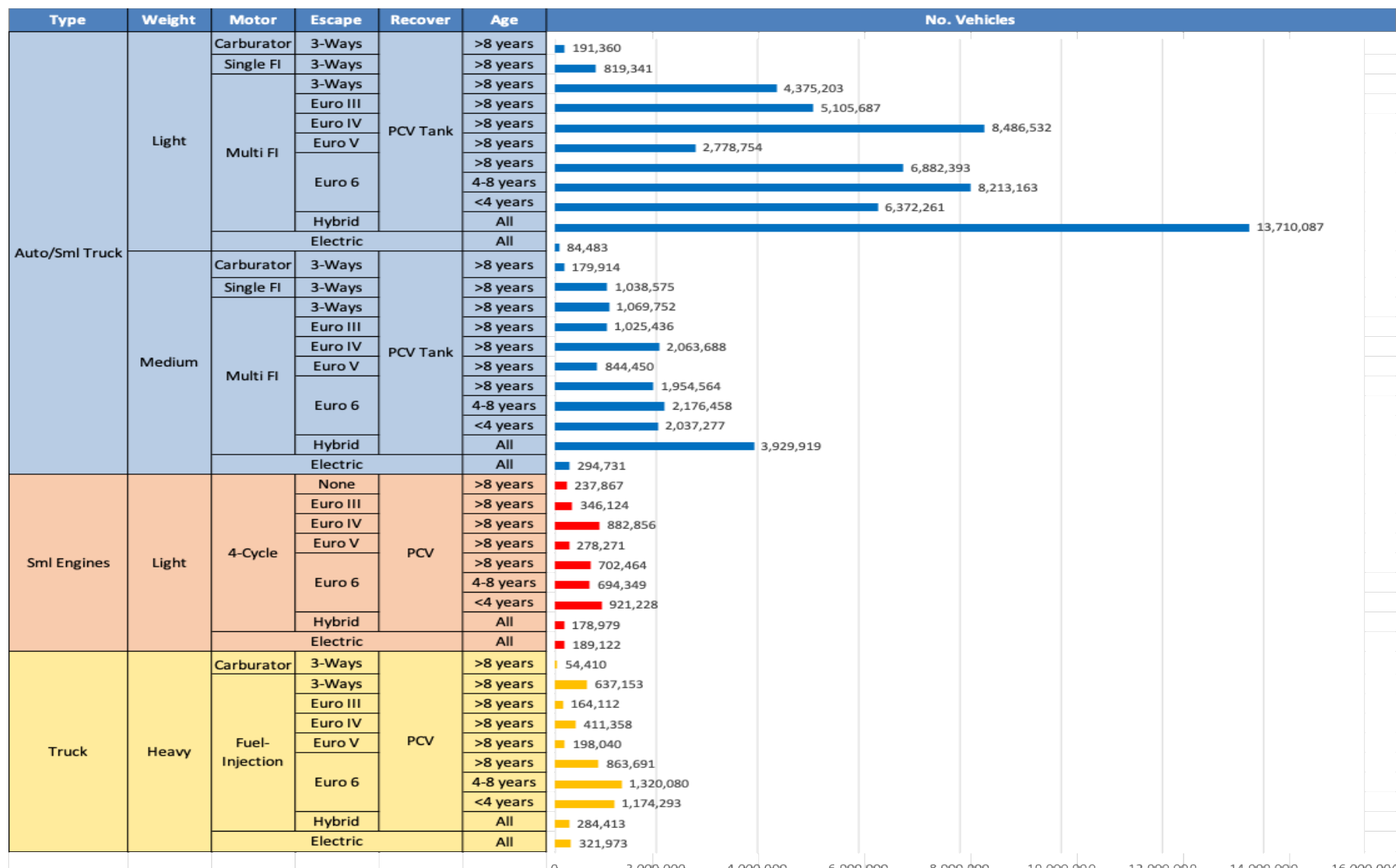


Average CIF 2024
Prices

Octane (RON)	96.0	97.6	99.1	100.5	102.0	103.2
Price (US\$/gal)	\$ 2.421	\$ 2.421	\$ 2.421	\$ 2.421	\$ 2.421	\$ 2.421

Source: Faro90

Gasoline Vehicle Fleet - Japan

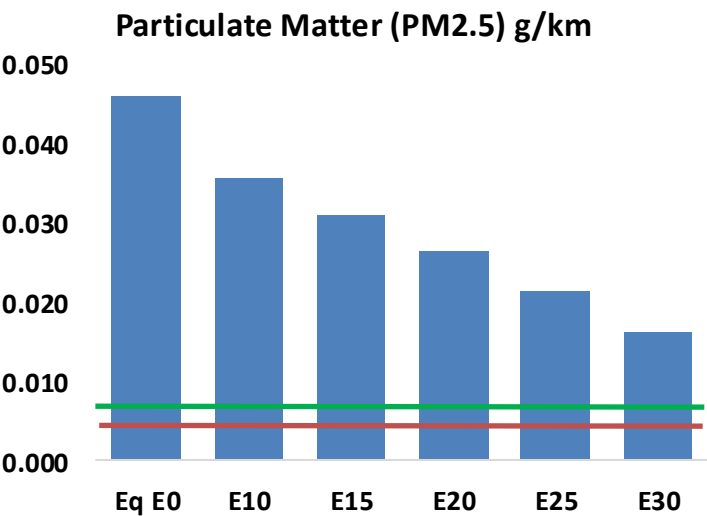
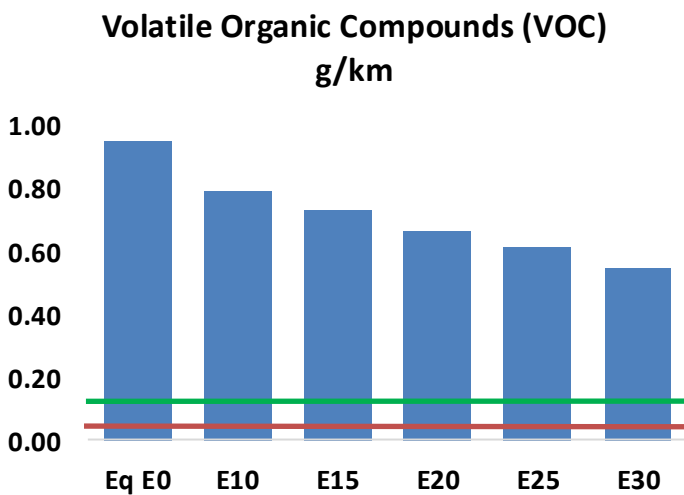
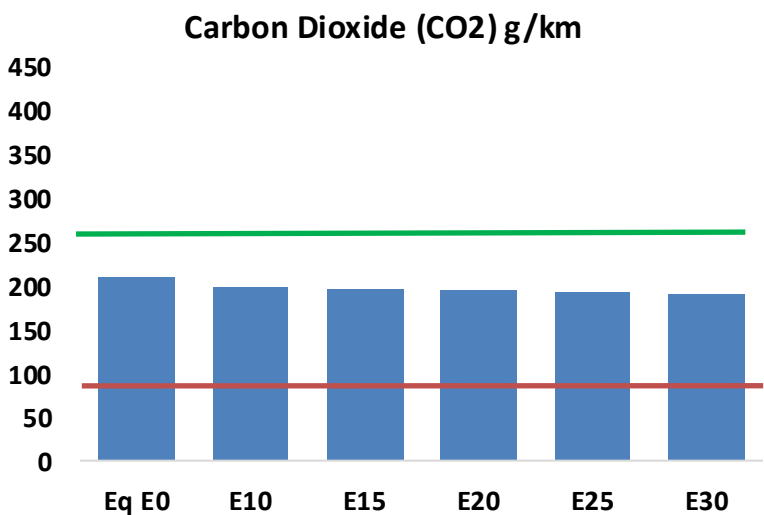
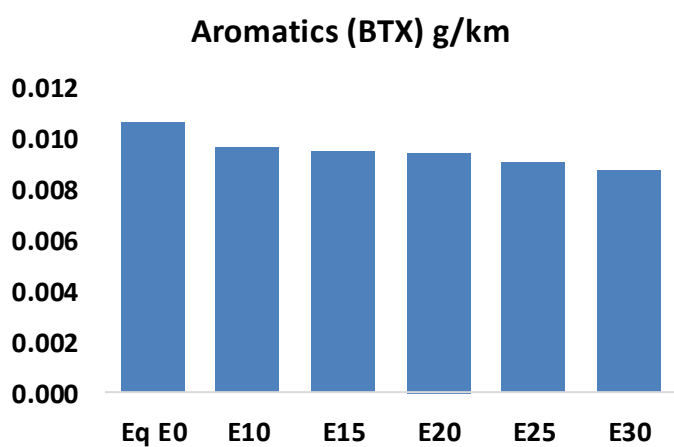
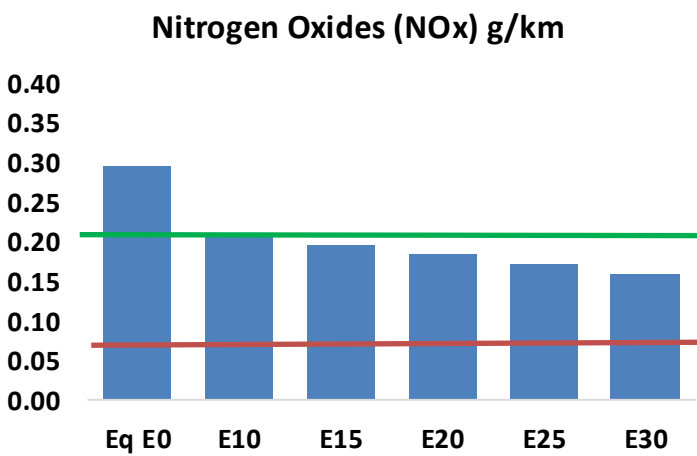
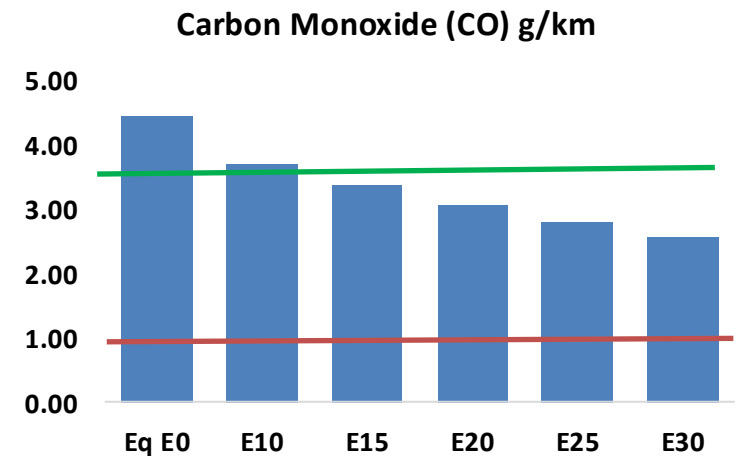
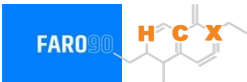


Vehicle Fleet: 83,494,811
Gasoline Vehicle Fleet: 76,775,958

Source: AIRIA

Japan – Vehicular Emissions

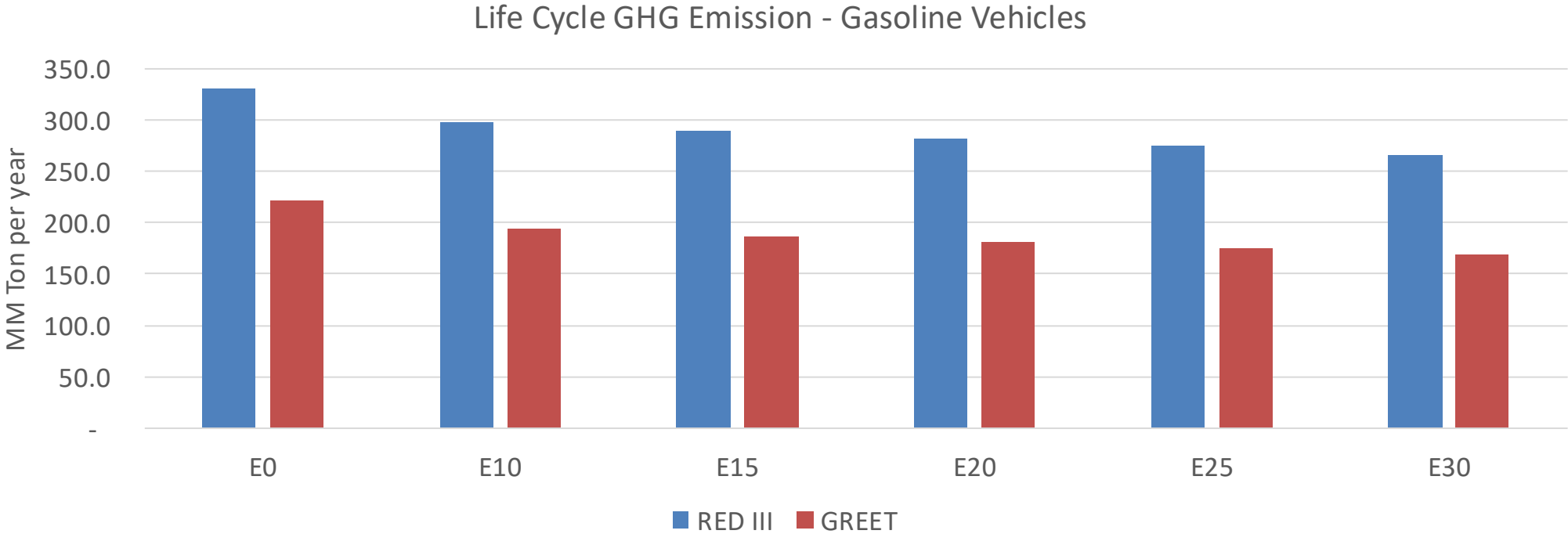
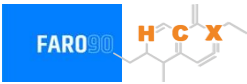
TIER III BIN 125 United States
Euro 6 Standard



Japan – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	7,996,709	7,320,077	6,643,444	6,076,898	5,501,211	5,056,863	4,584,871	-17%	-31%
CO2	375,859,078	366,194,131	357,066,124	349,887,216	346,334,811	344,557,634	338,206,446	-5%	-8%
VOC	1,702,687	1,563,867	1,425,046	1,309,743	1,192,997	1,103,012	983,689	-16%	-30%
NOx	529,045	396,784	370,331	349,037	329,172	307,269	284,097	-30%	-38%
SOx	4,260	3,938	3,616	3,343	3,082	2,799	2,501	-15%	-28%
NH3	146,209	144,761	143,314	143,994	145,614	146,753	147,700	-2%	0%
N2O	7,310	7,270	7,241	7,276	7,426	7,509	7,594	-1%	2%
CH4	345,533	345,533	345,533	350,716	358,317	364,882	371,102	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	6,553	6,230	5,906	5,675	5,460	5,299	5,067	-10%	-17%
Acetaldehyde	23,643	27,135	39,815	60,634	82,607	94,394	111,537	68%	249%
Formaldehyde	91,090	88,560	103,236	121,221	126,997	140,493	152,942	13%	39%
Benzene	19,076	18,332	17,370	17,095	16,953	16,214	15,690	-9%	-11%
PM2.5	23,472	20,841	18,211	15,814	13,429	10,897	8,258	-22%	-43%
PM10	59,053	52,435	45,817	39,787	33,786	27,417	20,776	-22%	-43%

Japan – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	1,102.2
Target @ 2035 (MMT/year)	570.4
Delta	531.8

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.8%	16.0%	18.5%	20.9%	24.2%
RED II/III	0.0%	9.8%	12.5%	14.7%	16.9%	19.5%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.3%	6.7%	7.7%	8.7%	10.1%
RED II/III	0.0%	6.1%	7.7%	9.1%	10.5%	12.1%

Source: Greet, RED II/III, climateactiontracker.org, F90

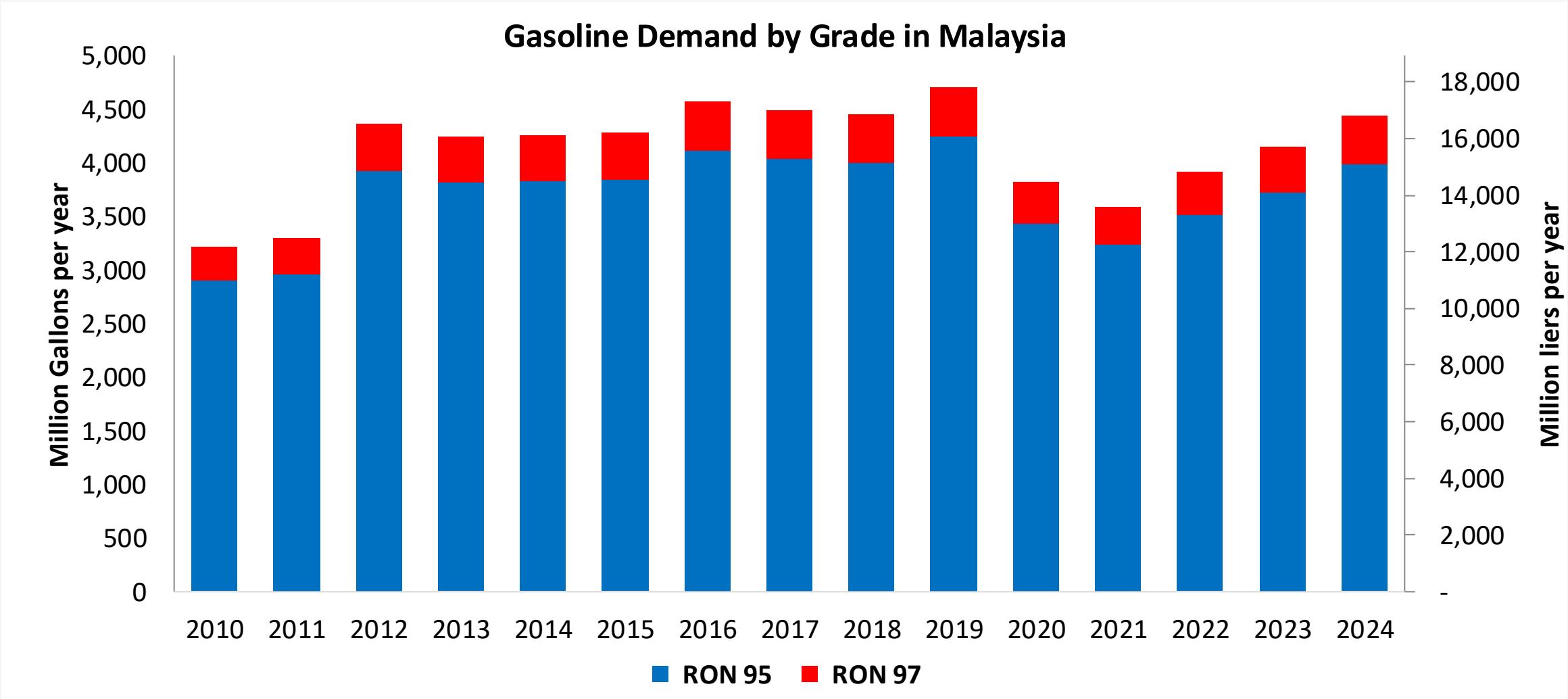
Ethanol Gasoline Blending in Malaysia



In 2024, gasoline consumption in Malaysia was 16,784 million liters (4,434 million gallons) and growth rate was -1.2%. Gasoline production and net imports were 5,436 and 11,348 million liters (1,436 and 2,998 million gallons) correspondingly. Malaysia offers two main grades: RON 95 and RON 97) with an average octane of 95.2 RON.

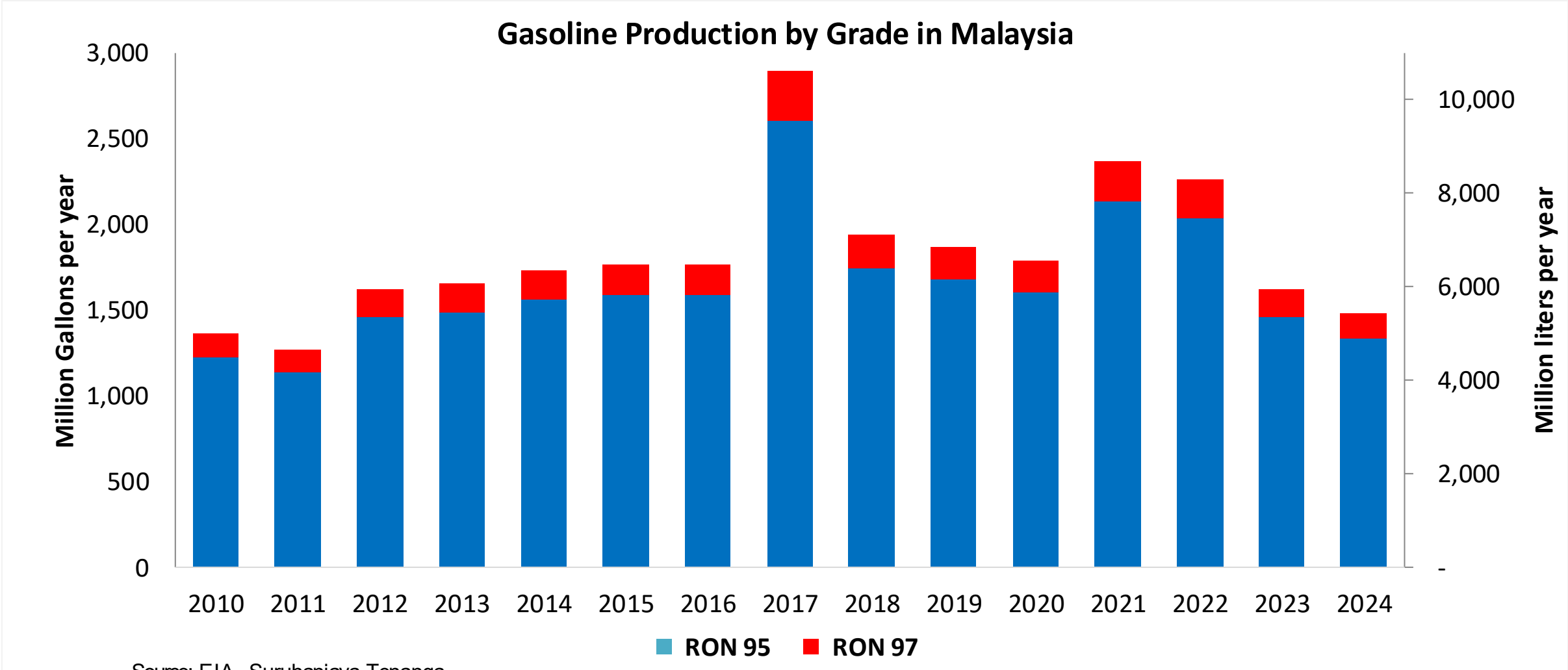
Government support for biofuels has so far targeted biodiesel. The Biofuel Industry Act of 2007 excluded ethanol as the source of alternative fuels under the National Biofuels Policy of 2006. As there is no policy support for ethanol, there is no production of fuel ethanol and minimal consumption. In 2024, ethanol consumption in Malaysia was 16 million liters (4 million gallons) representing 0.1% of gasoline demand.

Gasoline Demand in Malaysia



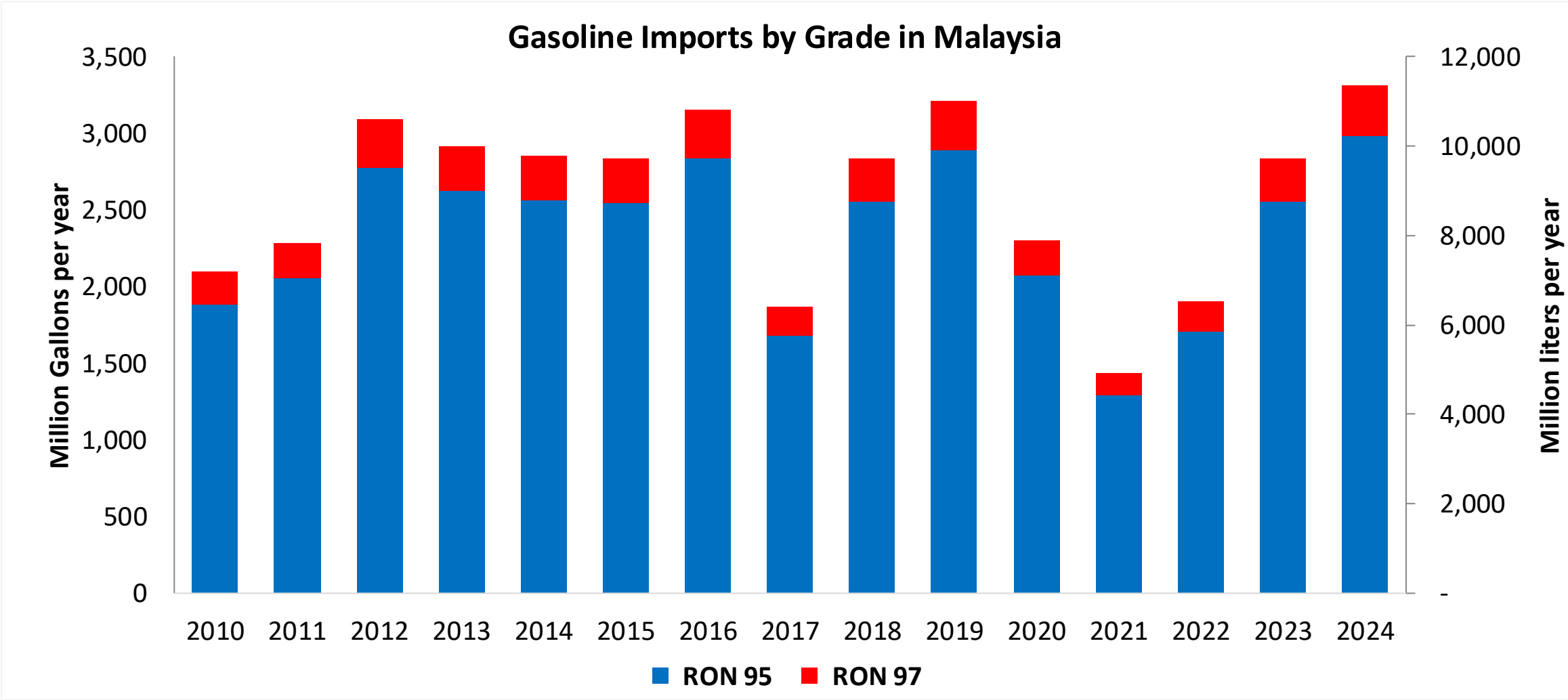
Source: EIA, Suruhanjaya Tenaga

Gasoline Production in Malaysia



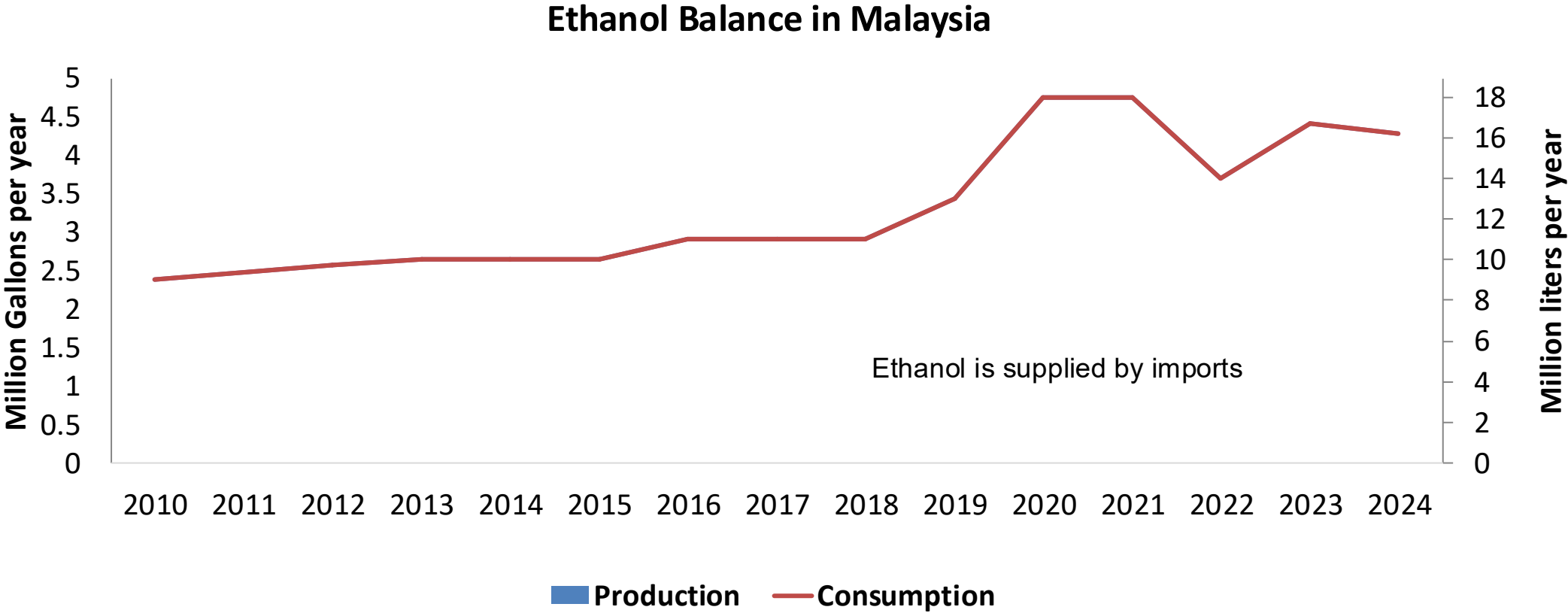
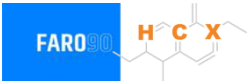
Source: EIA, Suruhanjaya Tenaga

Gasoline Net Imports in Malaysia



Source: EIA, Suruhanjaya Tenaga

Ethanol Balance in Malaysia



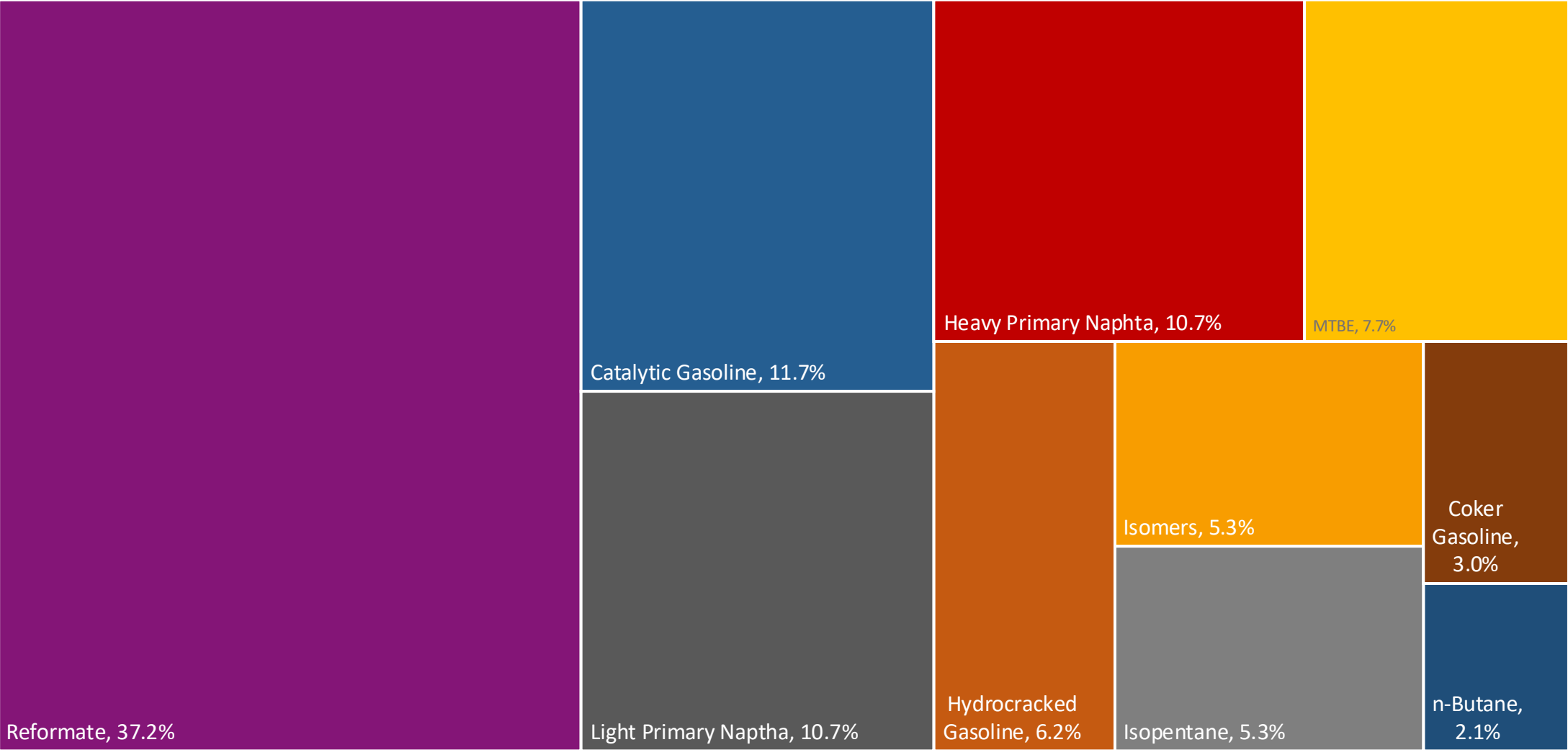
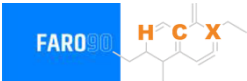
Source: USDA, USGrains

Gasoline Quality in Malaysia

Name	MS 118-3:2011			
Implementation Year	2011			
Applicability	Nationwide			
Grade	RON 95		RON 97	
	Min	Max	Min	Max
RON	95.0		97.0	
MON				
AKI				
Benzene (%vol)		1.0		1.0
Aromatics (% vol)		35.0		35.0
Olefins (%vol)		18.0		18.0
Lead (g/L)		0.013		0.013
Manganese (mg/L)				
Sulfur (ppm)		50		50
Oxygen (%v)				
Ethanol (%vol)				
MTBE (%vol)				
RVP (psi)	60.0	70.0	60.0	70.0

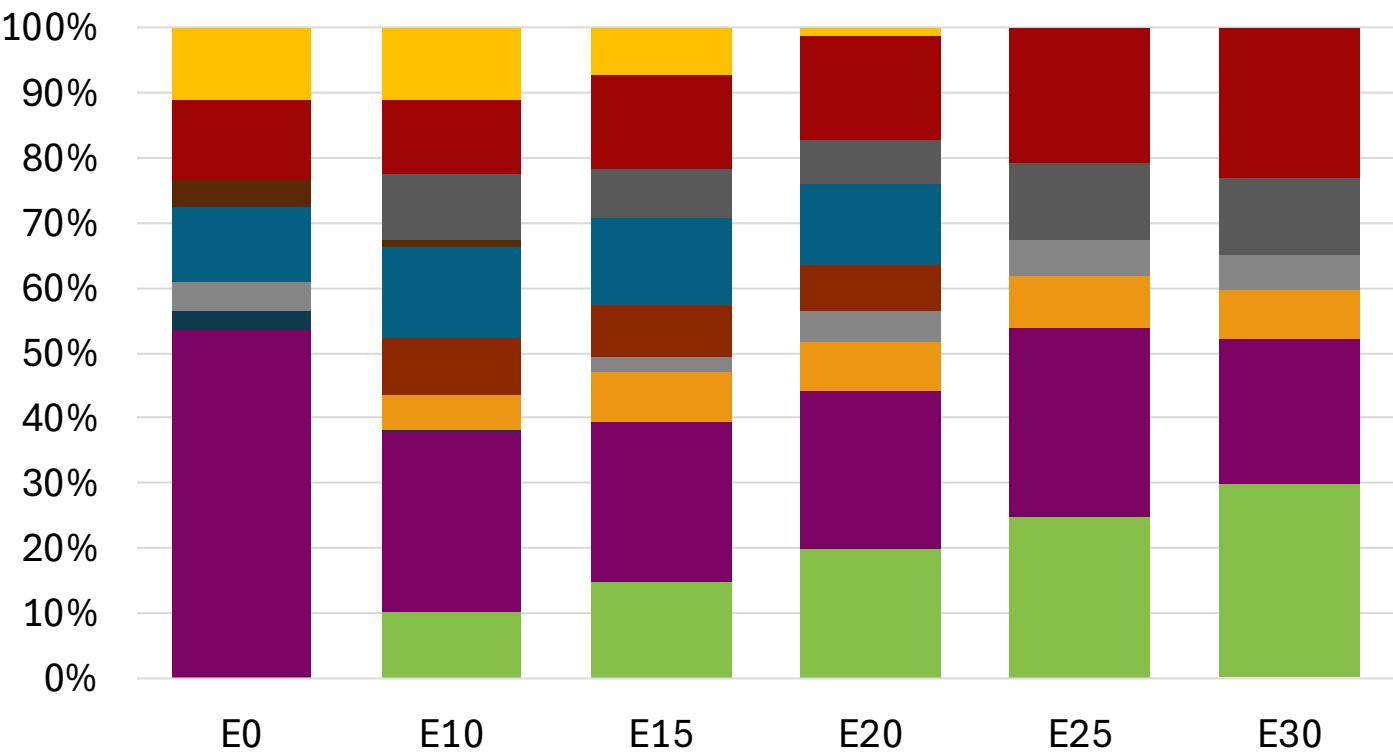
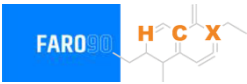
Source: Department of Standards Malaysia, Euro 4 Spec

Malaysia Available Blending Components



Source: Faro90

Malaysia – Gasoline 95 – Constant Octane

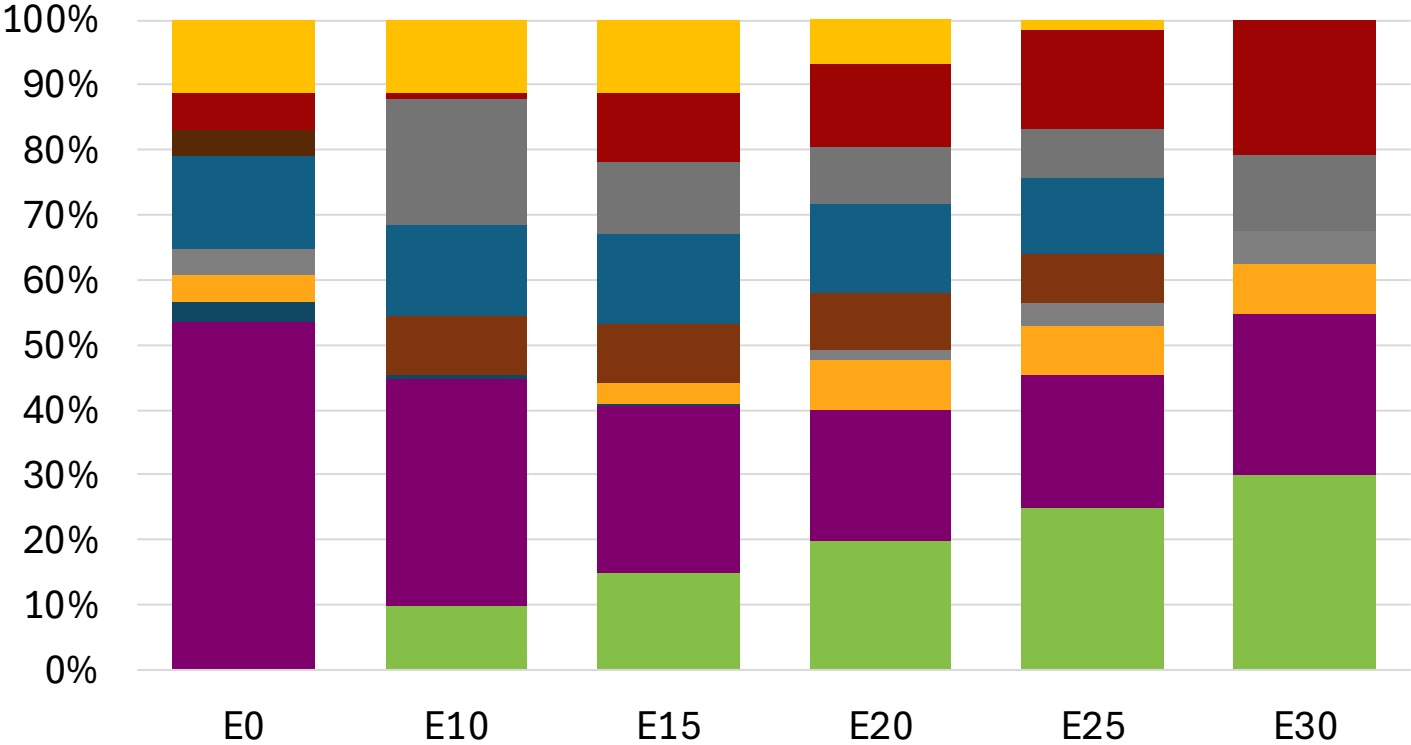
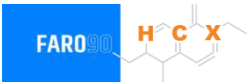


Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.6	96.6
Price (US\$/gal)	\$ 2.360	\$ 2.232	\$ 2.164	\$ 2.102	\$ 2.025	\$ 1.975

Source: Faro90

Malaysia - Gasoline 97 - Constant Octane

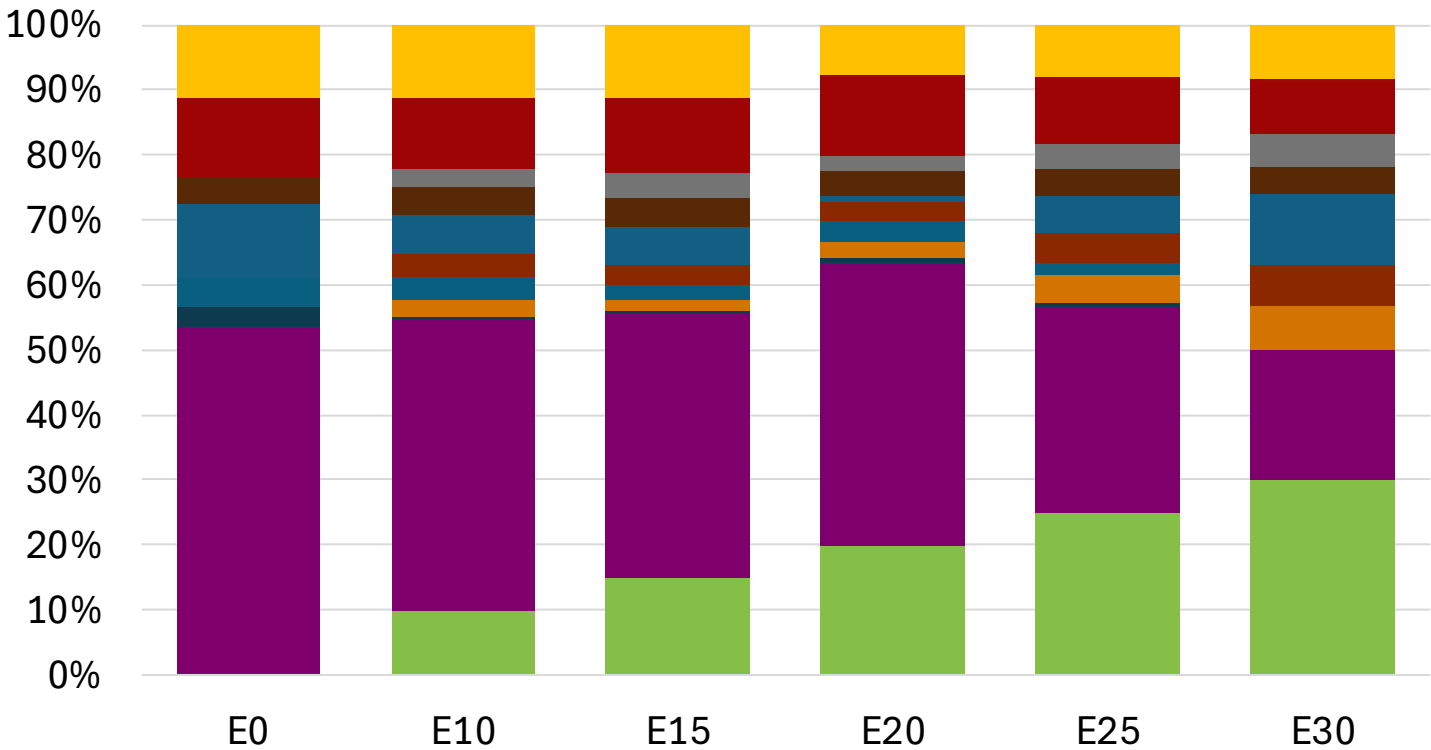


Average CIF 2024 Prices

Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.5
Price (US\$/gal)	\$ 2.466	\$ 2.345	\$ 2.249	\$ 2.187	\$ 2.116	\$ 2.033

Source: Faro90

Malaysia – Gasoline 95 – Increased Octane

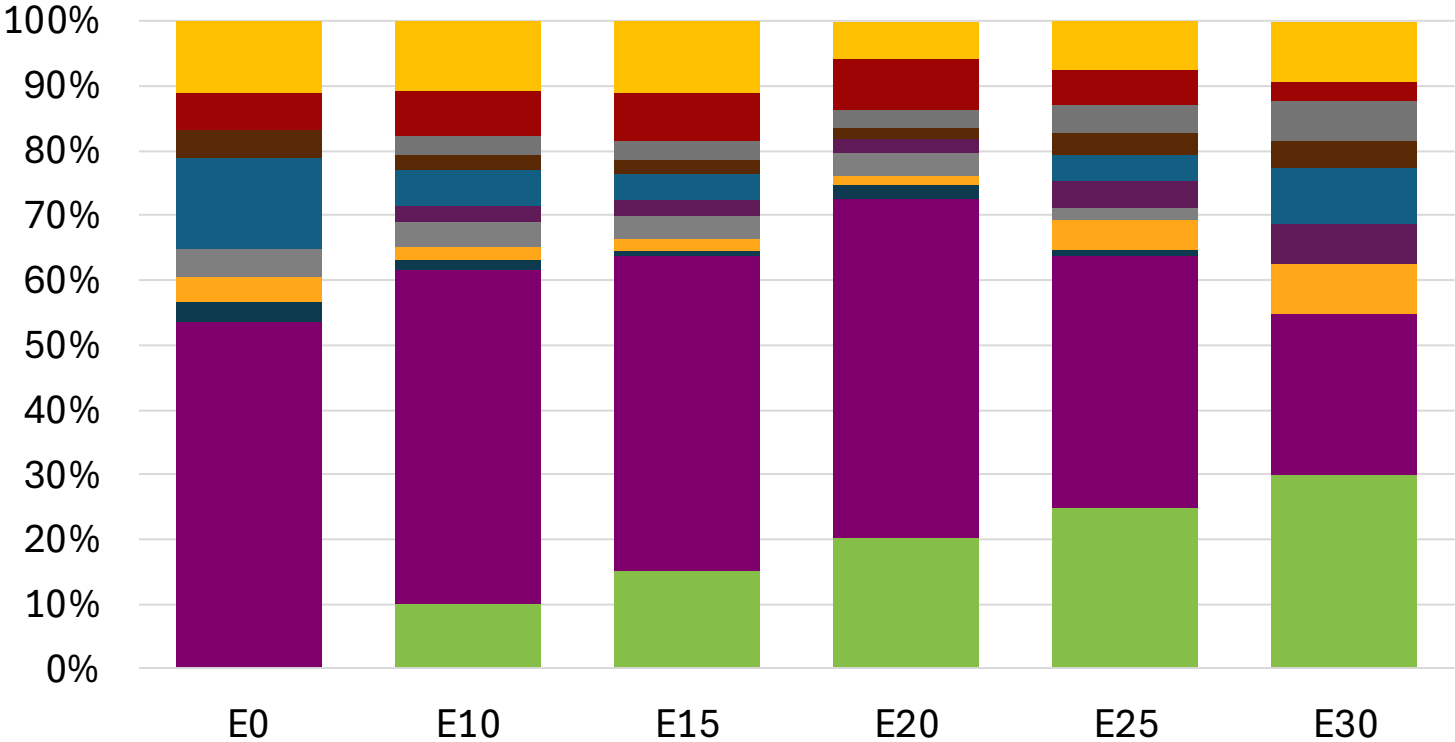


Average CIF 2024
Prices

Octane (RON)	95.0	97.8	99.4	100.5	101.2	103.2
Price (US\$/gal)	\$ 2.360	\$ 2.360	\$ 2.360	\$ 2.360	\$ 2.360	\$ 2.360

Source: Faro90

Malaysia – Gasoline 97 – Increased Octane

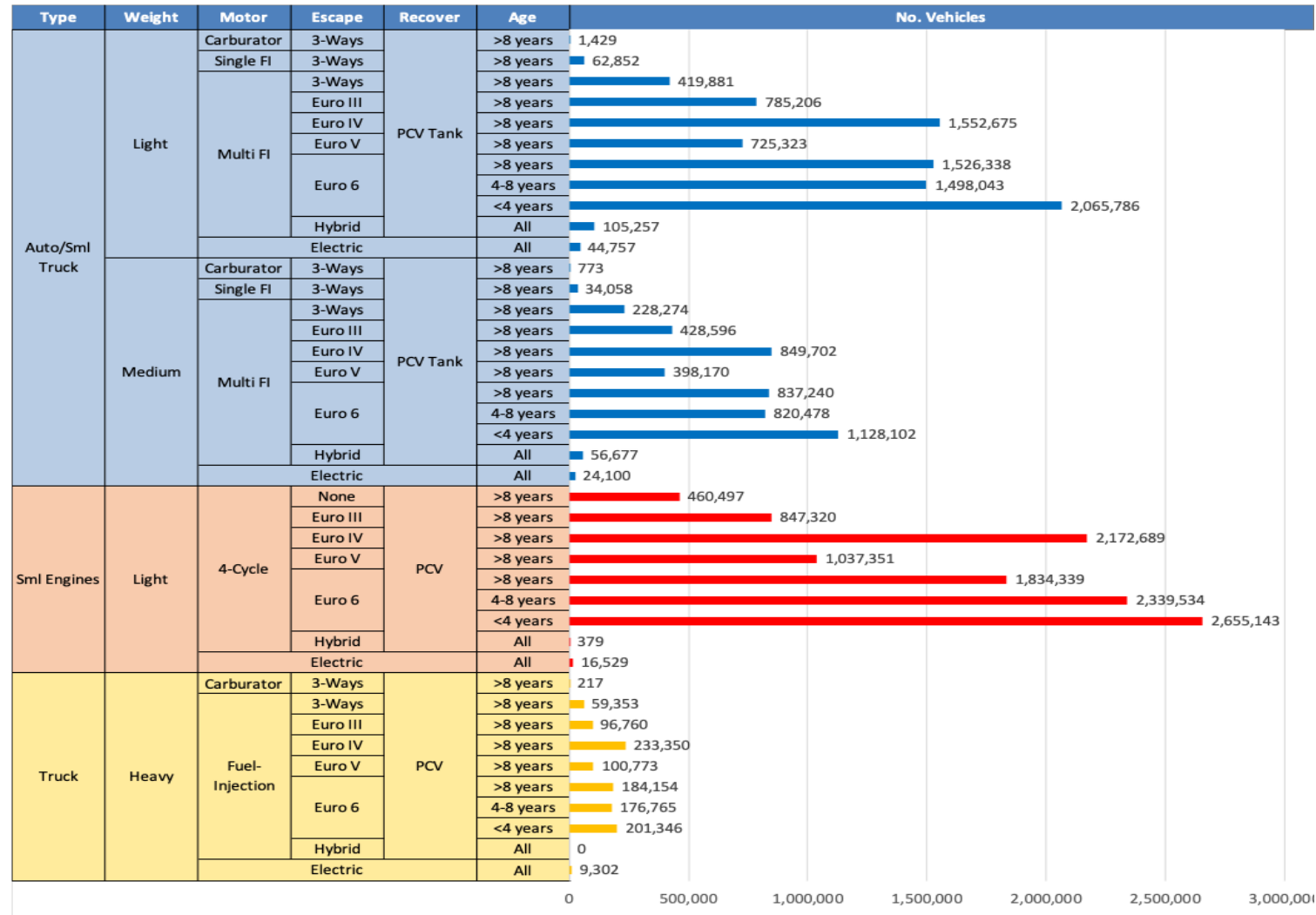


Average CIF 2024
Prices

Octane (RON)		97.0	99.8	101.3	102.1	102.9	104.9
Price (US\$/gal)	\$	2.466	\$ 2.466	\$ 2.466	\$ 2.466	\$ 2.466	\$ 2.466

Source: Faro90

Gasoline Vehicle Fleet - Malaysia

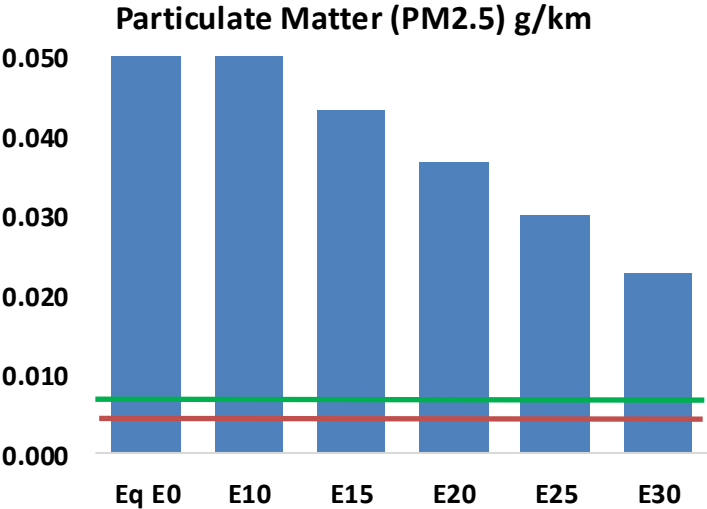
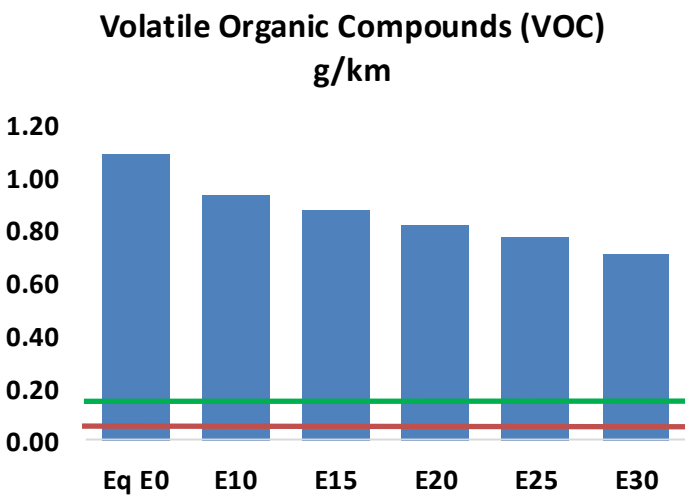
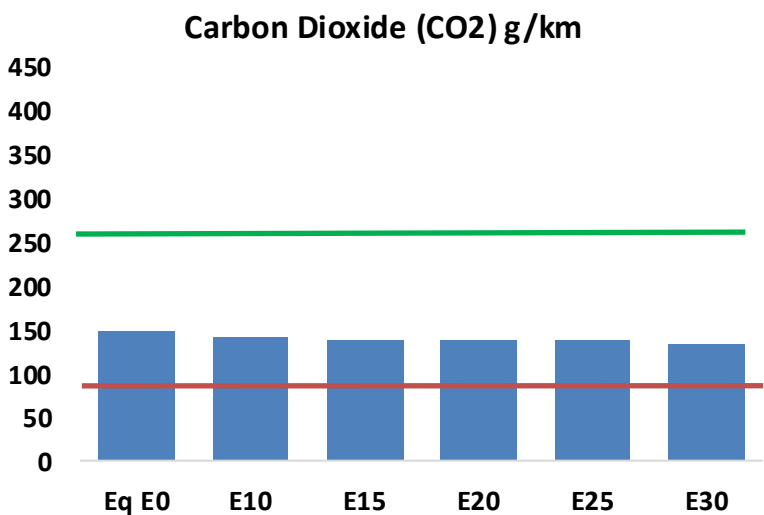
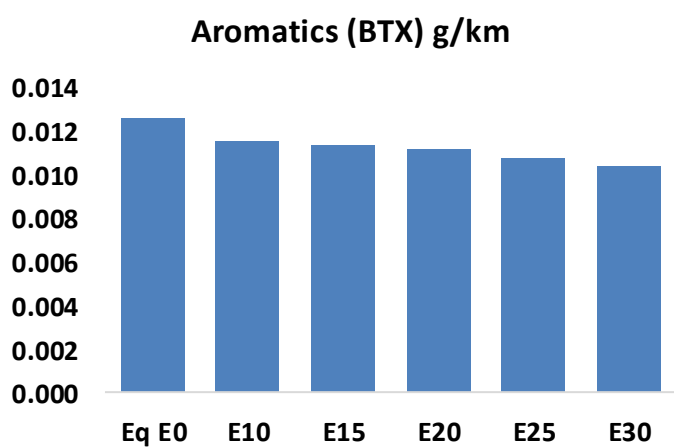
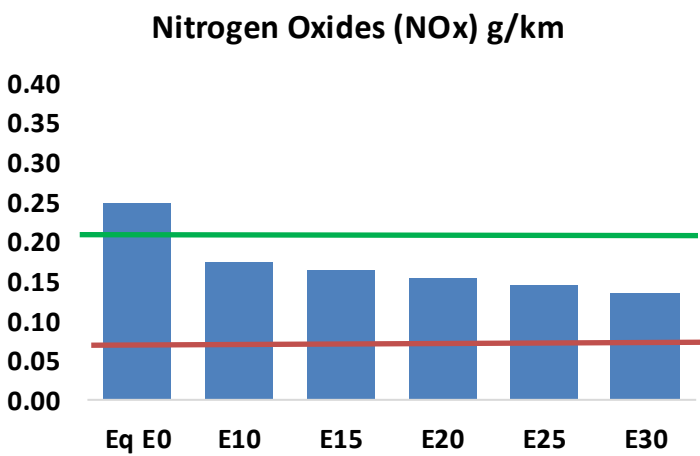
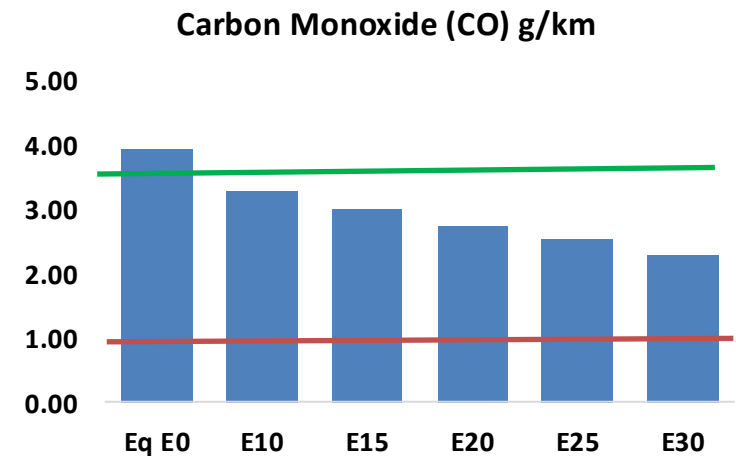
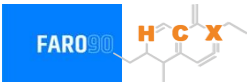


Vehicle Fleet: 26,019,519
Gasoline Vehicle Fleet: 22,285,447

Source: Data Gov Malaysia

Malaysia – Vehicular Emissions

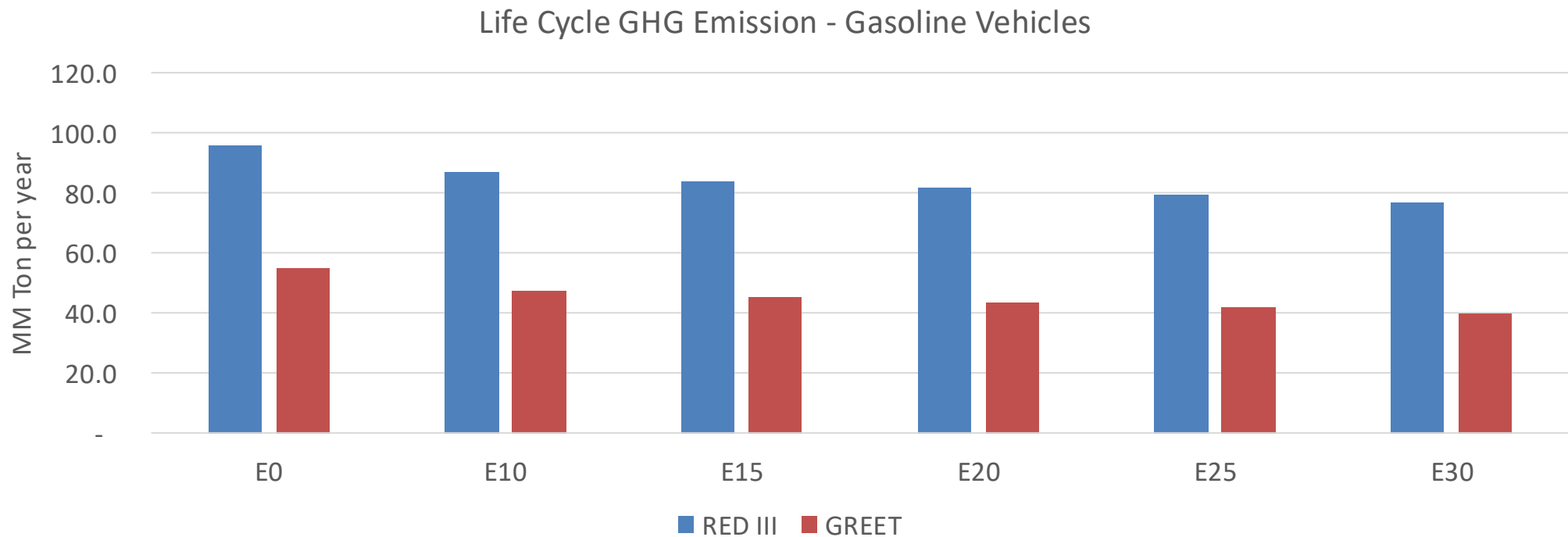
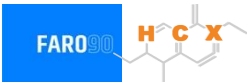
TIER III BIN 125 United States
Euro 6 Standard



Malaysia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,144,339	1,967,241	1,790,144	1,642,629	1,493,082	1,377,756	1,247,742	-17%	-30%
CO2	80,754,602	78,678,055	76,716,872	75,174,459	74,411,212	74,065,375	72,700,137	-5%	-8%
VOC	594,330	553,078	511,826	478,887	446,139	421,077	387,209	-14%	-25%
NOx	135,754	101,816	95,028	89,564	84,466	78,846	72,900	-30%	-38%
SOx	1,829	1,691	1,553	1,436	1,324	1,202	1,074	-15%	-28%
NH3	36,345	35,986	35,626	35,795	36,197	36,481	36,716	-2%	0%
N2O	1,108	1,102	1,098	1,103	1,126	1,139	1,151	-1%	2%
CH4	130,499	130,499	130,499	132,457	135,328	137,807	140,156	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,805	2,654	2,503	2,393	2,287	2,207	2,095	-11%	-18%
Acetaldehyde	10,337	11,864	17,408	26,510	36,116	41,270	48,765	68%	249%
Formaldehyde	43,189	41,989	48,948	57,475	60,214	66,612	72,515	13%	39%
Benzene	6,902	6,633	6,285	6,185	6,134	5,866	5,677	-9%	-11%
PM2.5	5,686	5,049	4,412	3,831	3,253	2,640	2,000	-22%	-43%
PM10	29,526	26,217	22,908	19,893	16,893	13,708	10,388	-22%	-43%

Malaysia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	407.7
Target @ 2035 (MMT/year)	285.4
Delta	122.3

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	13.7%	17.4%	20.5%	23.4%	27.2%
RED II/III	0.0%	9.5%	12.3%	14.8%	17.2%	20.0%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	6.1%	7.8%	9.1%	10.5%	12.2%
RED II/III	0.0%	7.4%	9.6%	11.6%	13.5%	15.7%

Source: Greet, RED II/III, climateactiontracker.org, F90

Ethanol Gasoline Blending in South Korea

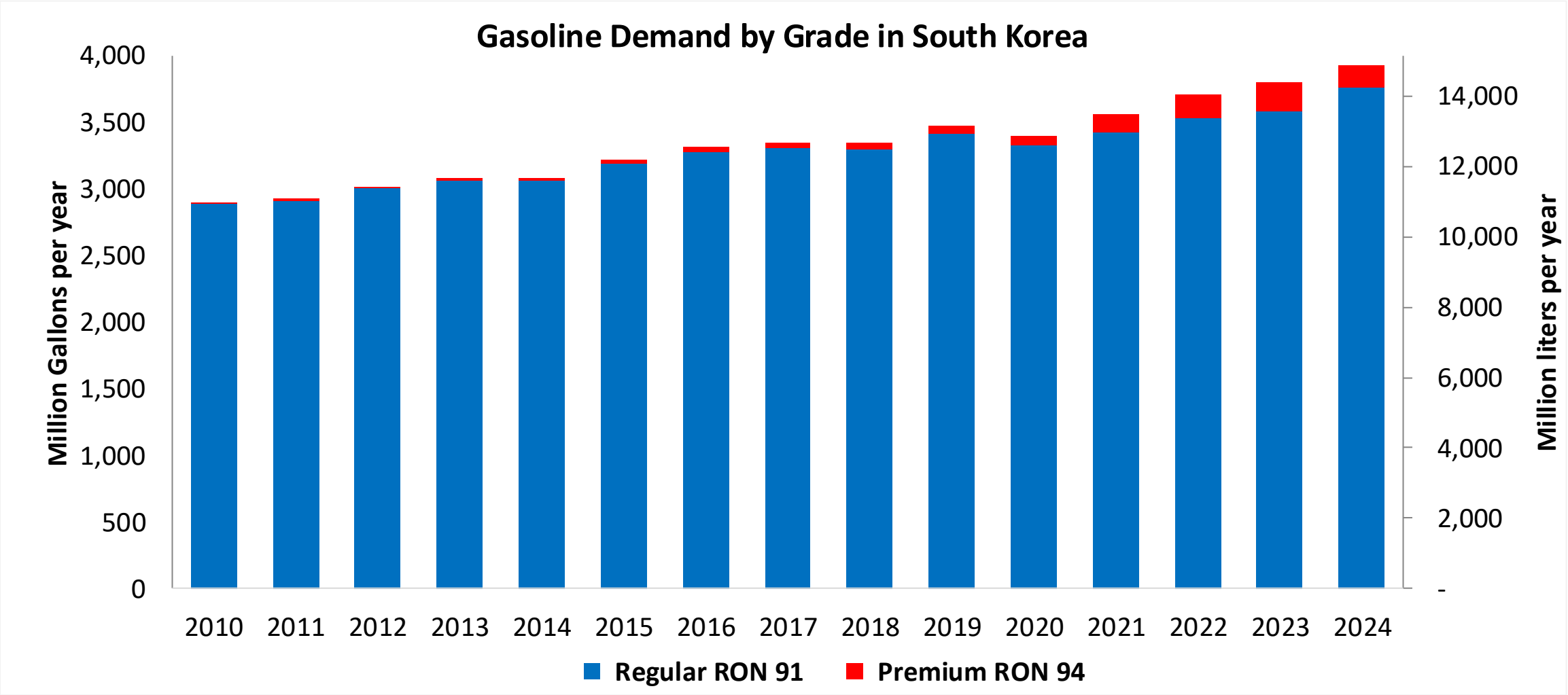


In 2024, gasoline consumption in South Korea was 14,841 million liters (3,920 million gallons) and growth rate was 2.4%. Gasoline production and net imports were 30,840 and -16,000 million liters (8,147 and -4,227 million gallons) correspondingly. South Korea offers two main grades: RON 91 and RON 94) with an average octane of 91.1 RON.

Ethanol is blended in South Korea to offer non mandated E3-E10 gasoline. Ethanol consumption in South Korea last year was 175 million liters (46 million gallons) representing 1.2% of gasoline demand. Due to a lack of domestic production, South Korea is dependent on imports for its bioethanol supply. The government plans to increase the blending ratio, aiming for a 4% bioethanol-gasoline blend by 2025 and 8% by 2030 under the Renewable Fuel Standard (RFS).

Source: Petronet Korea, IEA

Gasoline Demand in South Korea

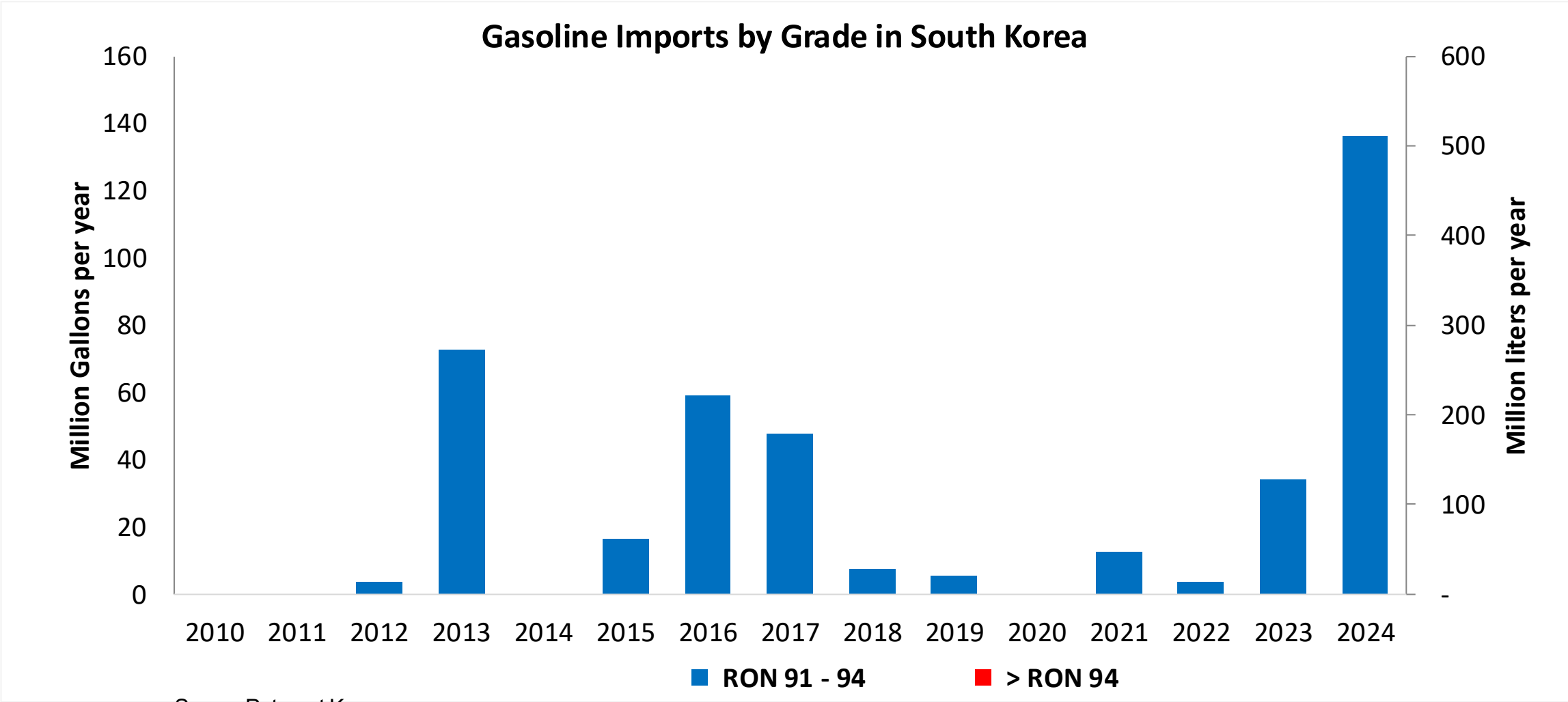


Source: Petronet Korea

The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexadiene. The structure features a six-membered ring with two double bonds. Substituents include a hydrogen atom (H), a carbon atom (C), and a group labeled 'X' (which is a vinyl group, -CH=CH₂). The ring also has methyl groups at the 3 and 5 positions.

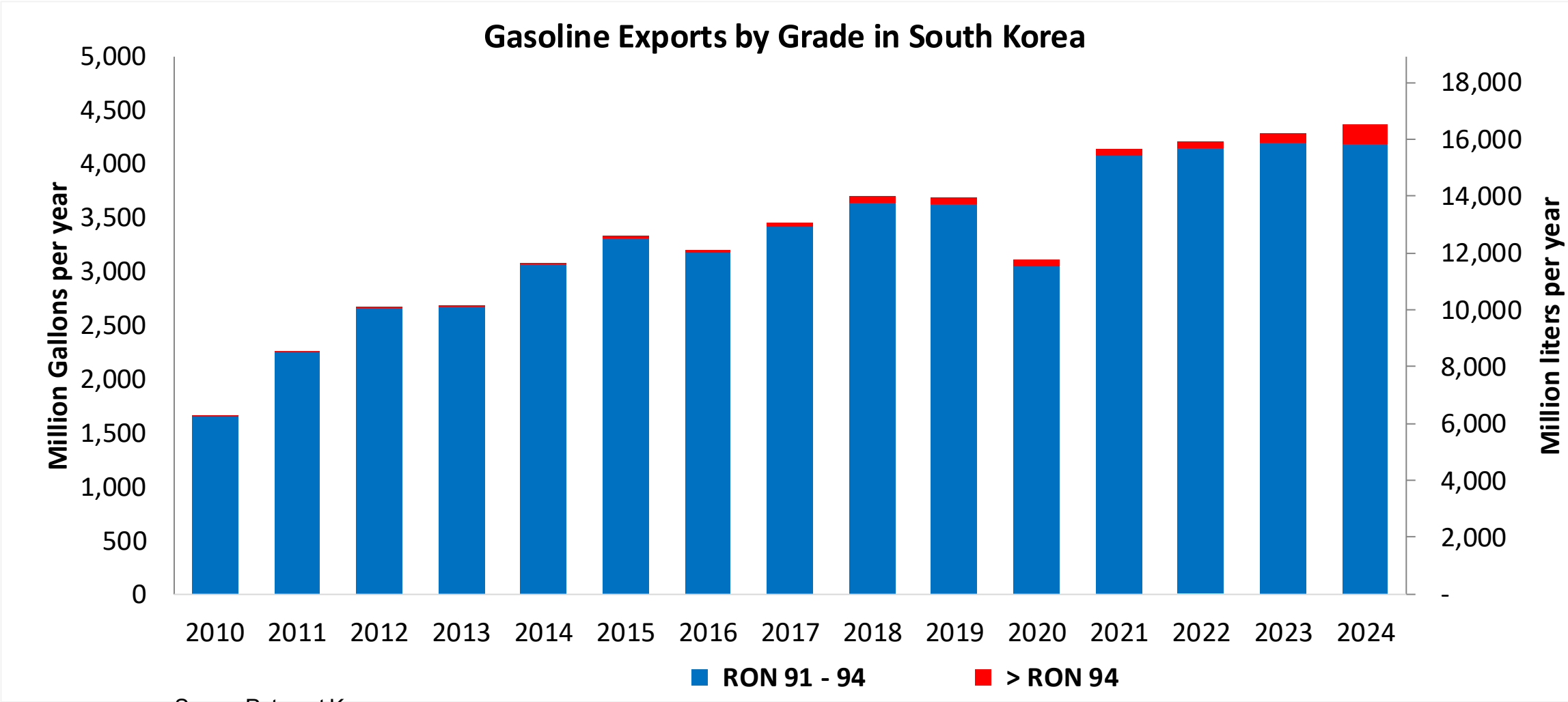


Gasoline Imports in South Korea



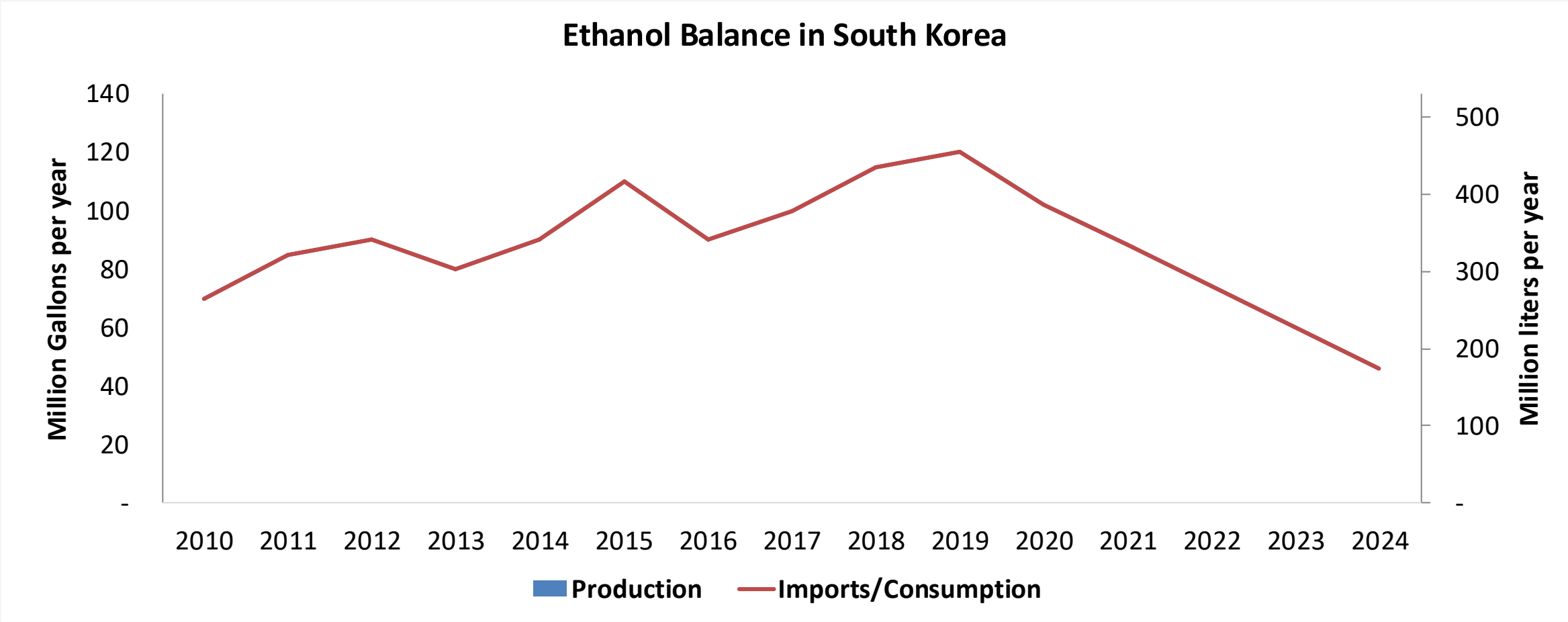
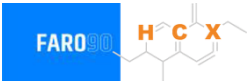
Source: Petronet Korea

Gasoline Exports in South Korea



Source: Petronet Korea

Ethanol Balance in South Korea



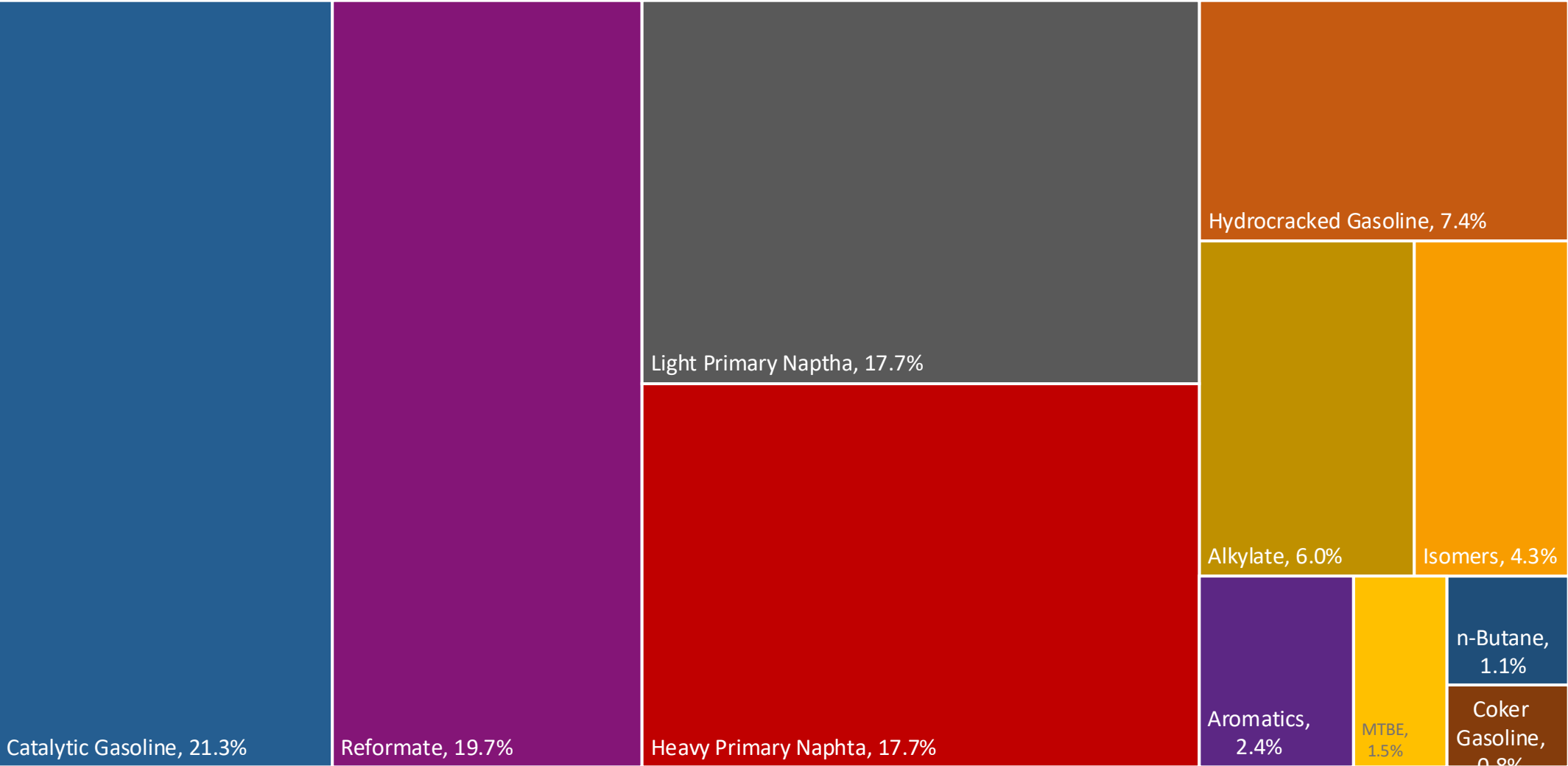
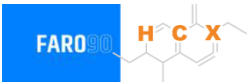
Source: Petronet Korea, IEA

Gasoline Quality in South Korea

Name	Clean Air Conservation Act			
Implementation Year	2009			
Applicability	Nationwide			
Grade	RON 91		RON 94	
	Min	Max	Min	Max
RON	91.0		97.0	
MON				
AKI				
Benzene (%vol)		0.7		0.7
Aromatics (% vol)		24.0		24.0
Olefins (%vol)		18.0		18.0
Lead (g/L)		0.013		0.013
Manganese (mg/L)		-		-
Sulfur (ppm)		10		10
Oxygen (%v)	0.5	2.3	0.5	2.3
Ethanol (%vol)		-		-
MTBE (%vol)		-		-
RVP (psi)		8.7		8.7

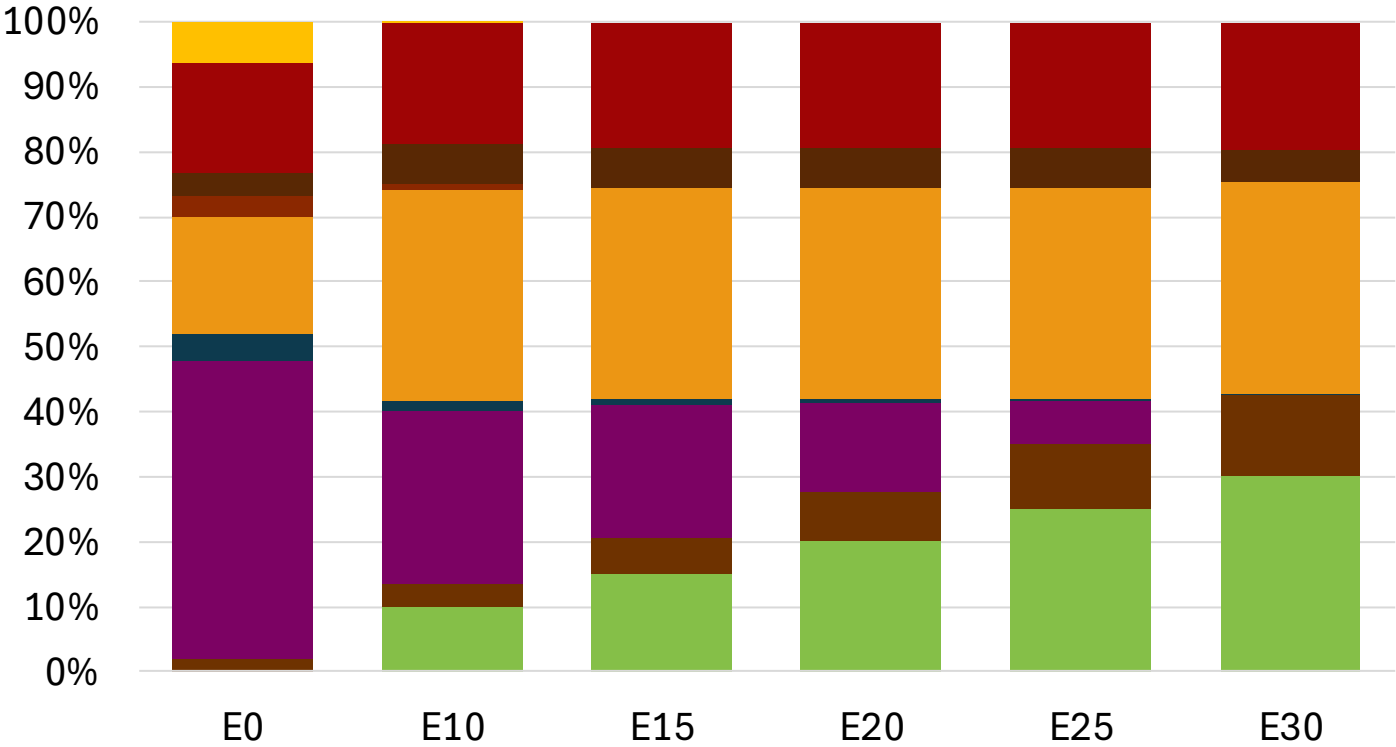
Source: xxx

South Korea Available Blending Components



Source: Faro90

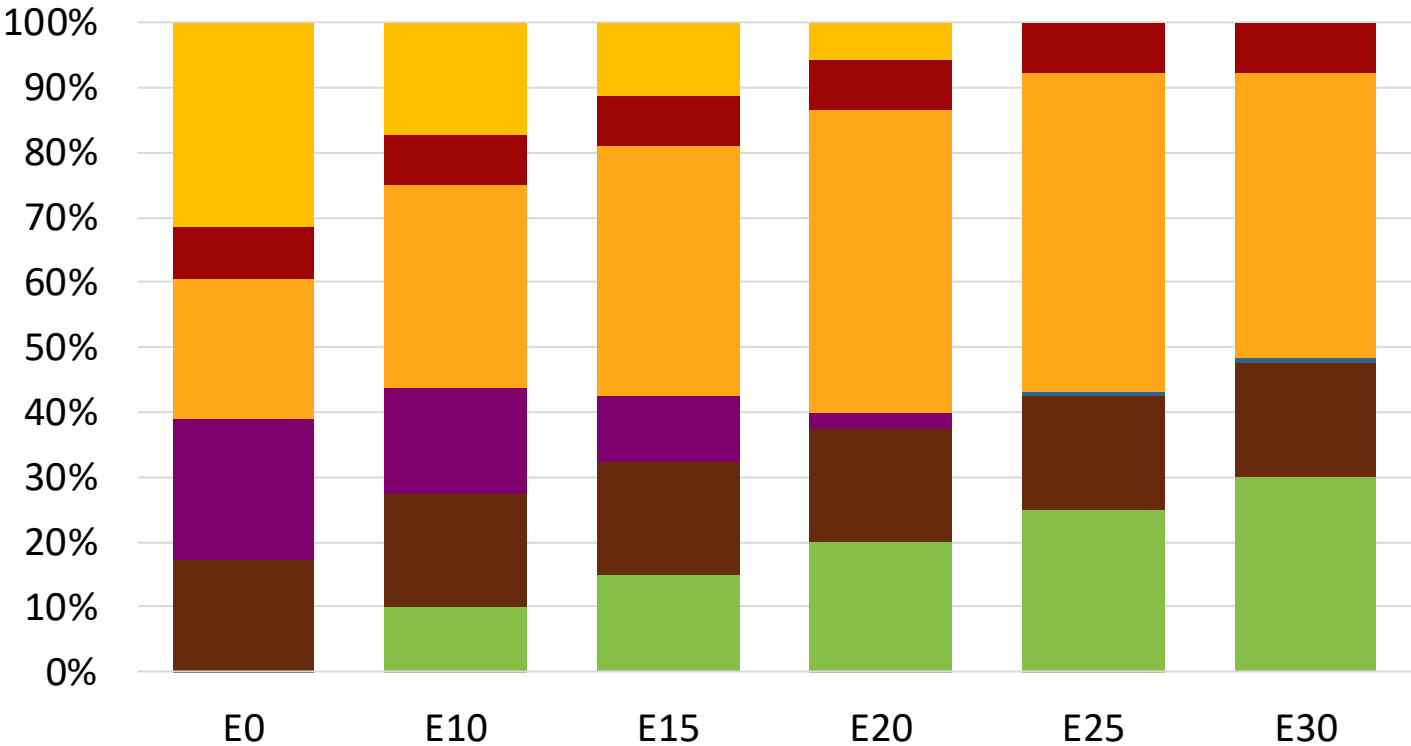
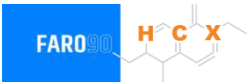
South Korea – Gasoline 91 – Constant Octane



Average CIF 2024 Prices

Source: Faro90

South Korea - Gasoline 94 - Constant Octane

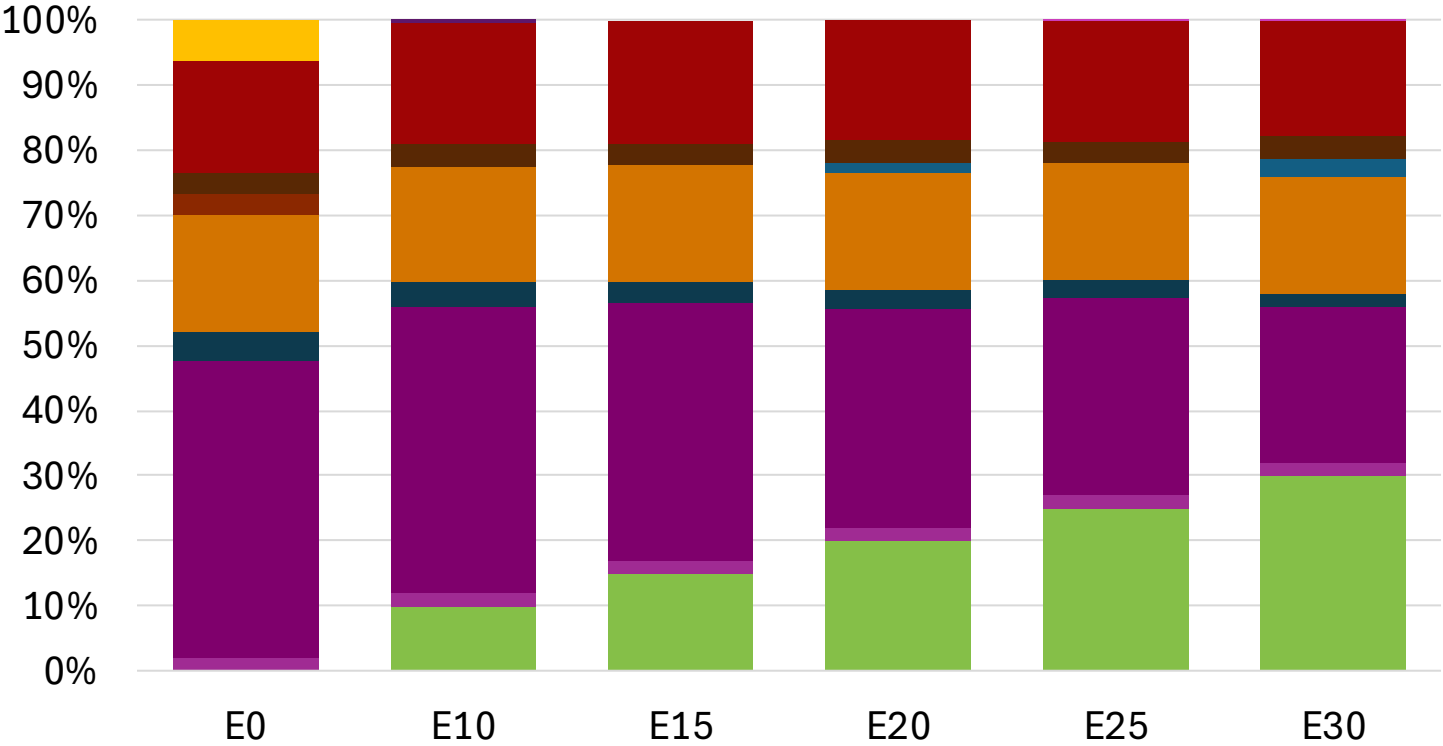


Average CIF 2024 Prices

Octane (RON)	94.0	94.0	94.0	94.0	94.7	96.9
Price (US\$/gal)	\$ 2.240	\$ 2.185	\$ 2.140	\$ 2.086	\$ 2.060	\$ 2.080

Source: Faro90

South Korea – Gasoline 91 – Increased Octane

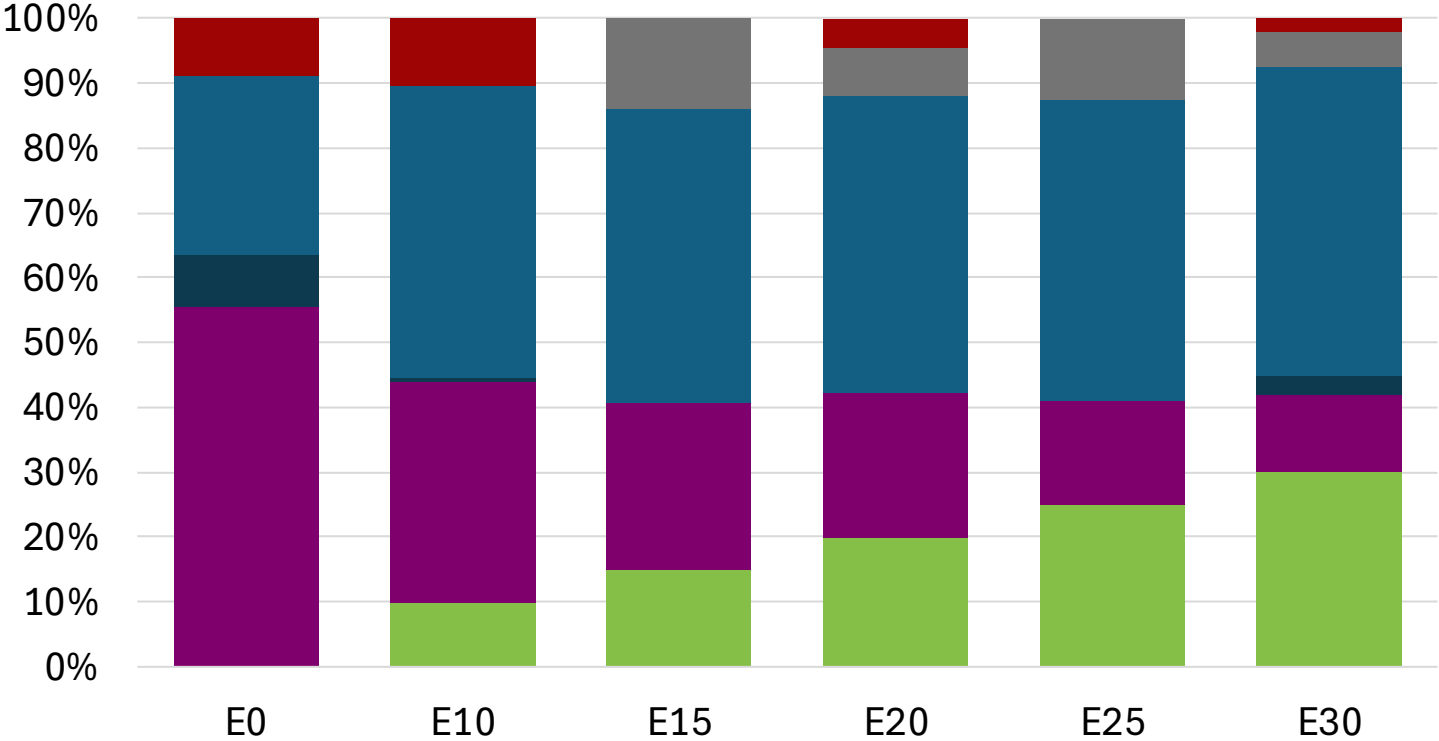
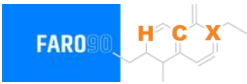


Average CIF 2024
Prices

Octane (RON)	91.0	93.7	95.4	97.1	98.8	100.3
Price (US\$/gal)	\$ 2.257	\$ 2.257	\$ 2.257	\$ 2.257	\$ 2.257	\$ 2.257

Source: Faro90

South Korea – Gasoline 94 – Increased Octane

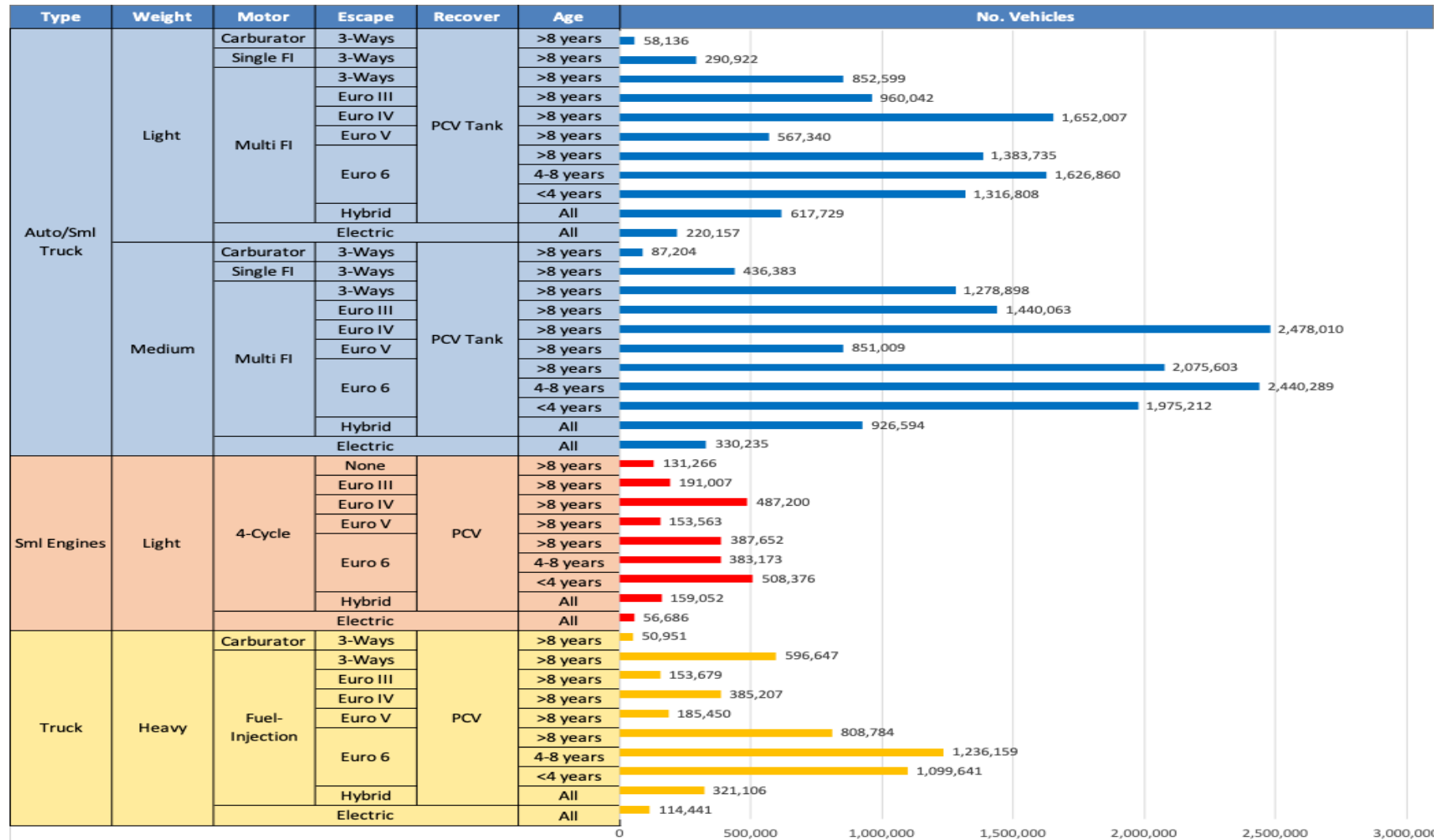


Average CIF 2024 Prices

Octane (RON)	94.0	94.0	94.2	95.6	97.3	99.5
Price (US\$/gal)	\$ 2.240	\$ 2.240	\$ 2.240	\$ 2.240	\$ 2.240	\$ 2.240

Source: Faro90

Gasoline Vehicle Fleet - South Korea

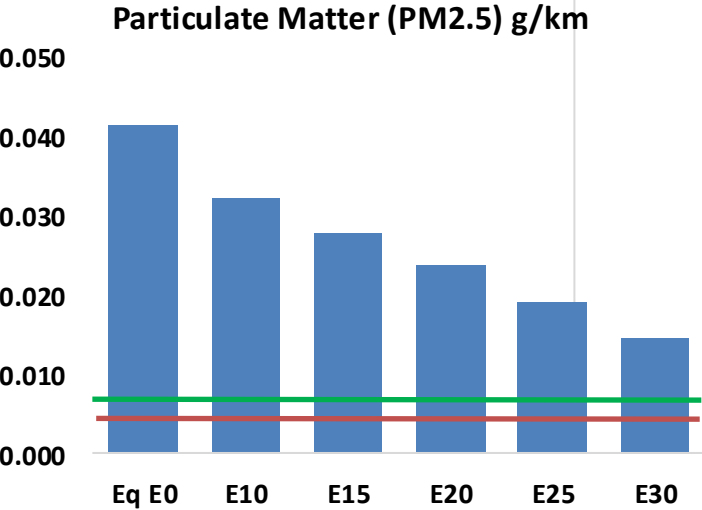
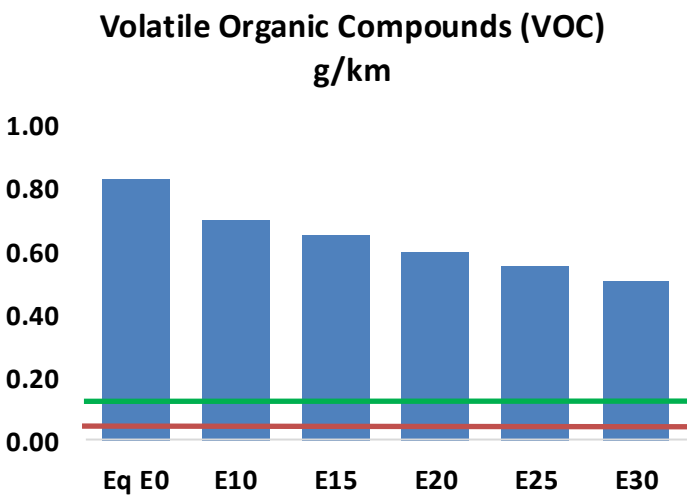
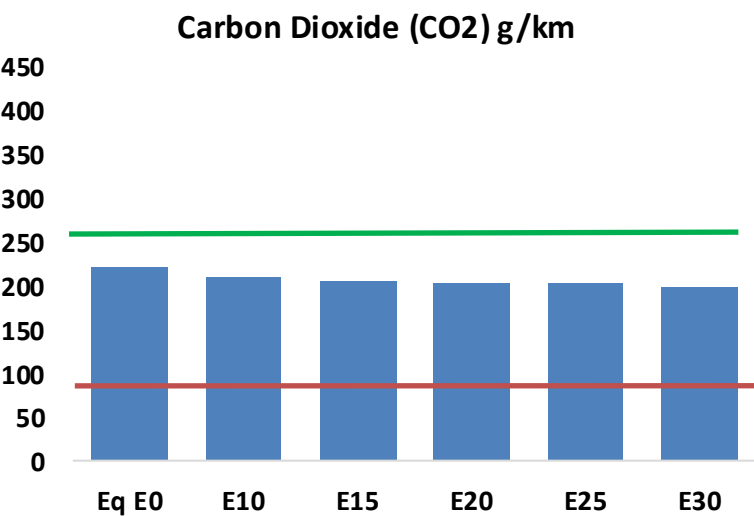
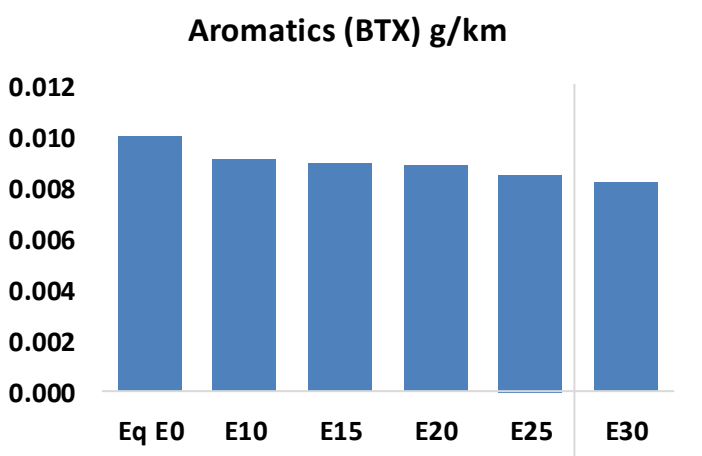
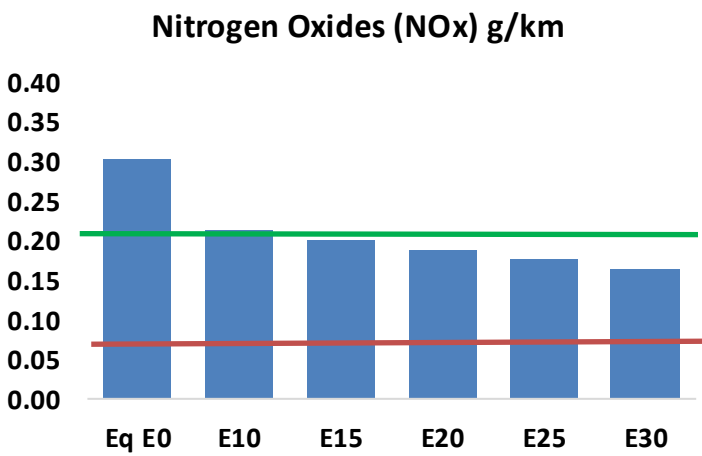
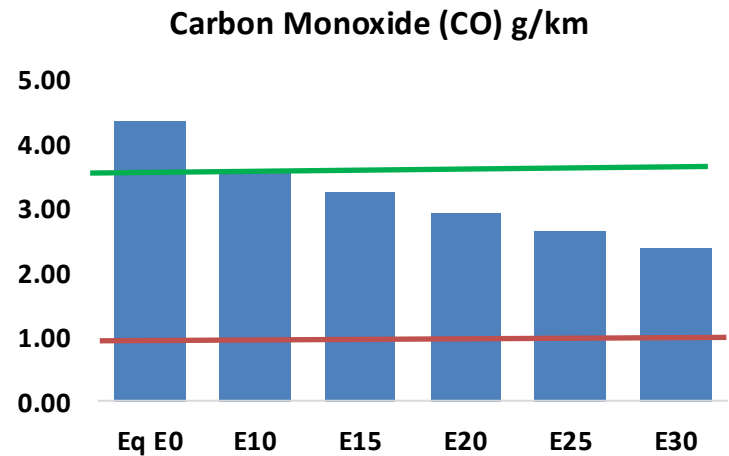
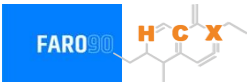


Vehicle Fleet: 31,275,876
Gasoline Vehicle Fleet: 25,599,696

Source: MOLIT

South Korea – Vehicular Emissions

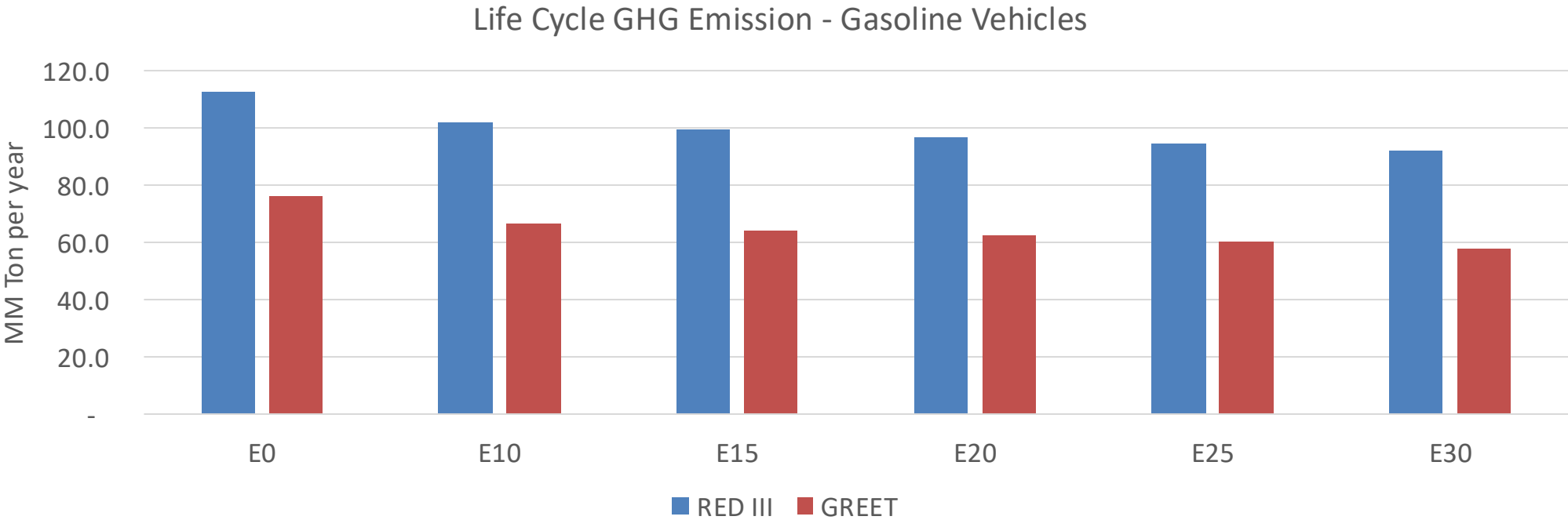
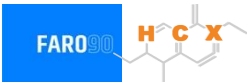
TIER III BIN 125 United States
Euro 6 Standard



South Korea – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,591,317	2,359,846	2,128,374	1,932,397	1,732,283	1,577,533	1,414,428	-18%	-33%
CO2	130,279,254	126,929,216	123,765,291	121,276,958	120,045,633	119,459,096	117,257,122	-5%	-8%
VOC	489,471	452,129	414,788	384,244	353,532	329,925	298,394	-15%	-28%
NOx	180,005	135,004	126,004	118,758	111,999	104,547	96,663	-30%	-38%
SOx	1,520	1,405	1,290	1,193	1,100	999	892	-15%	-28%
NH3	46,891	46,427	45,962	46,181	46,700	47,065	47,369	-2%	0%
N2O	2,724	2,709	2,698	2,712	2,768	2,798	2,830	-1%	2%
CH4	101,103	101,103	101,103	102,619	104,844	106,765	108,584	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,005	1,895	1,785	1,704	1,626	1,567	1,484	-11%	-19%
Acetaldehyde	6,582	7,554	11,084	16,879	22,996	26,277	31,049	68%	249%
Formaldehyde	25,164	24,465	28,519	33,488	35,083	38,812	42,251	13%	39%
Benzene	5,947	5,715	5,415	5,330	5,285	5,055	4,891	-9%	-11%
PM2.5	7,723	6,858	5,992	5,204	4,419	3,586	2,717	-22%	-43%
PM10	16,819	14,934	13,049	11,332	9,623	7,808	5,917	-22%	-43%

South Korea – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	682.3
Target @ 2035 (MMT/year)	332.3
Delta	349.9

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.7%	15.9%	18.4%	20.8%	24.1%
RED II/III	0.0%	9.4%	11.9%	14.0%	15.9%	18.4%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.8%	3.5%	4.0%	4.5%	5.2%
RED II/III	0.0%	3.0%	3.8%	4.5%	5.1%	5.9%

Source: Greet, RED II/III, climateactiontracker.org, F90

Ethanol Gasoline Blending in Taiwan

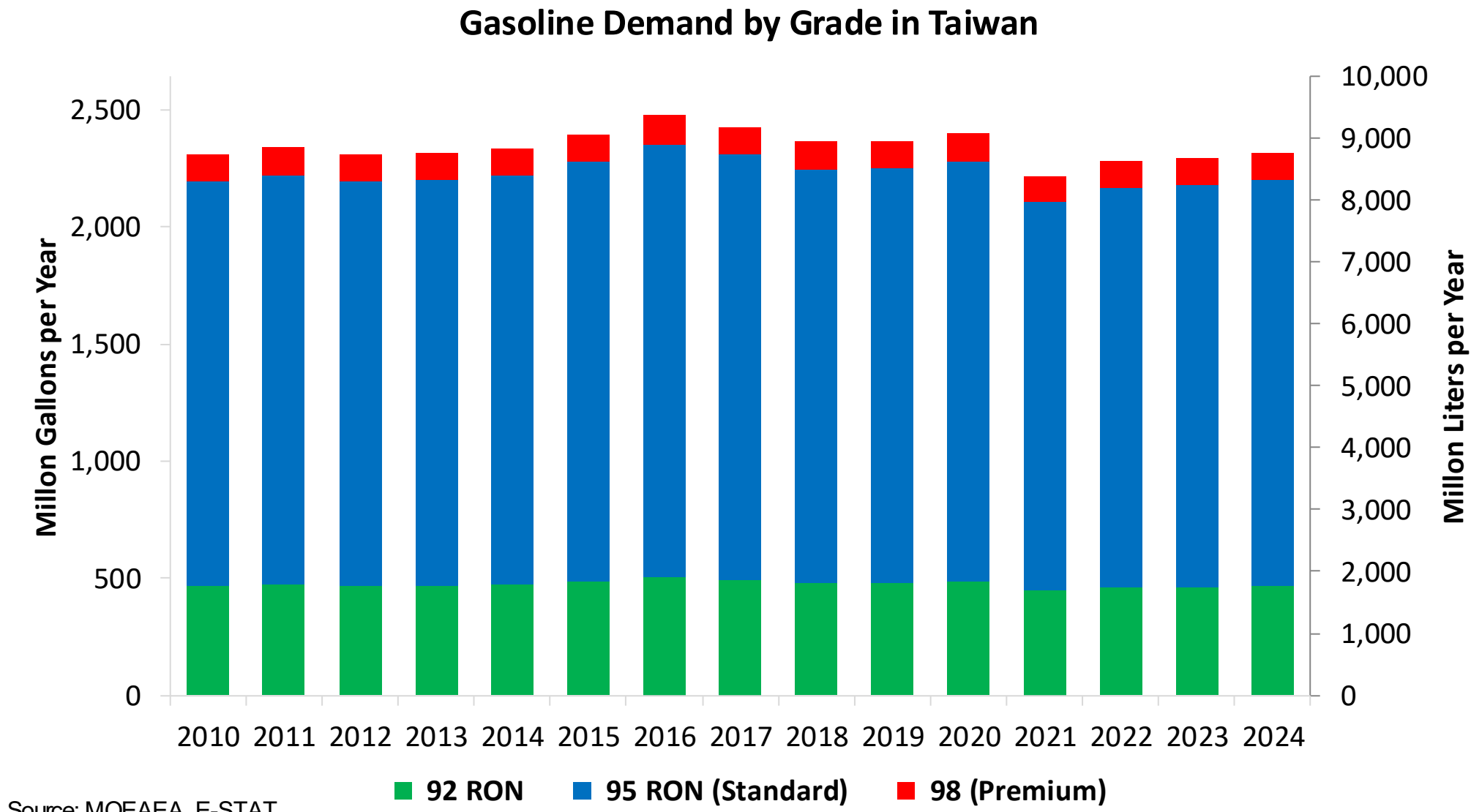


In 2024, gasoline consumption in Taiwan was 8,761 million liters (2,314 million gallons) and growth rate was -0.4%. Gasoline production and net imports were 12,450 and -3,689 million liters (3,289 and -975 million gallons) correspondingly. Taiwan offers three main grades: RON 92, RON 95 and RON 98) with an average octane of 94.5 RON.

Taiwan has policies to promote bioethanol in transportation to reduce imported fuel dependency and lower carbon emissions, using it as an oxygenate additive. While there was a limited trial of E3 gasohol in the mid-2000s, the lack of domestic production and insufficient incentives led to its discontinuation. Only a few gas stations offer it. Ethanol is supplied by imports and corresponds to an average content of 0.2%. Future bioethanol use depends on developing profitable cellulosic ethanol production from non-food waste like rice straw and food waste, rather than imported feedstocks.

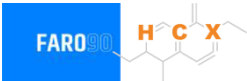
Source: MOEAEA, E-STAT, MOENV, USDA

Gasoline Demand in Taiwan

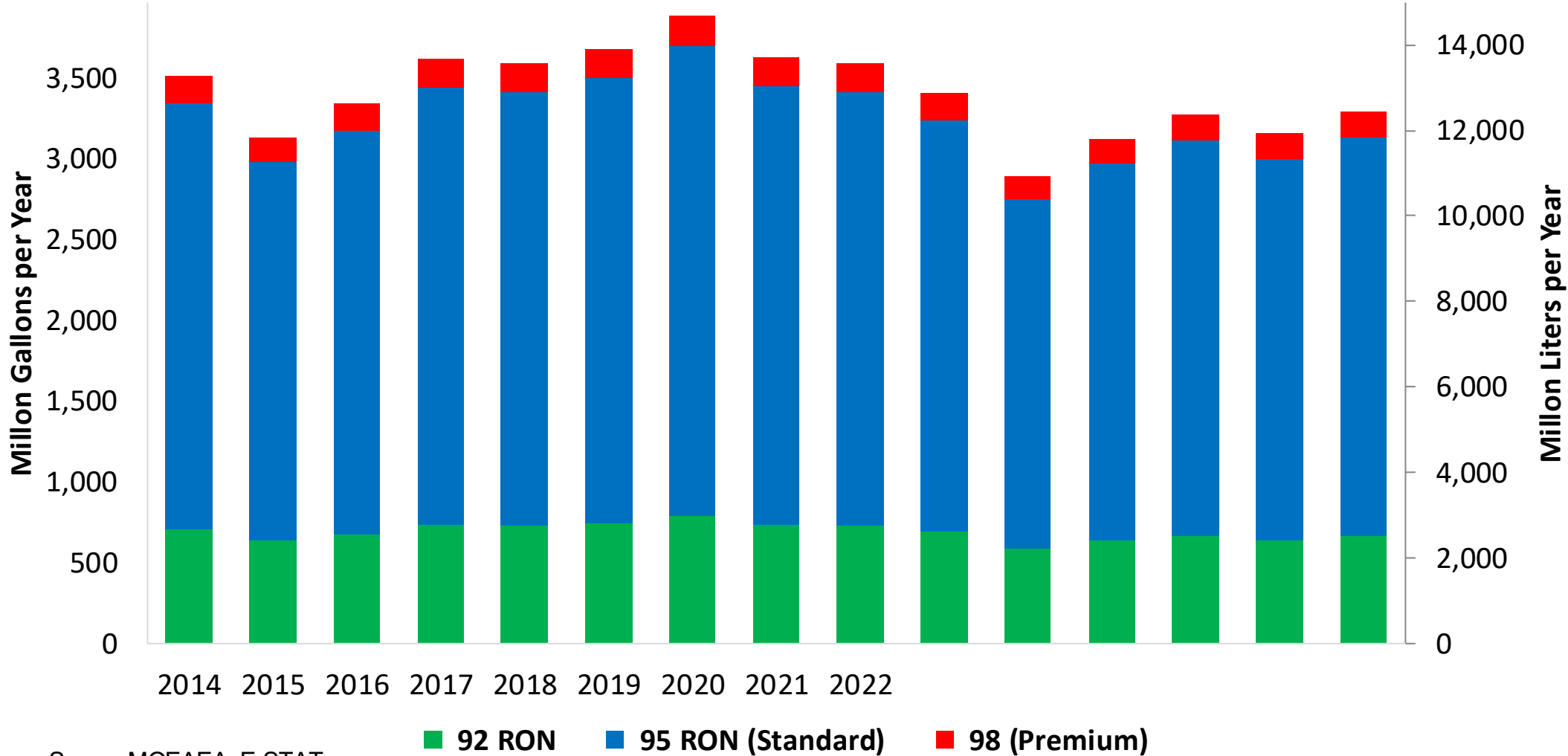


Source: MOEAEA, E-STAT

Gasoline Production in Taiwan



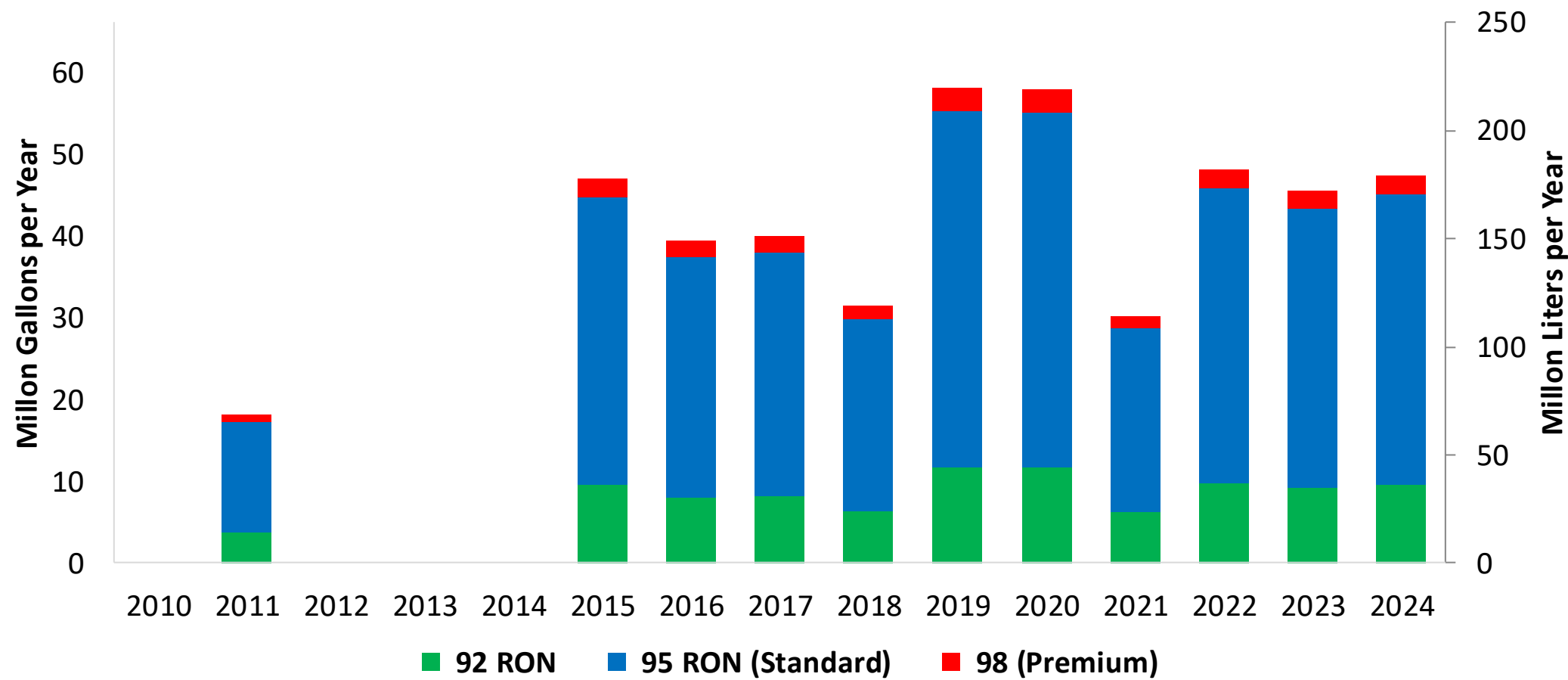
Gasoline Productionby Grade in Taiwan



Source: MOEAEA, E-STAT

Gasoline Imports in Taiwan

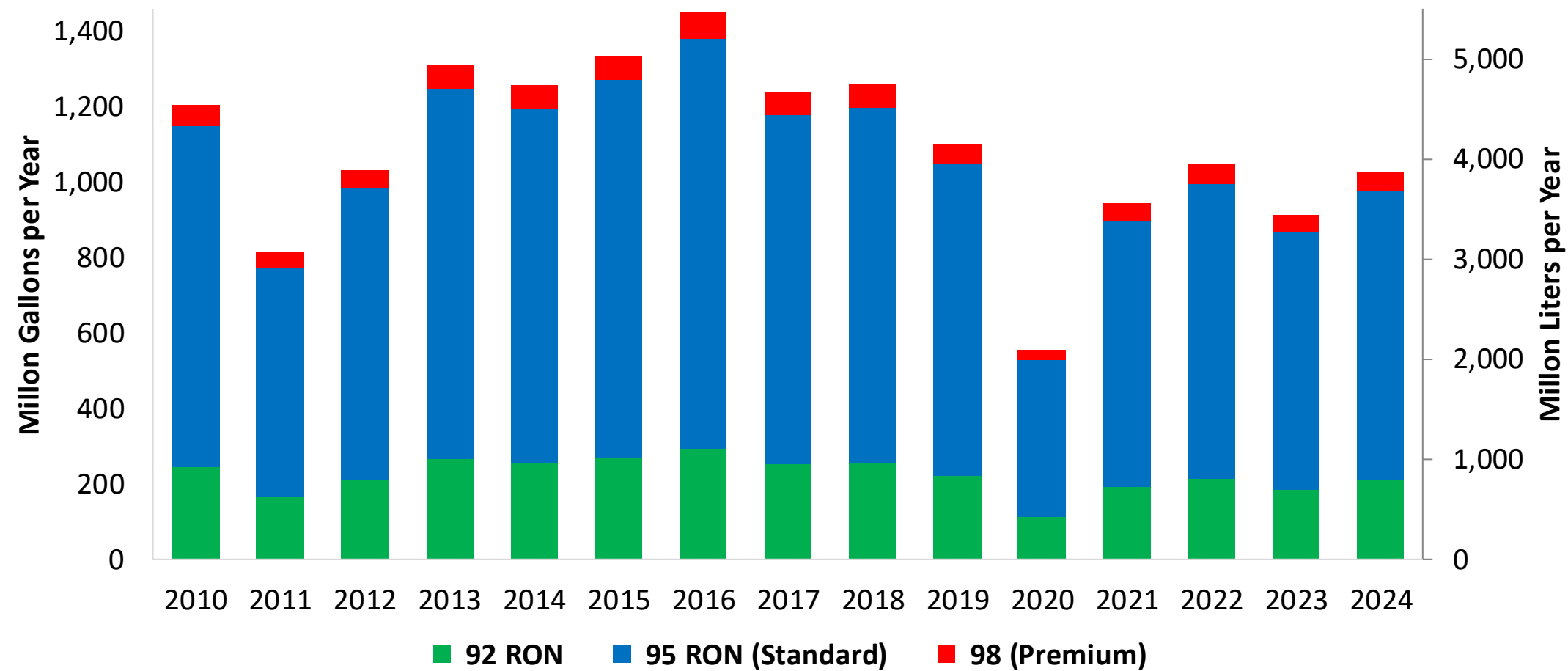
Gasoline Imports by Grade in Taiwan



Source: MOEAEA, E-STAT

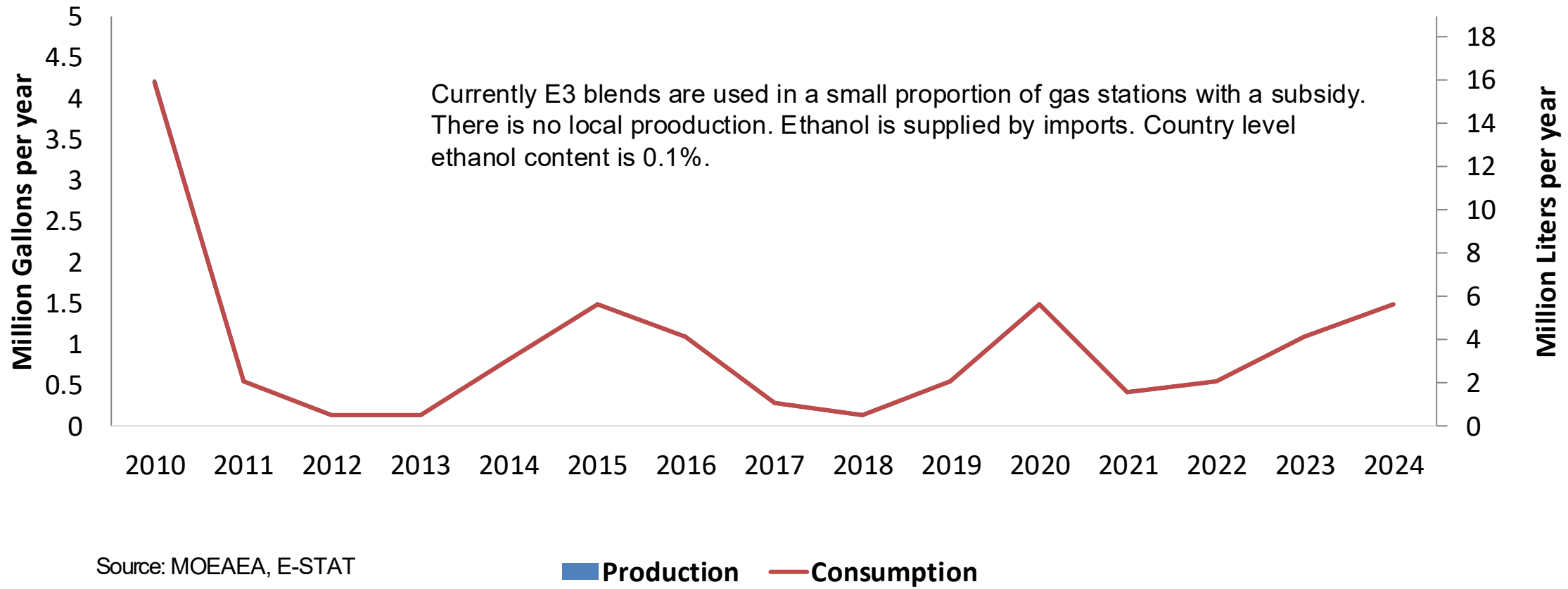
Gasoline Exports in Taiwan

Gasoline Exports by Grade in Taiwan



Source: MOEAEA, E-STAT

Ethanol Balance in Taiwan

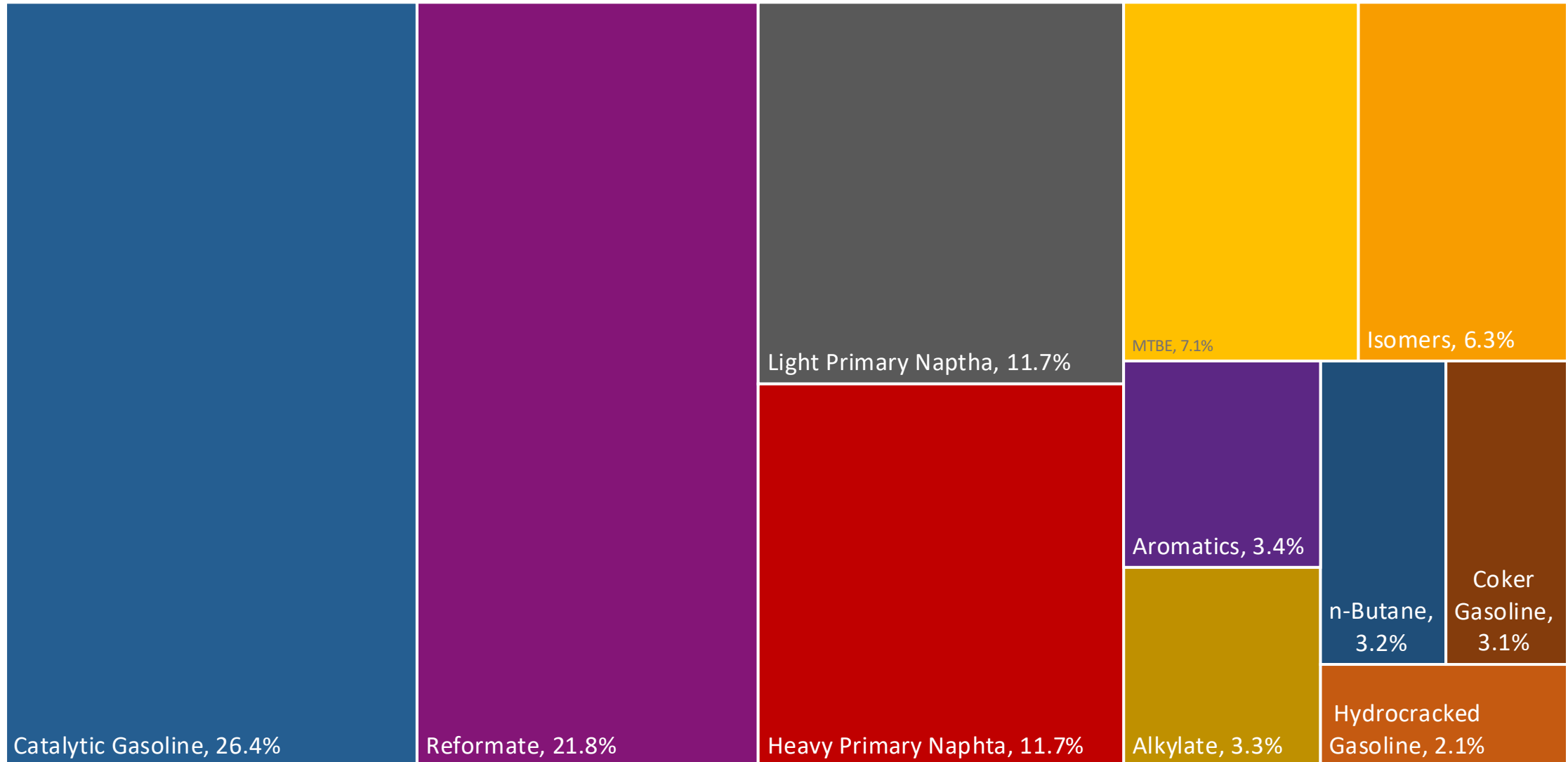


Gasoline Quality in Taiwan

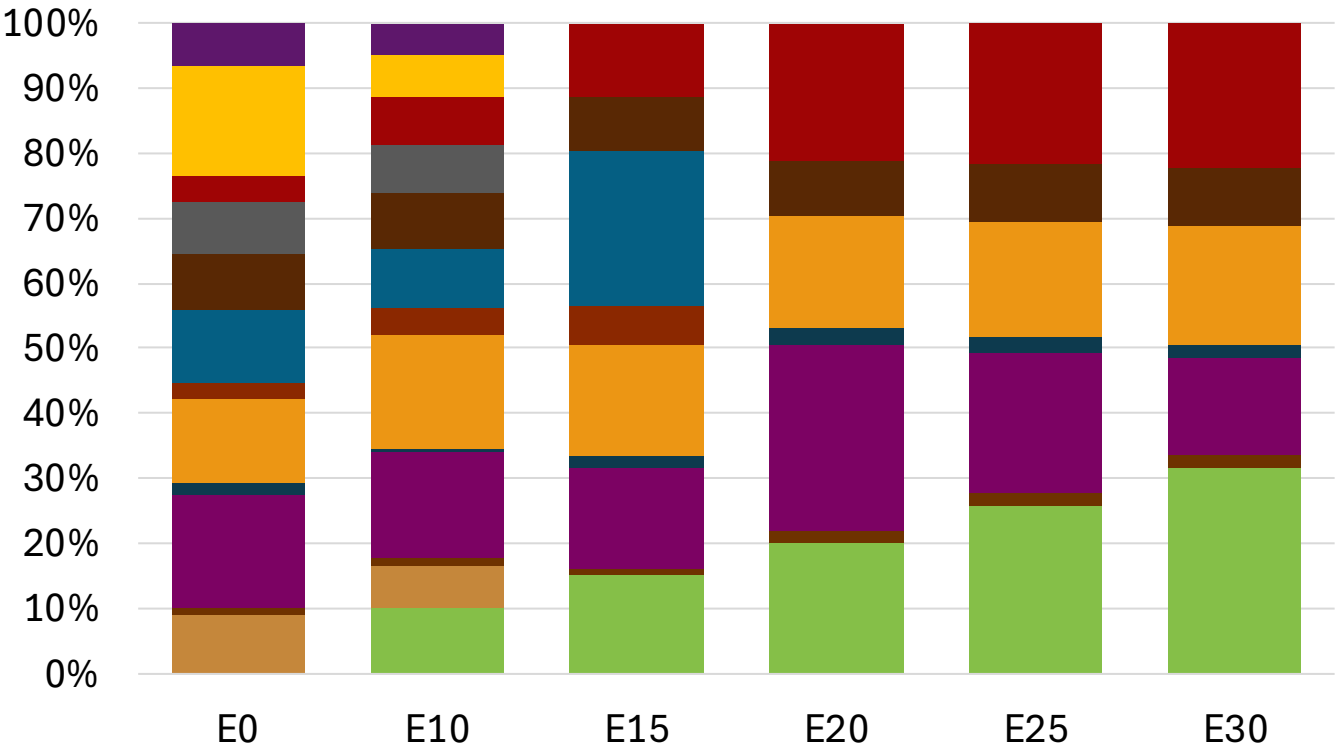
Name	Mobile Pollution Source Fuel Composition Control Standard Regulation					
Implementation Year	2024					
Applicability	Nationwide					
Grade	92 RON		95 RON (Standard)		98 RON (Premium)	
	Min	Max	Min	Max	Min	Max
RON	92.0		95.0		98.0	
MON						
AKI						
Benzene (%vol)		0.8		0.8		0.8
Aromatics (% vol)		35.0		35.0		35.0
Olefins (%vol)		18.0		18.0		18.0
Lead (g/L)						
Manganese (mg/L)						
Sulfur (ppm)		10		10		10
Oxygen (%v)		3.2		3.2		3.2
Ethanol (%vol)		10.0		10.0		10.0
MTBE (%vol)						
RVP (kPa)	45.0	66.9	45.0	66.9	45.0	66.9

Source: MOEAEA, MOENV, Formosa Petrochemical

Taiwan Available Blending Components



Taiwan – Gasoline 95 – Constant Octane

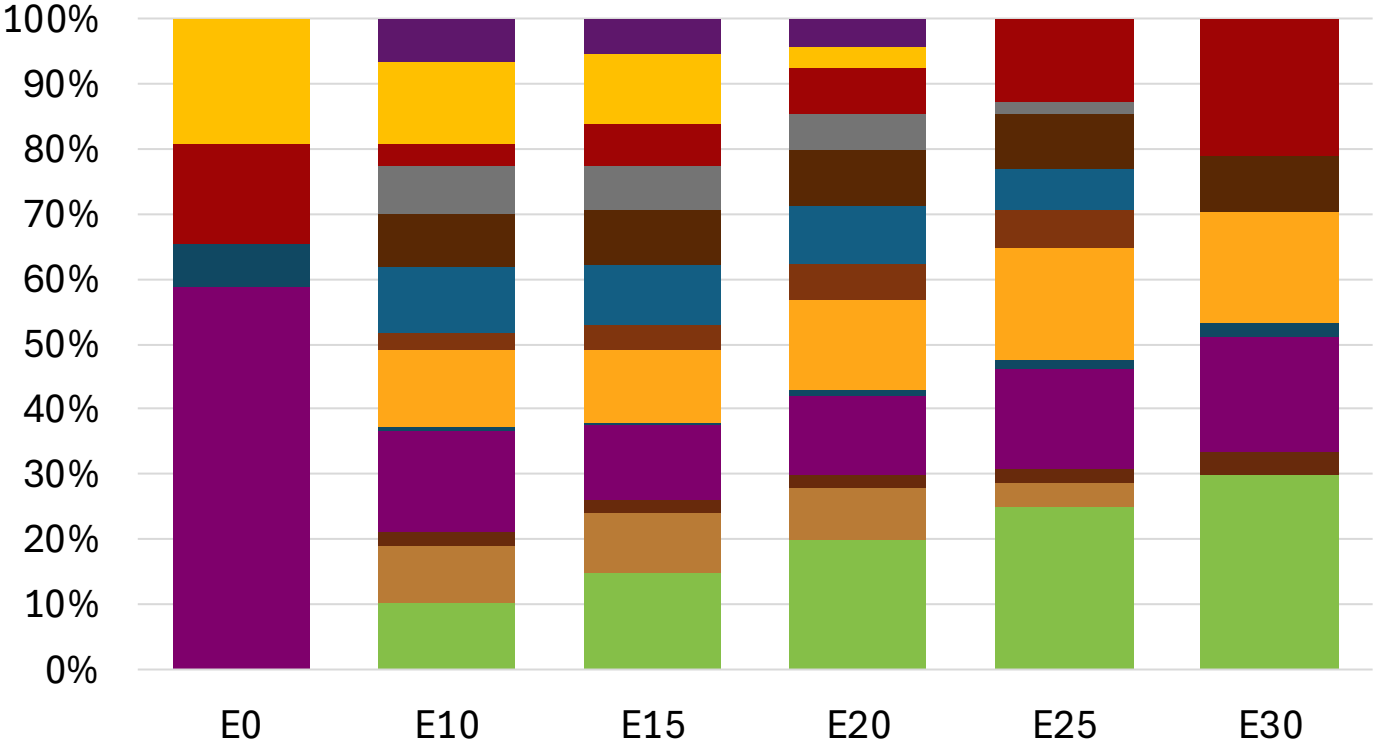
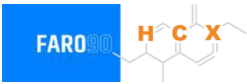


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.8	95.0	95.0
Price (US\$/gal)	\$ 2.388	\$ 2.307	\$ 2.223	\$ 2.149	\$ 2.061	\$ 1.995

Source: Faro90

Taiwan - Gasoline 98 - Constant Octane

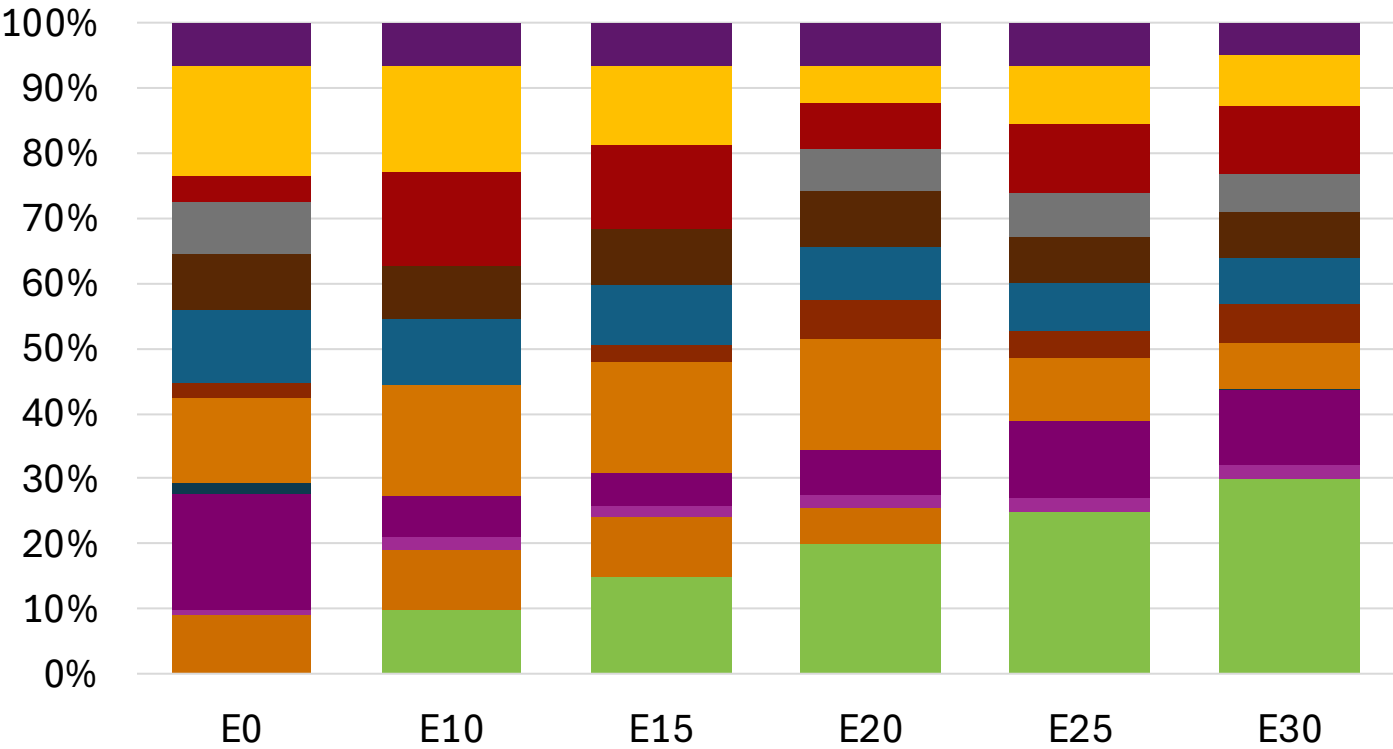


Average CIF 2024 Prices

Octane (RON)	98.0		98.0		98.0		98.0		98.0		98.6	
Price (US\$/gal)	\$	2.445	\$	2.418	\$	2.346	\$	2.305	\$	2.210	\$	2.117

Source: Faro90

Taiwan – Gasoline 95 – Increased Octane

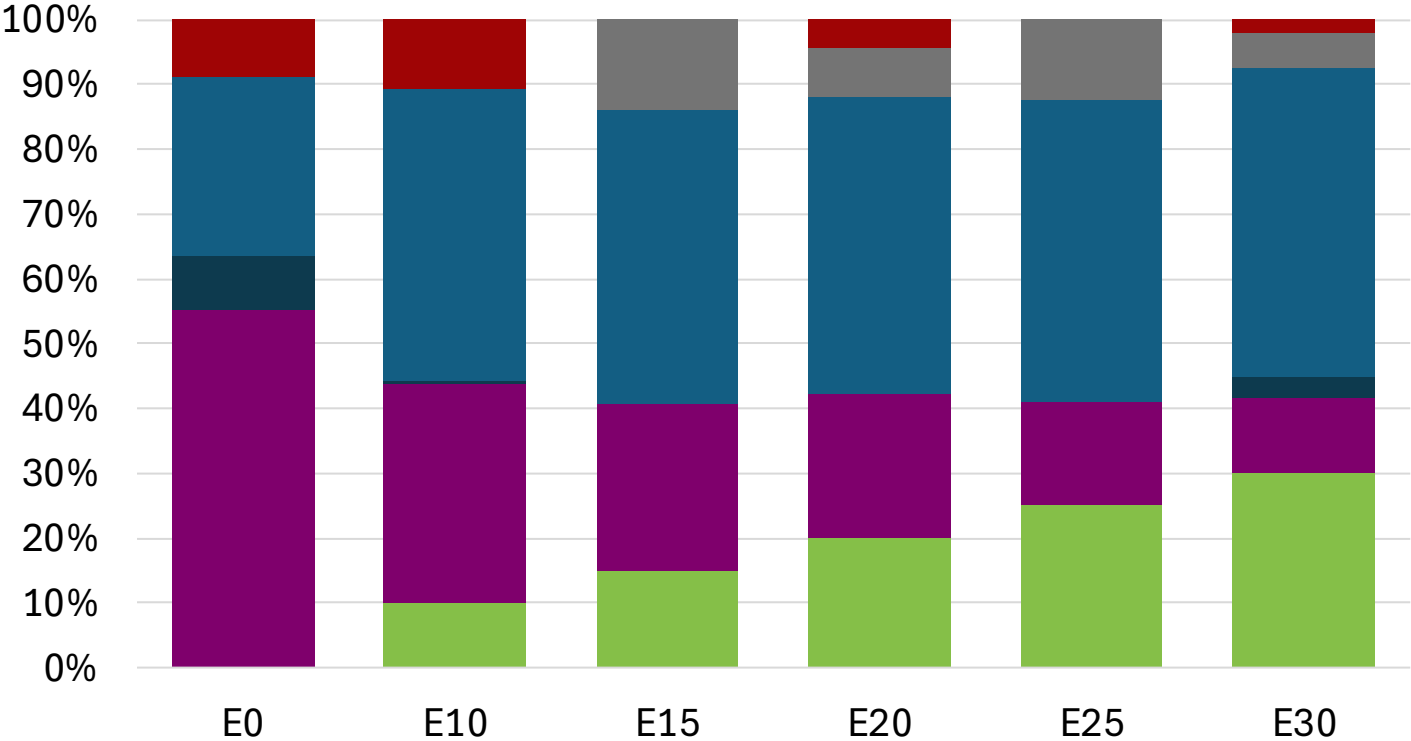


Average CIF 2024
Prices

Octane (RON)	95.0	96.4	97.4	98.1	100.6	102.0
Price (US\$/gal)	\$ 2.388	\$ 2.388	\$ 2.388	\$ 2.388	\$ 2.388	\$ 2.388

Source: Faro90

Taiwan – Gasoline 98 – Increased Octane

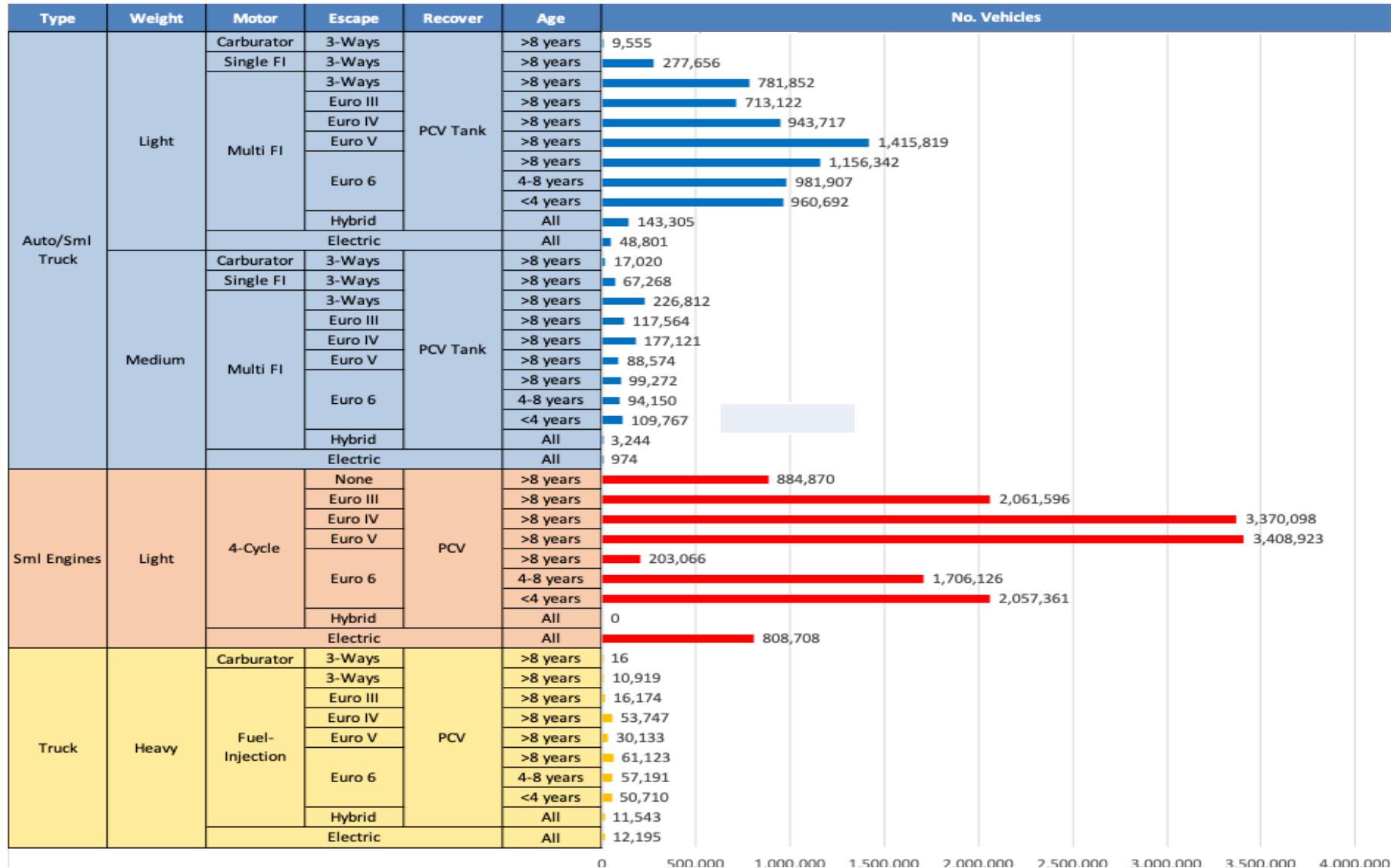


Average CIF 2024
Prices

Octane (RON)	98.0	99.8	99.9	101.3	102.6	103.9
Price (US\$/gal)	\$ 2.445	\$ 2.445	\$ 2.445	\$ 2.445	\$ 2.445	\$ 2.445

Source: Faro90

Gasoline Vehicle Fleet - Taiwan

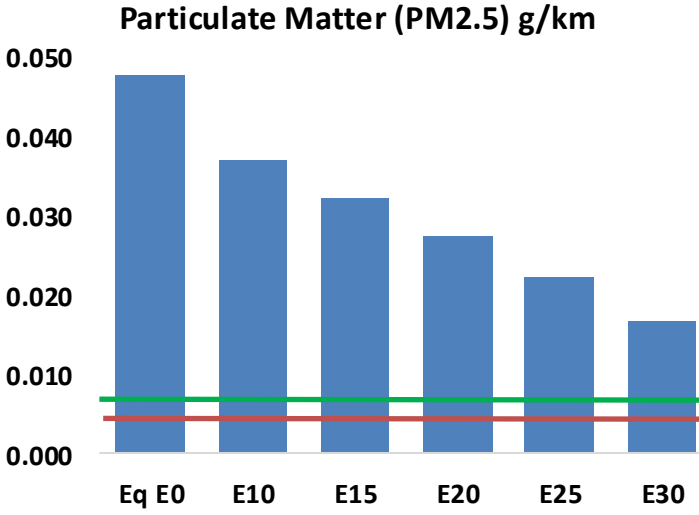
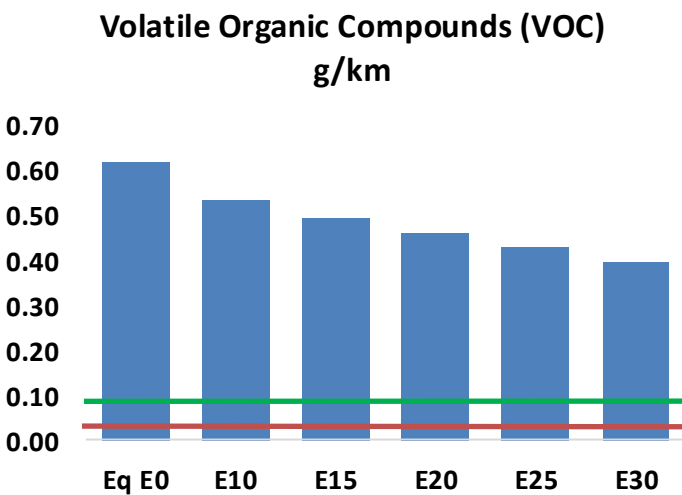
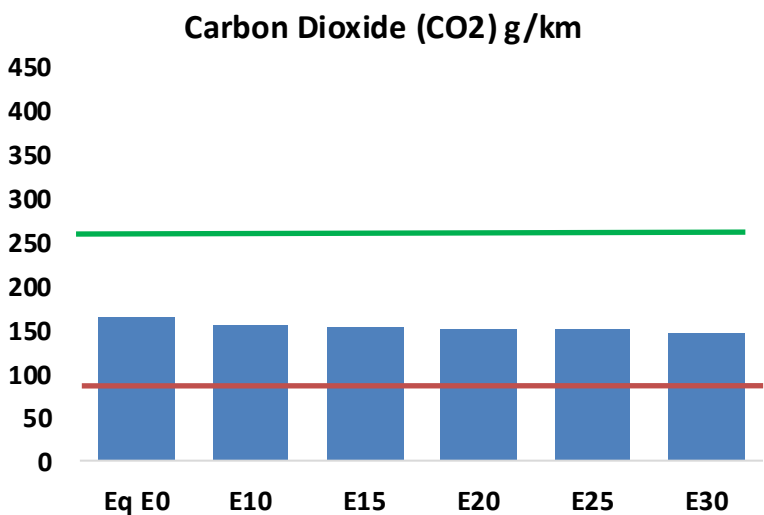
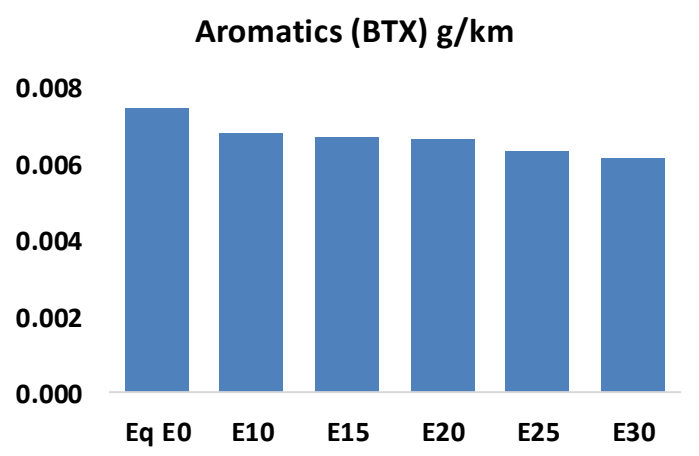
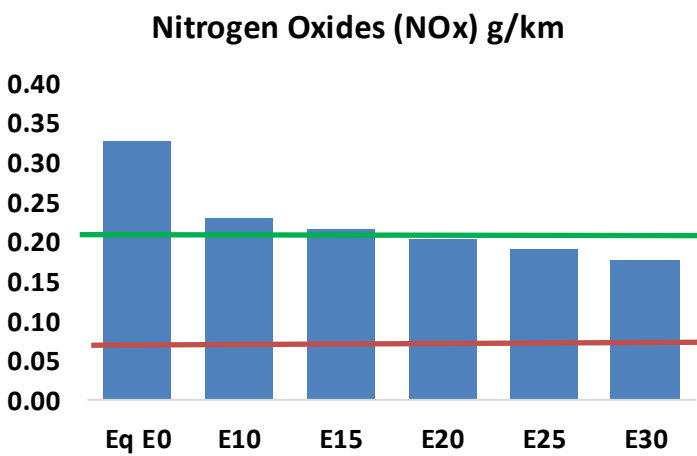
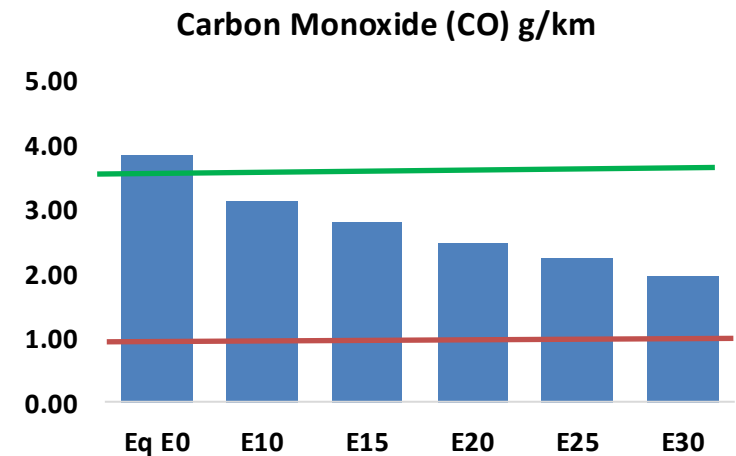
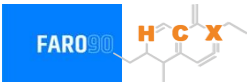


Vehicle Fleet: 23,239,031
Gasoline Vehicle Fleet: 20,069,210

Source: DATA-GOV.tw, CIECData

Taiwan – Vehicular Emissions

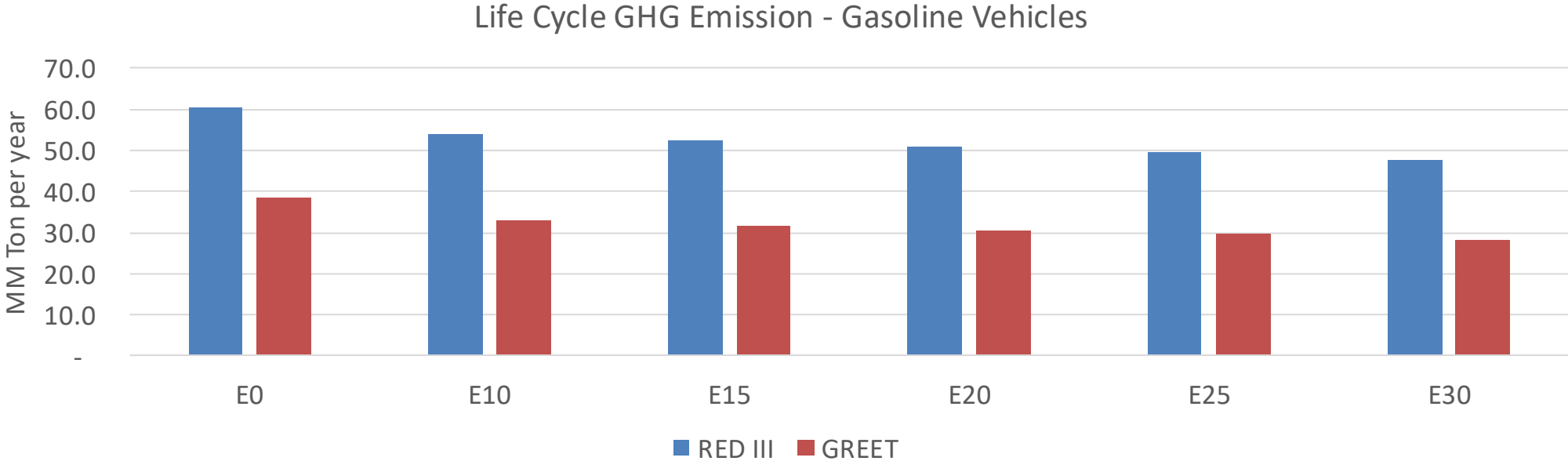
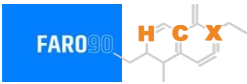
TIER III BIN 125 United States
Euro 6 Standard



Taiwan – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,360,185	1,230,696	1,101,207	990,234	876,319	788,050	691,909	-19%	-36%
CO2	57,083,319	55,615,462	54,229,153	53,138,861	52,599,342	52,408,803	51,442,758	-5%	-8%
VOC	218,167	202,516	186,866	174,260	161,676	152,029	139,049	-14%	-26%
NOx	114,877	86,158	80,414	75,790	71,477	66,721	61,689	-30%	-38%
SOx	636	588	540	499	460	418	373	-15%	-28%
NH3	23,536	23,303	23,070	23,179	23,440	23,623	23,776	-2%	0%
N2O	766	762	759	763	779	787	796	-1%	2%
CH4	54,499	54,499	54,499	55,317	56,516	57,551	58,532	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	1,595	1,496	1,397	1,321	1,247	1,190	1,112	-12%	-22%
Acetaldehyde	4,628	5,312	7,794	11,869	16,170	18,477	21,833	68%	249%
Formaldehyde	18,061	17,559	20,469	24,035	25,180	27,856	30,325	13%	39%
Benzene	2,632	2,529	2,397	2,359	2,339	2,237	2,165	-9%	-11%
PM2.5	3,932	3,491	3,051	2,649	2,250	1,826	1,383	-22%	-43%
PM10	12,864	11,422	9,980	8,667	7,360	5,972	4,526	-22%	-43%

Taiwan – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	291.7
Target @ 2035 (MMT/year)	204.2
Delta	87.5

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	14.6%	18.0%	20.9%	23.6%	27.2%
RED II/III	0.0%	10.6%	13.4%	15.8%	18.0%	20.8%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	6.4%	8.0%	9.2%	10.4%	12.0%
RED II/III	0.0%	7.3%	9.2%	10.9%	12.4%	14.4%

Source: Greet, RED II/III, climateactiontracker.org, F90

Ethanol Gasoline Blending in Thailand



In 2024, gasoline consumption in Thailand was 11,452 million liters (3,025 million gallons) and growth rate was -0.5%. Gasoline production and net imports were 13,337 and -1,885 million liters (3,523 and -498 million gallons) correspondingly. Thailand offers two main grades: RON 91 (E10) and RON 95 (E10 and E20) with an average octane of 94.1 RON.

Thailand's bioethanol production primarily uses sugarcane and cassava as feedstocks, with approximately 30 facilities producing over 1 billion liters annually. The promotion of bio-based products like ethanol is part of Thailand's bioeconomy policy, supporting both the energy sector and domestic economic growth. In 2024, ethanol consumption supplied by local production in Thailand was 1,289 million liters (341 million gallons) representing 11.3% of gasoline demand.

Source: Eppo, USDA, Department of Energy Business

The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexadiene. The structure features a six-membered ring with two double bonds. Substituents include a hydrogen atom (H), a carbon atom (C), and a group labeled 'X' (which is a vinyl group, -CH=CH₂). There are also methyl groups attached to the ring.

Source: Eppo

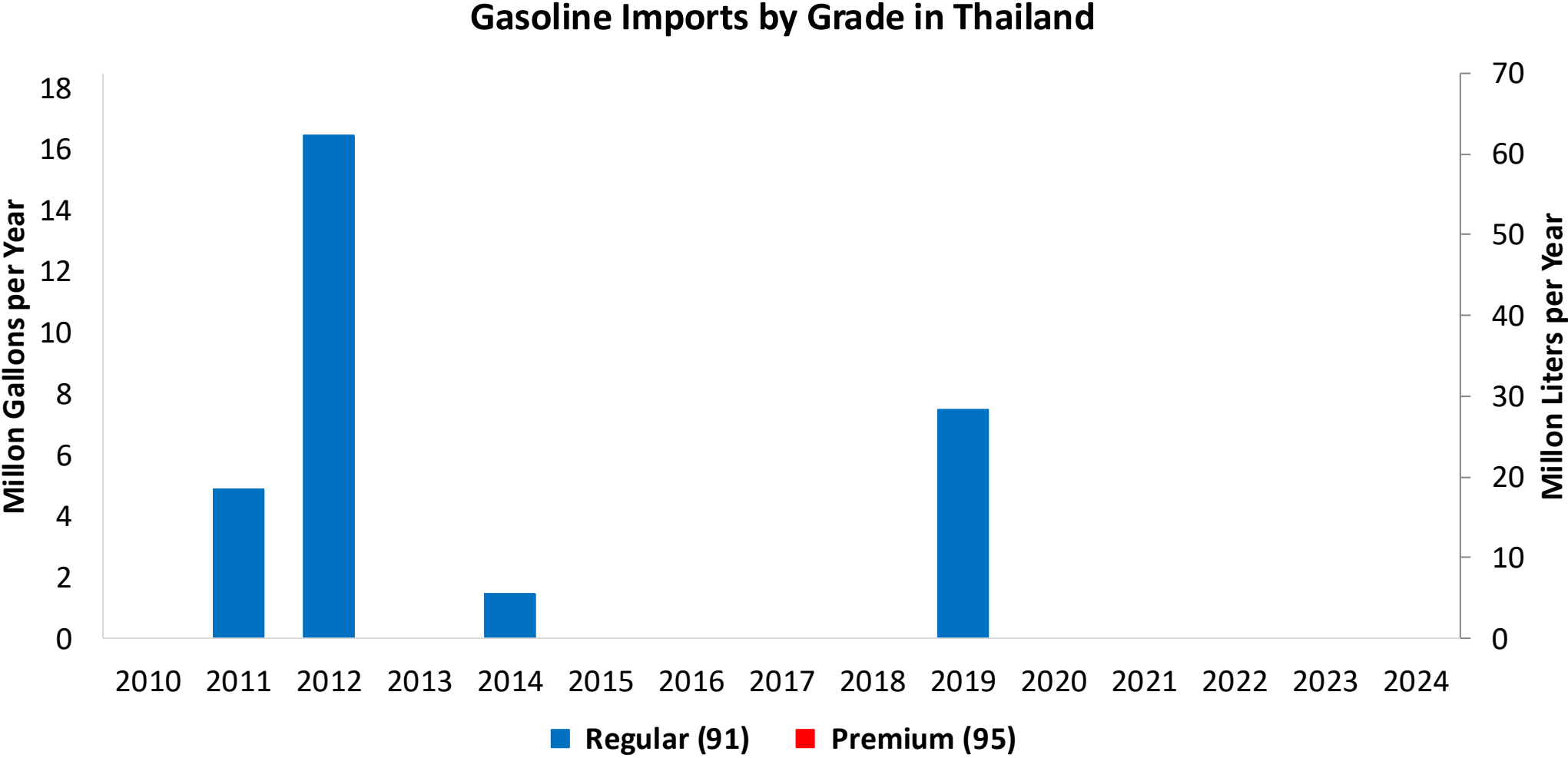
Year	Gasohol E10 (91) (Million Gallons)	Gasohol E10 (95) (Million Gallons)	Gasohol E20 (95) (Million Gallons)	Total (Million Gallons)
2010	1,200	550	50	1,800
2011	1,300	380	50	1,730
2012	1,400	280	100	1,780
2013	900	780	120	1,800
2014	950	770	150	1,870
2015	1,050	950	140	2,140
2016	1,050	1,150	150	2,350
2017	1,020	930	250	2,200
2018	950	1,100	250	2,300
2019	900	1,050	300	2,250
2020	780	1,450	220	2,450
2021	650	1,450	180	2,280
2022	680	1,400	220	2,300
2023	650	1,900	100	2,650
2024	700	1,950	100	2,750

The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexadiene. The structure features a six-membered ring with two double bonds. Substituents include a hydrogen atom (H), a methyl group (C), and an ethyl group (X) on one carbon, and a methyl group on an adjacent carbon. The ring is connected to a side chain.

This stacked bar chart displays the annual sales of Regular (91) and Premium (95) gasoline from 2010 to 2024. The left Y-axis measures sales in Million Gallons per Year (0 to 3,500), and the right Y-axis measures sales in Million Liters per Year (0 to 14,000). The legend indicates that blue represents Regular (91) and red represents Premium (95).

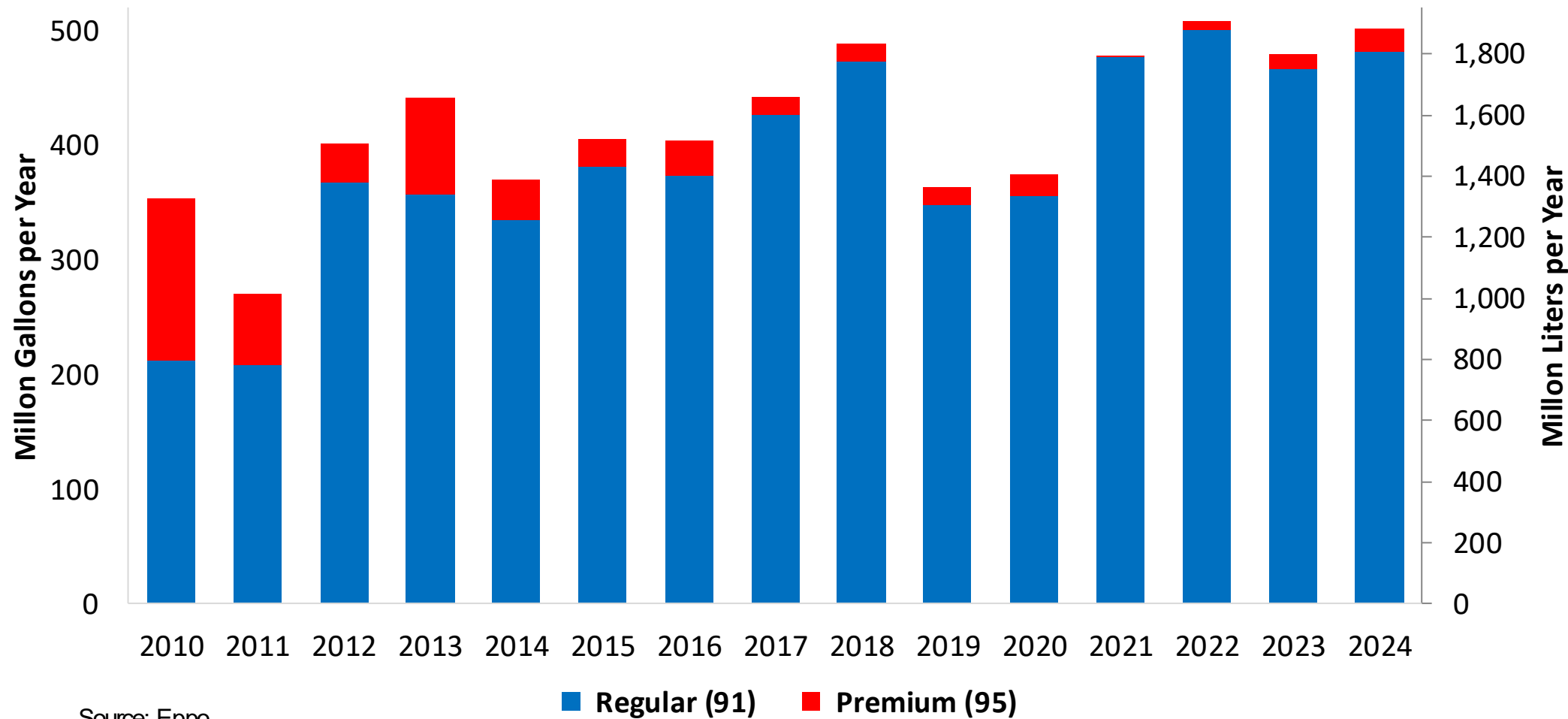
Year	Regular (91) (Million Gallons)	Premium (95) (Million Gallons)	Total (Million Gallons)
2010	1,400	900	2,300
2011	1,500	700	2,200
2012	1,750	650	2,400
2013	1,250	1,350	2,600
2014	1,280	1,320	2,600
2015	1,450	1,500	2,950
2016	1,450	1,750	3,200
2017	1,450	1,850	3,300
2018	1,450	2,050	3,500
2019	1,250	2,200	3,450
2020	1,150	2,300	3,450
2021	1,150	2,100	3,250
2022	1,180	2,220	3,400
2023	1,150	2,350	3,500
2024	1,200	2,300	3,500

2



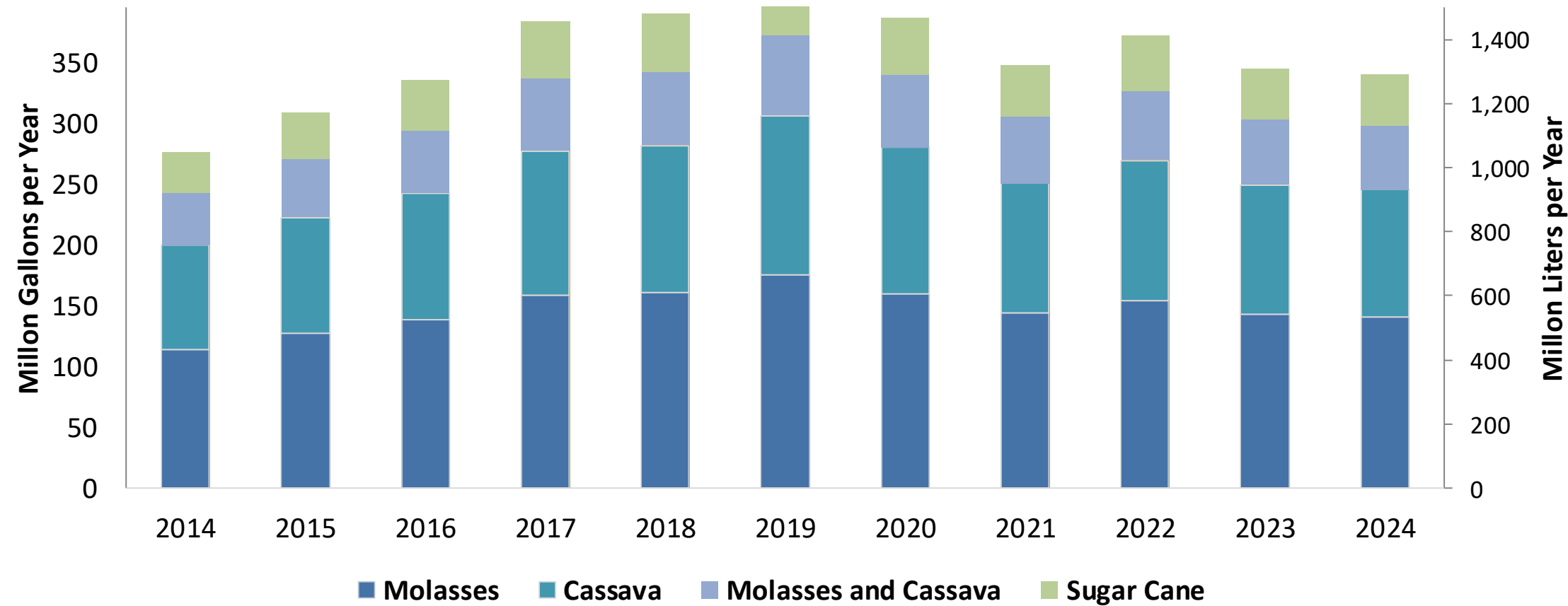
Source: Eppo

Gasoline Exports by Grade in Thailand



Source: Eppo

Ethanol Production and Demand in Thailand



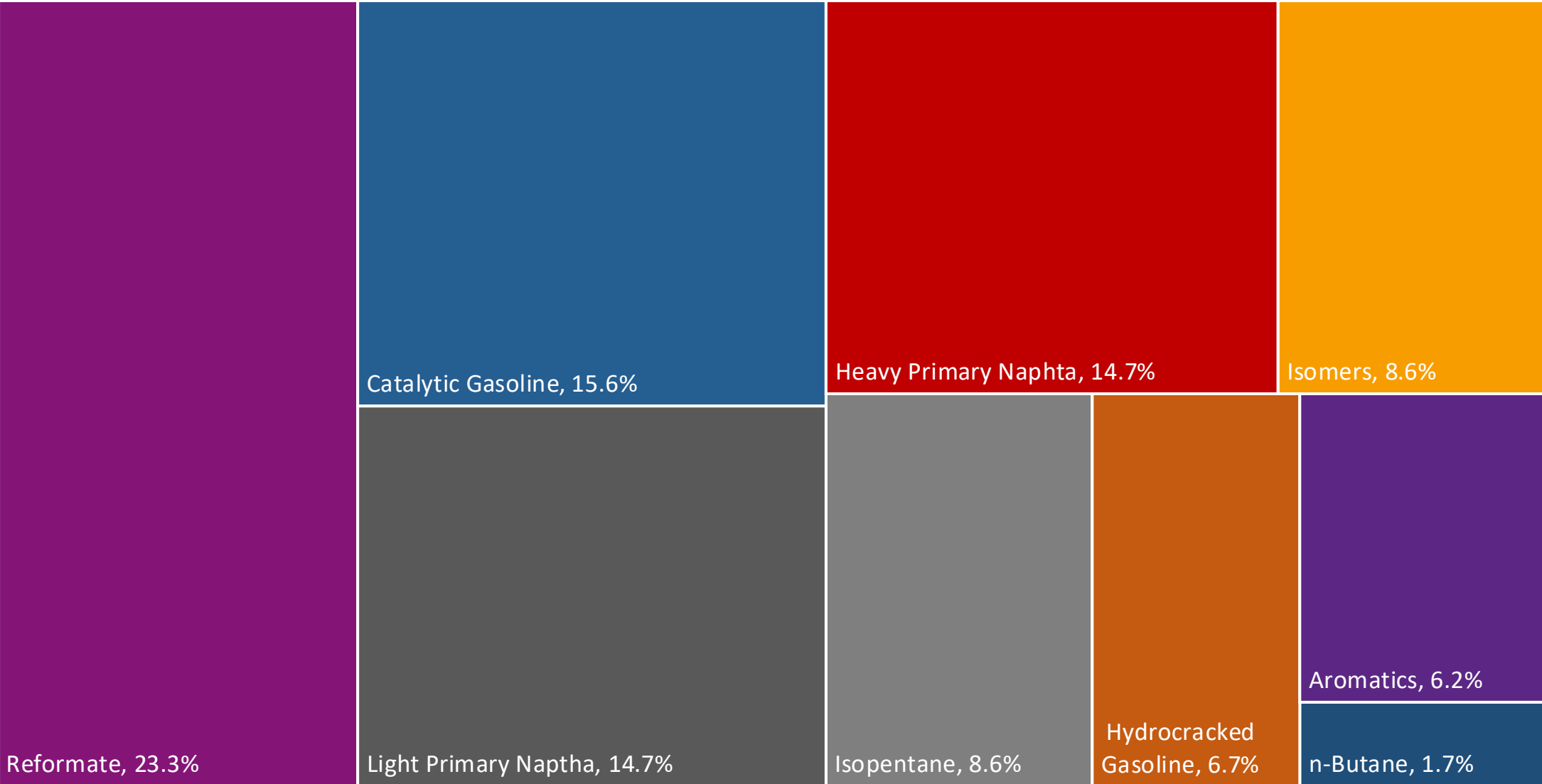
Source: Eppo, USDA

Gasoline Quality in Thailand

Name	B.E. 2557 (2014)					
Implementation Date	2014					
Applicability	Nationwide					
Grade	Gasohol E10 - 91		Gasohol E10 - 95		Gasohol E20	
	Min	Max	Min	Max	Min	Max
RON	91.0		95.0		95.0	
MON	80.0		84.0		84.0	
AKI						
Benzene (%vol)		1.0		1.0		1.0
Aromatics (% vol)		35.0		35.0		35.0
Olefins (%vol)		18.0		18.0		18.0
Lead (g/L)		0.005		0.005		0.005
Phosphorous (g/L)		0.013		0.013		0.013
Sulfur (ppm)		50		50		50
Oxygen (%w)						
Ethanol (%vol)		10.0		10.0		20.0
Other Oxygenates	The approval of the Director General of the Department of Energy Business					
PVR 37.8°C (kPa)		62		62		62

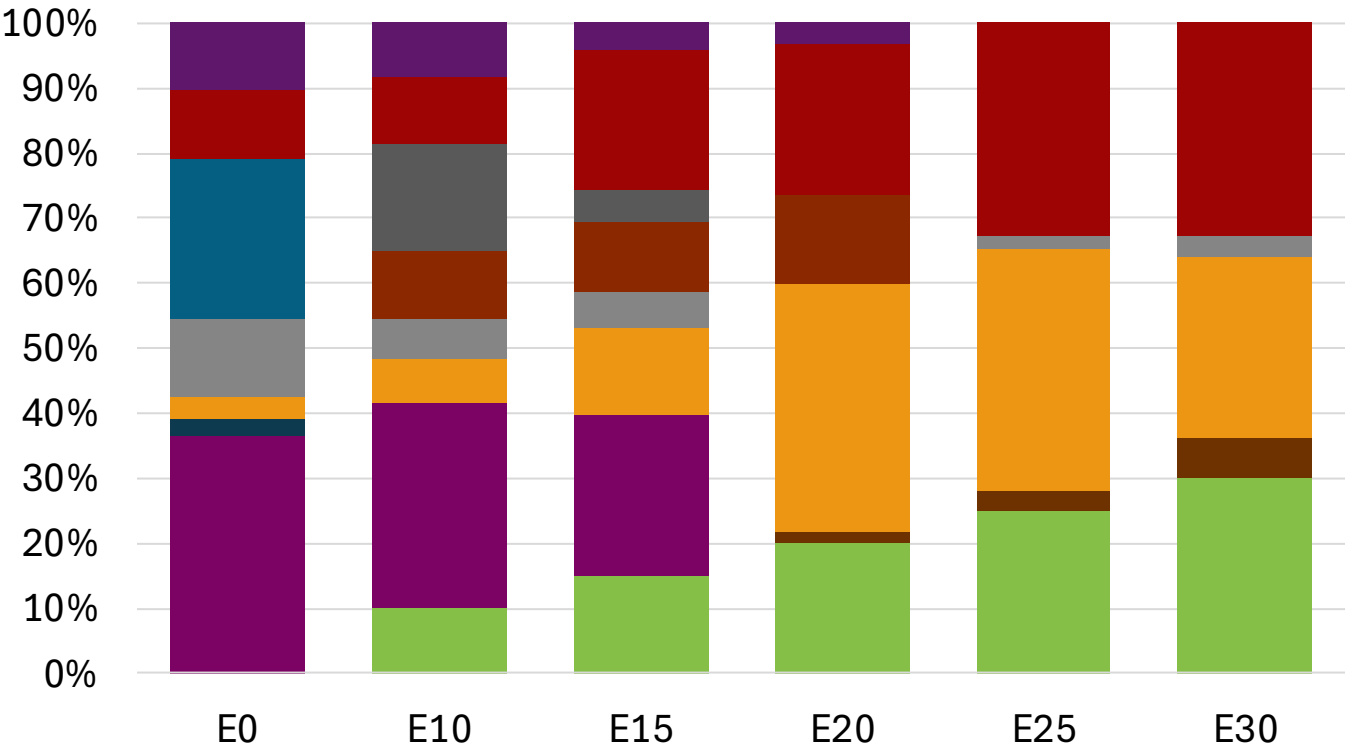
Source: Department of Energy Business

Thailand Available Blending Components



Source: Faro90

Thailand – Gasoline 91 – Constant Octane

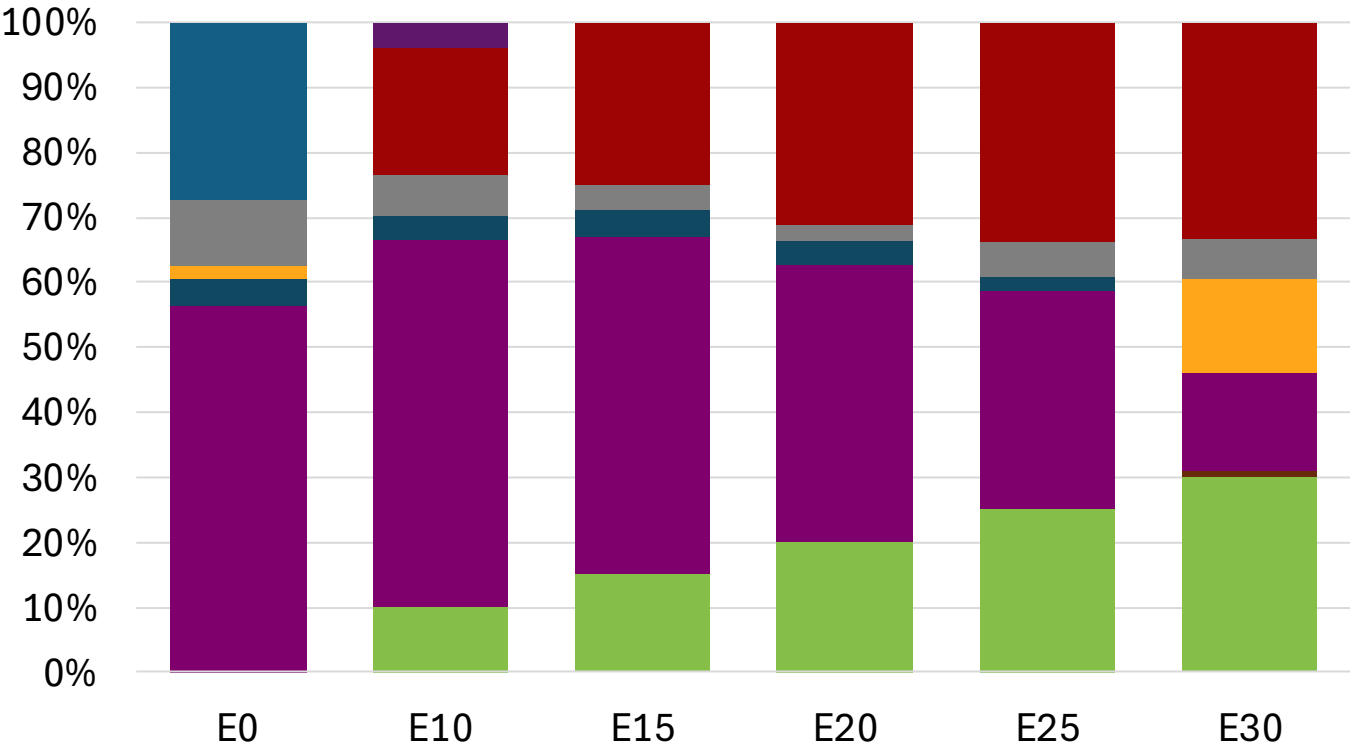
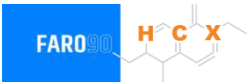


Average CIF 2024
Prices

Octane (RON)	91.0	91.0	91.0	91.4	91.6	93.1
Price (US\$/gal)	\$ 2.256	\$ 2.123	\$ 2.007	\$ 1.947	\$ 1.808	\$ 1.766

Source: Faro90

Thailand - Gasoline 95 - Constant Octane

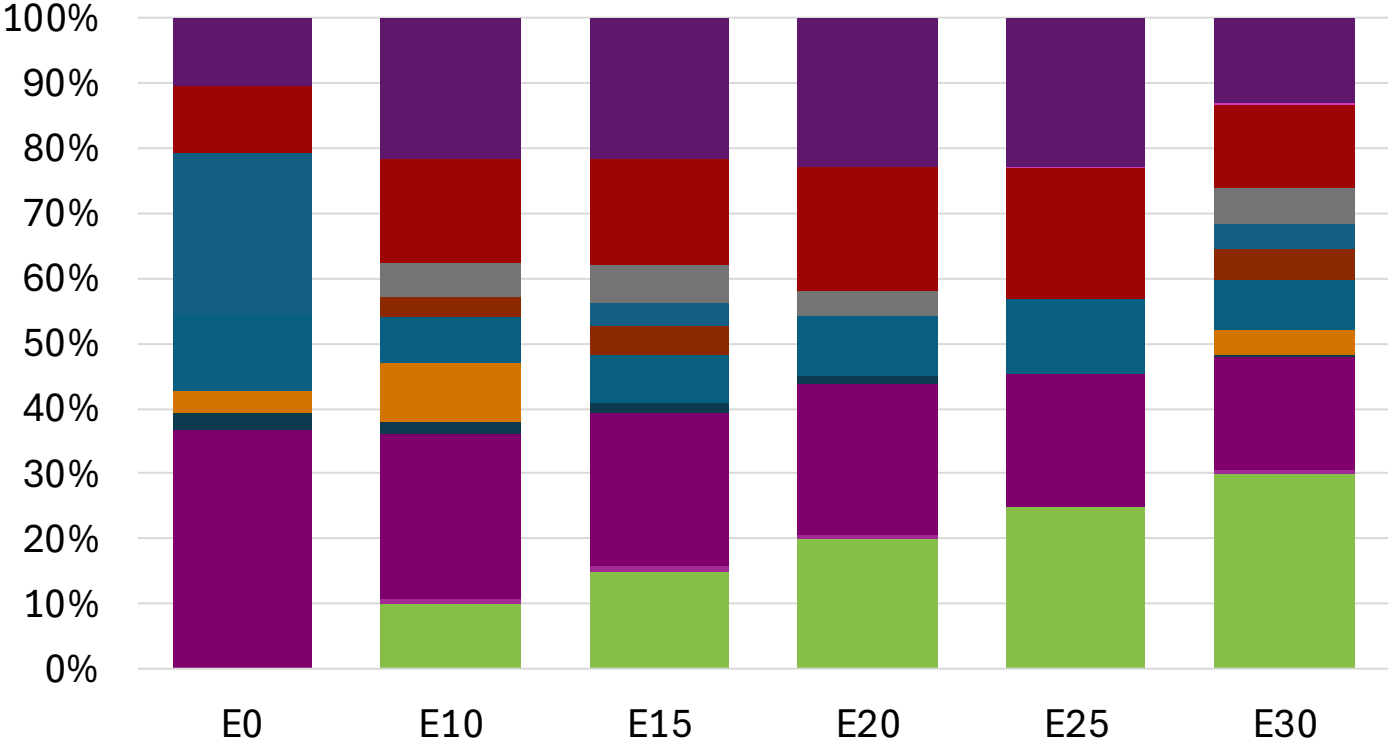
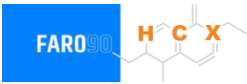


Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (US\$/gal)	\$ 2.505	\$ 2.276	\$ 2.179	\$ 2.075	\$ 1.968	\$ 1.873

Source: Faro90

Thailand – Gasoline 91 – Increased Octane

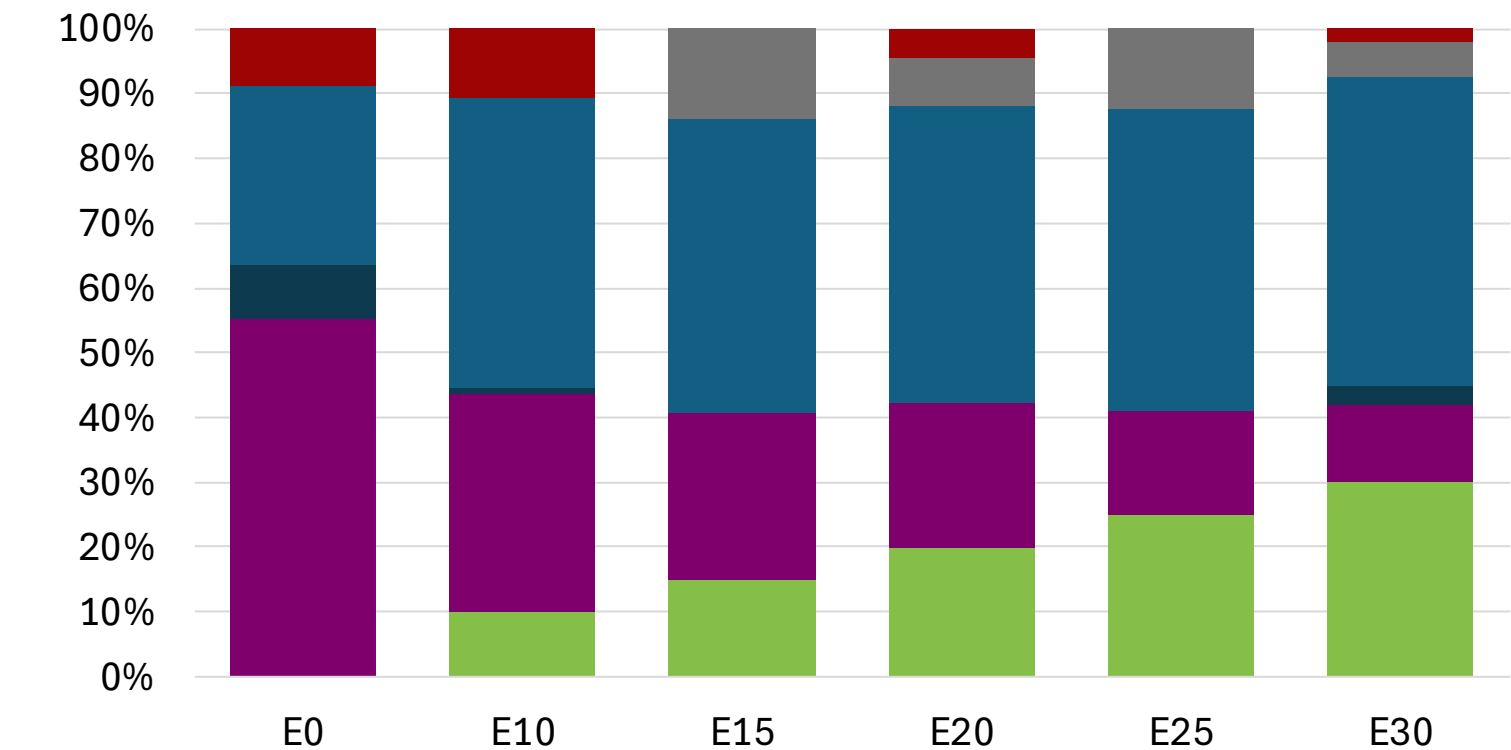


Average CIF 2024 Prices

Octane (RON)	91.0	93.8	95.7	97.5	99.2	100.7
Price (US\$/gal)	\$ 2.256	\$ 2.256	\$ 2.256	\$ 2.256	\$ 2.256	\$ 2.256

Source: Faro90

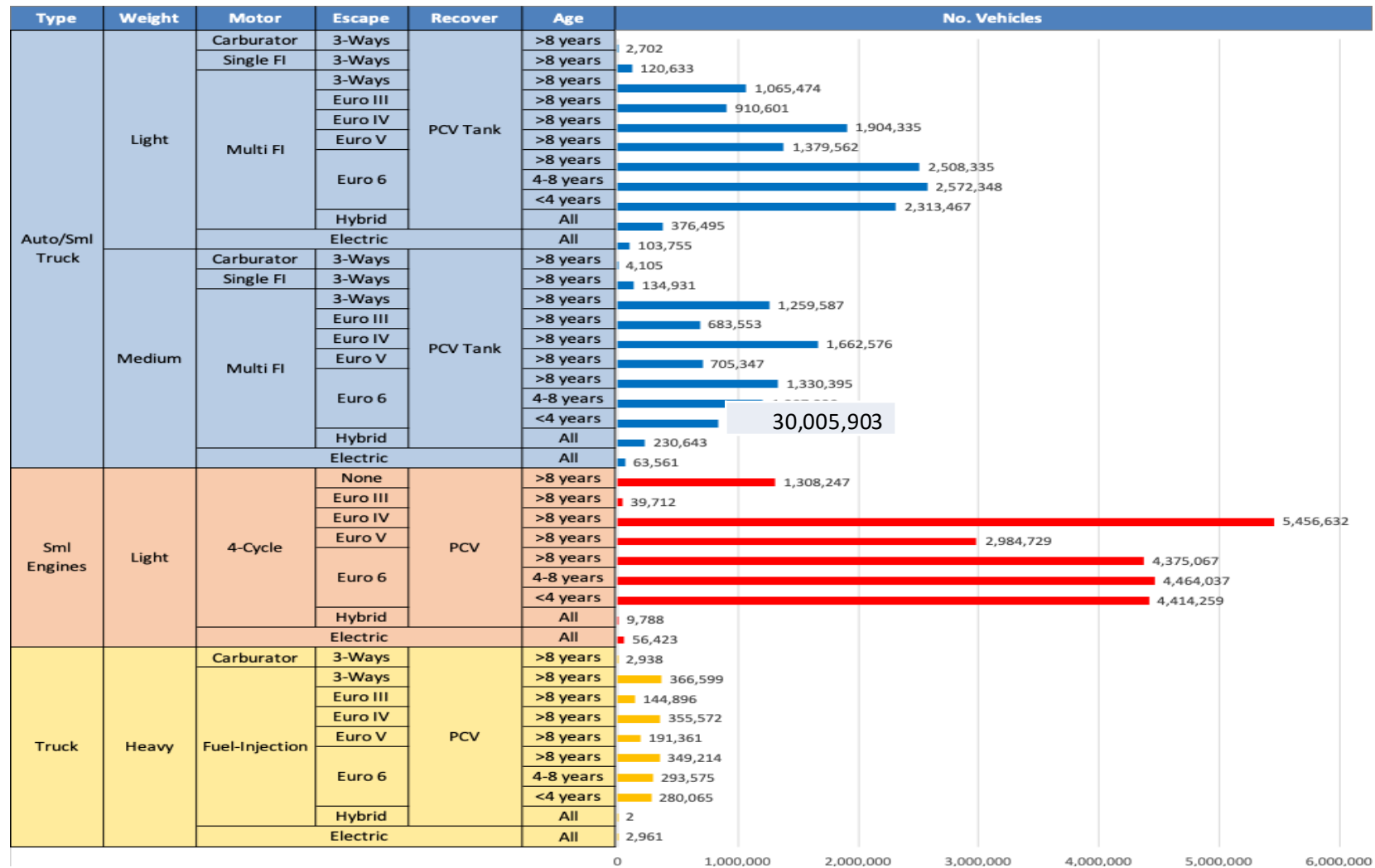
Thailand – Gasoline 95 – Increased Octane



Average CIF 2024
Prices

Source: Faro90

Gasoline Vehicle Fleet - Thailand

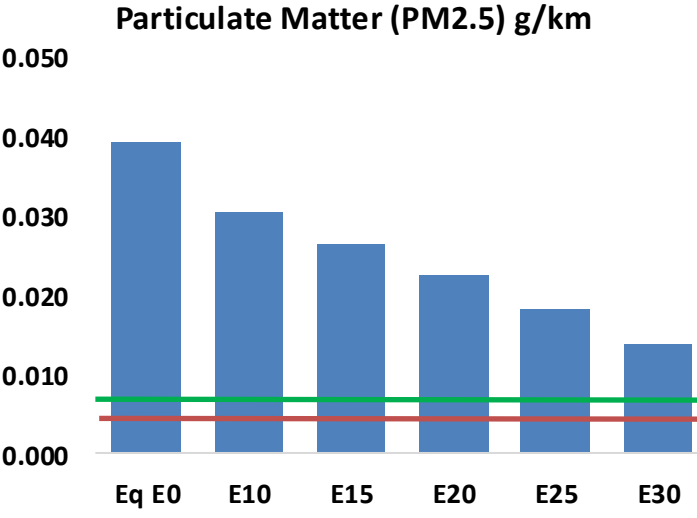
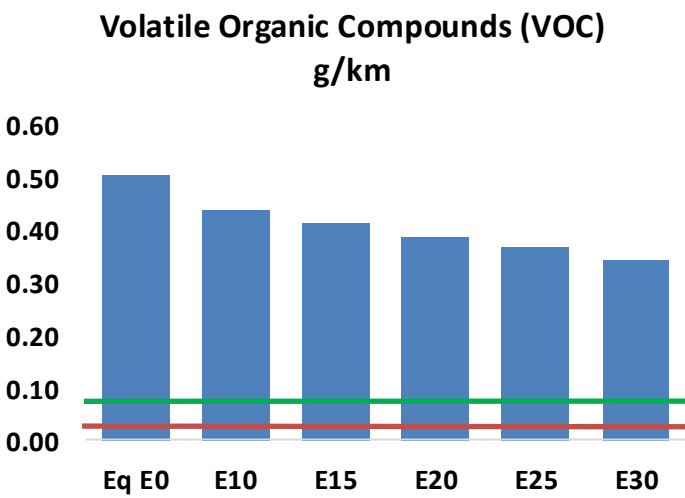
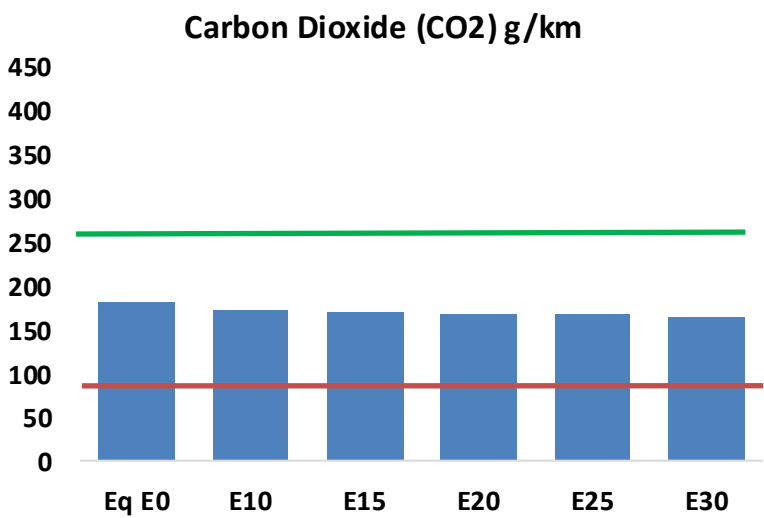
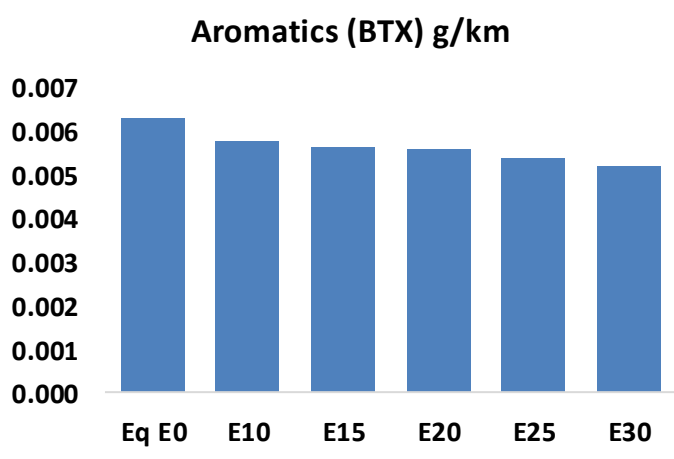
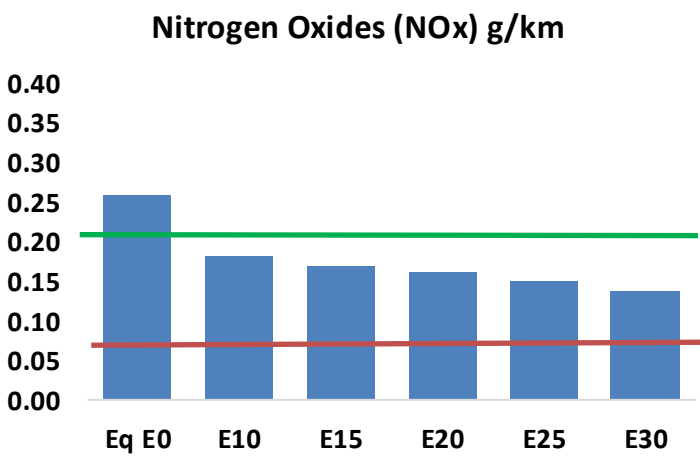
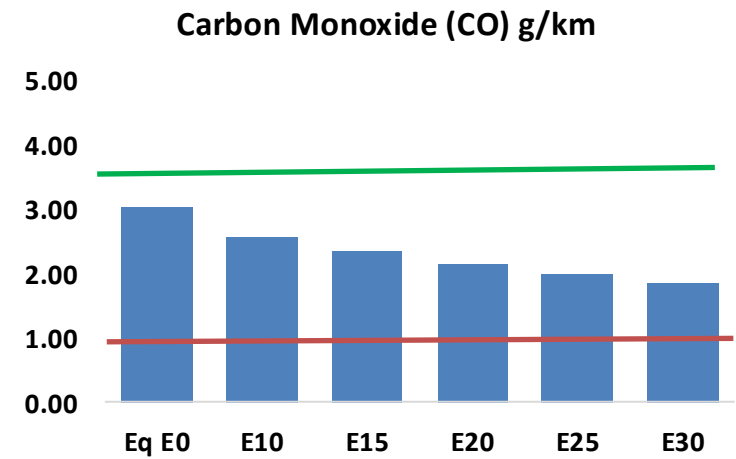
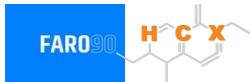


Vehicle Fleet: 46,475,523
Gasoline Vehicle Fleet: 30,005,903

Source: datagov.mot.go.th

Thailand – Vehicular Emissions

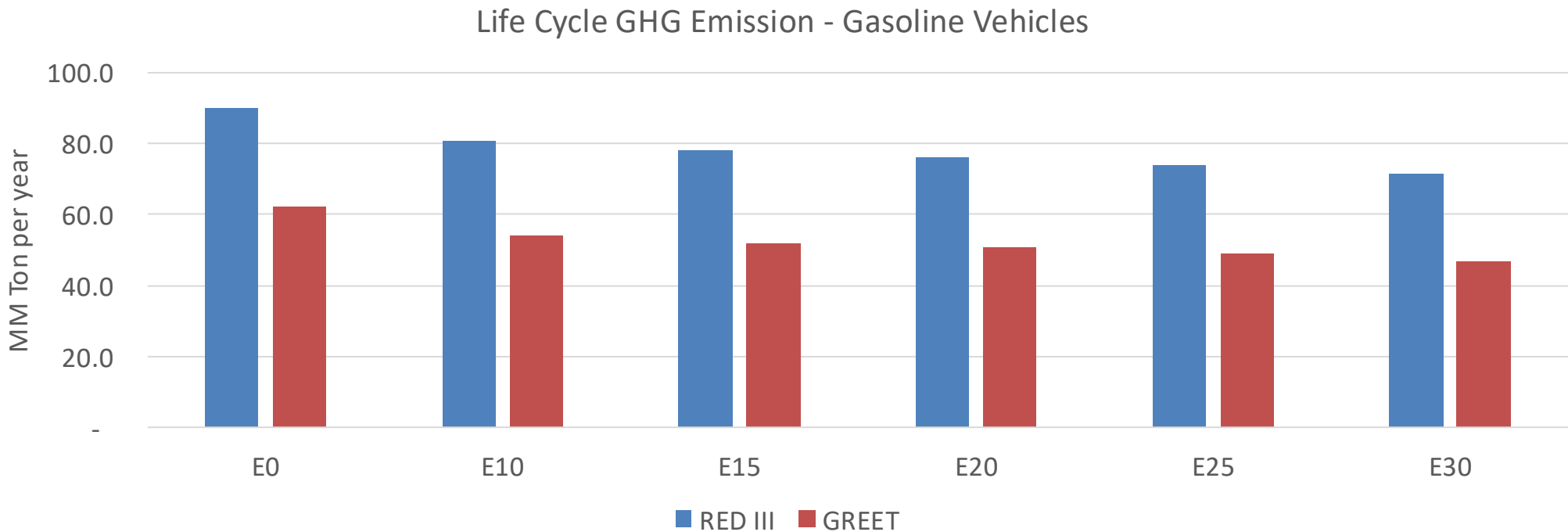
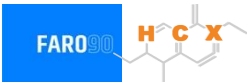
TIER III BIN 125 United States
Euro 6 Standard



Thailand – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,771,402	1,631,957	1,492,513	1,377,607	1,261,681	1,172,451	1,075,545	-16%	-29%
CO2	105,993,888	103,268,331	100,694,194	98,669,711	97,667,917	97,249,062	95,456,482	-5%	-8%
VOC	294,251	275,158	256,065	241,107	226,372	215,137	199,807	-13%	-23%
NOx	150,376	112,782	105,263	99,211	93,564	87,339	80,752	-30%	-38%
SOx	1,202	1,111	1,020	943	870	790	706	-15%	-28%
NH3	40,484	40,083	39,682	39,871	40,319	40,635	40,897	-2%	0%
N2O	1,710	1,701	1,694	1,703	1,738	1,757	1,777	-1%	2%
CH4	70,833	70,833	70,833	71,896	73,454	74,800	76,075	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	1,944	1,844	1,745	1,673	1,605	1,554	1,482	-10%	-17%
Acetaldehyde	5,608	6,437	9,444	14,383	19,595	22,391	26,457	68%	249%
Formaldehyde	21,725	21,121	24,621	28,911	30,288	33,507	36,476	13%	39%
Benzene	3,690	3,546	3,360	3,306	3,279	3,136	3,035	-9%	-11%
PM2.5	6,957	6,177	5,398	4,687	3,980	3,230	2,448	-22%	-43%
PM10	15,967	14,178	12,388	10,758	9,135	7,413	5,618	-22%	-43%

Thailand – Gasoline Vehicle Fleet Life Cycle GHG Emissions



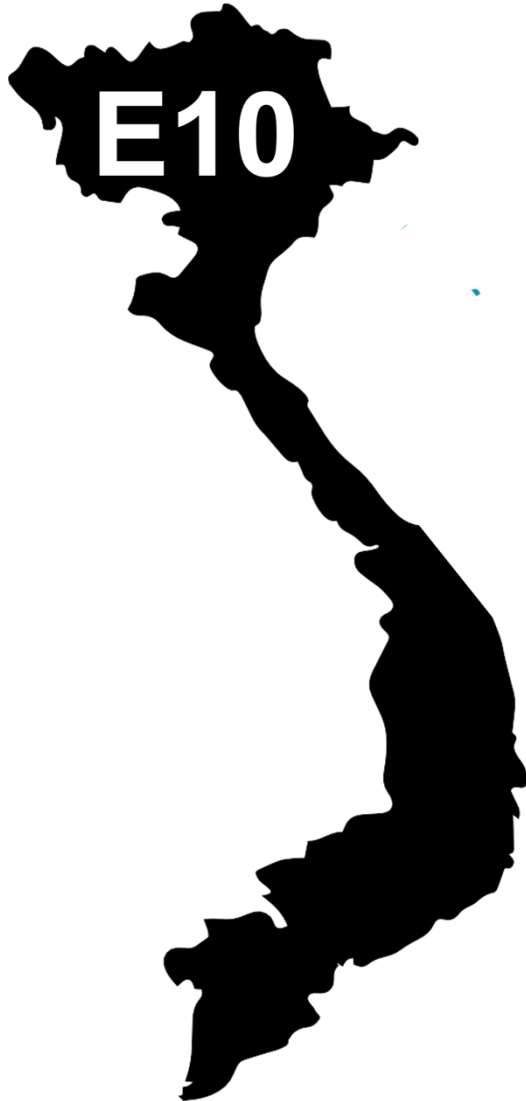
Country Emissions 2024 (MMT/año)	356.2
Target @ 2035 (MMT/year)	211.6
Delta	144.7

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.8%	16.0%	18.5%	20.8%	24.0%
RED II/III	0.0%	10.2%	13.1%	15.5%	17.8%	20.6%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.5%	6.8%	7.9%	8.9%	10.3%
RED II/III	0.0%	6.4%	8.2%	9.7%	11.1%	12.8%

Source: Greet, RED II/III, climateactiontracker.org, F90

Ethanol Gasoline Blending in Vietnam

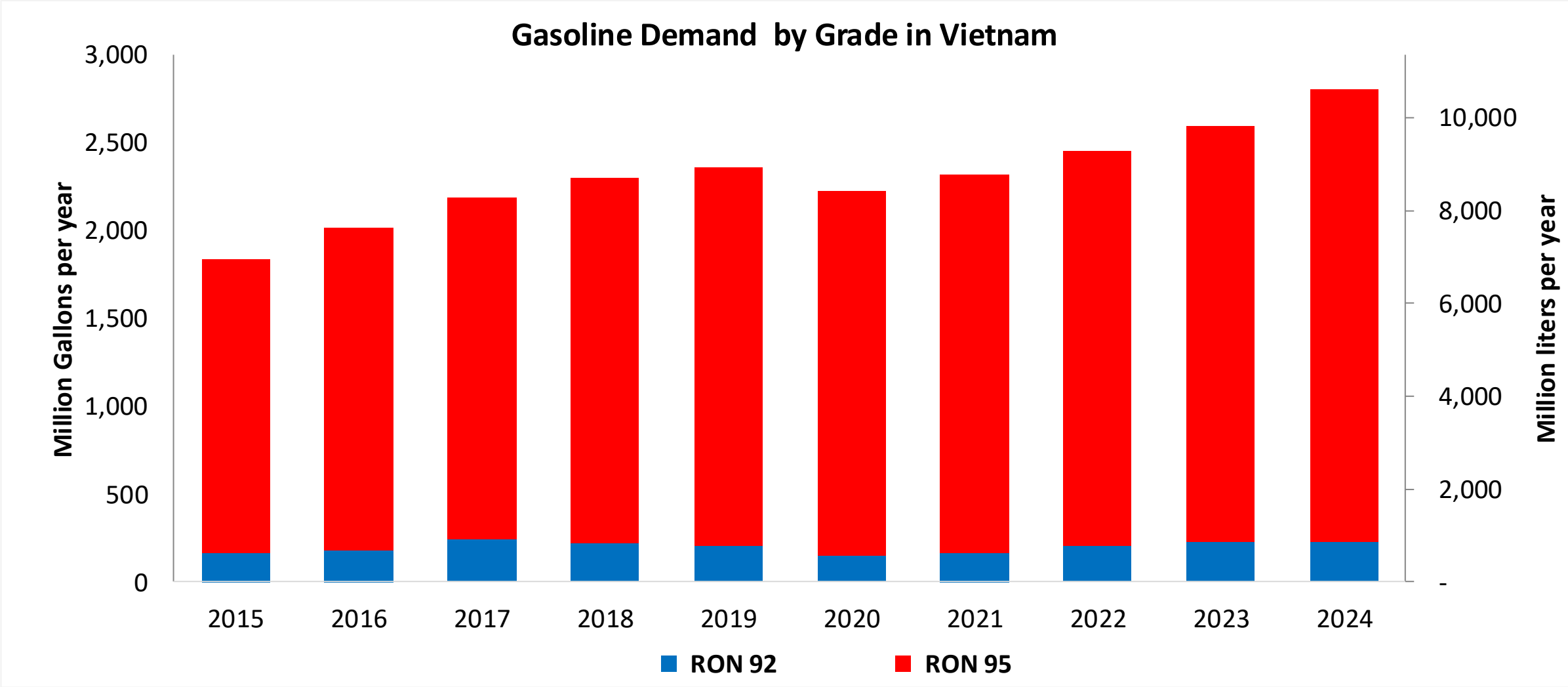


In 2024, gasoline consumption in Vietnam was 10,600 million liters (2,800 million gallons) and growth rate was 3.8%. Gasoline production and net imports were 7,870 and 2,730 million liters (2,079 and 721 million gallons) correspondingly. Vietnam complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 94.8 RON.

Vietnam's bioethanol demand is growing rapidly, driven by government policies like a planned nationwide E10 gasoline mandate set to begin in 2026. Ethanol consumption in Vietnam was 236 million liters (62 million gallons) representing 2.5% of gasoline demand. Ethanol production and Net Imports were 200 and 36 million liters (62 and 9 million gallons) correspondingly.

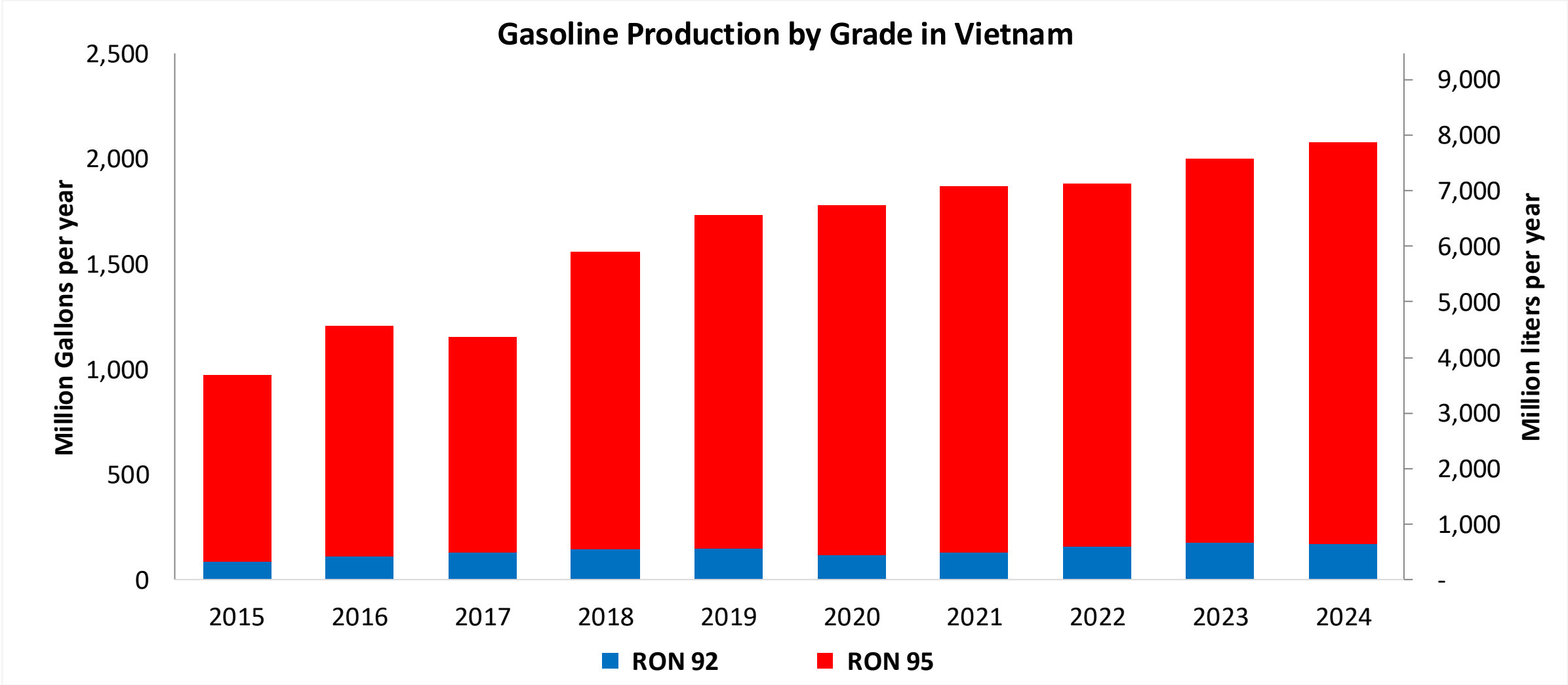
Source: Petro Vietnam, EIA,, EITdata, USDA

Gasoline Demand in Vietnam



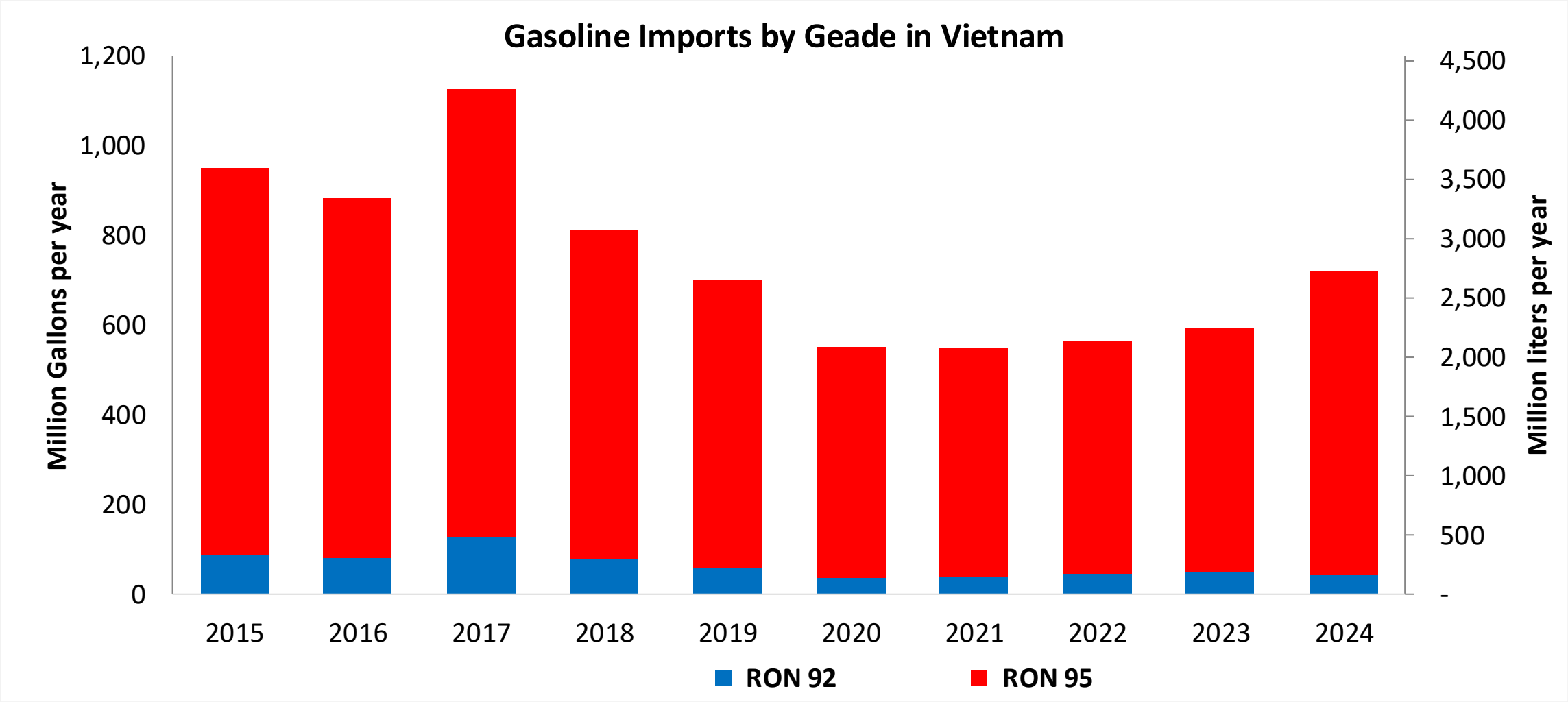
Source: Petro Vietnam, EIA,, EITdata

Gasoline Production in Vietnam



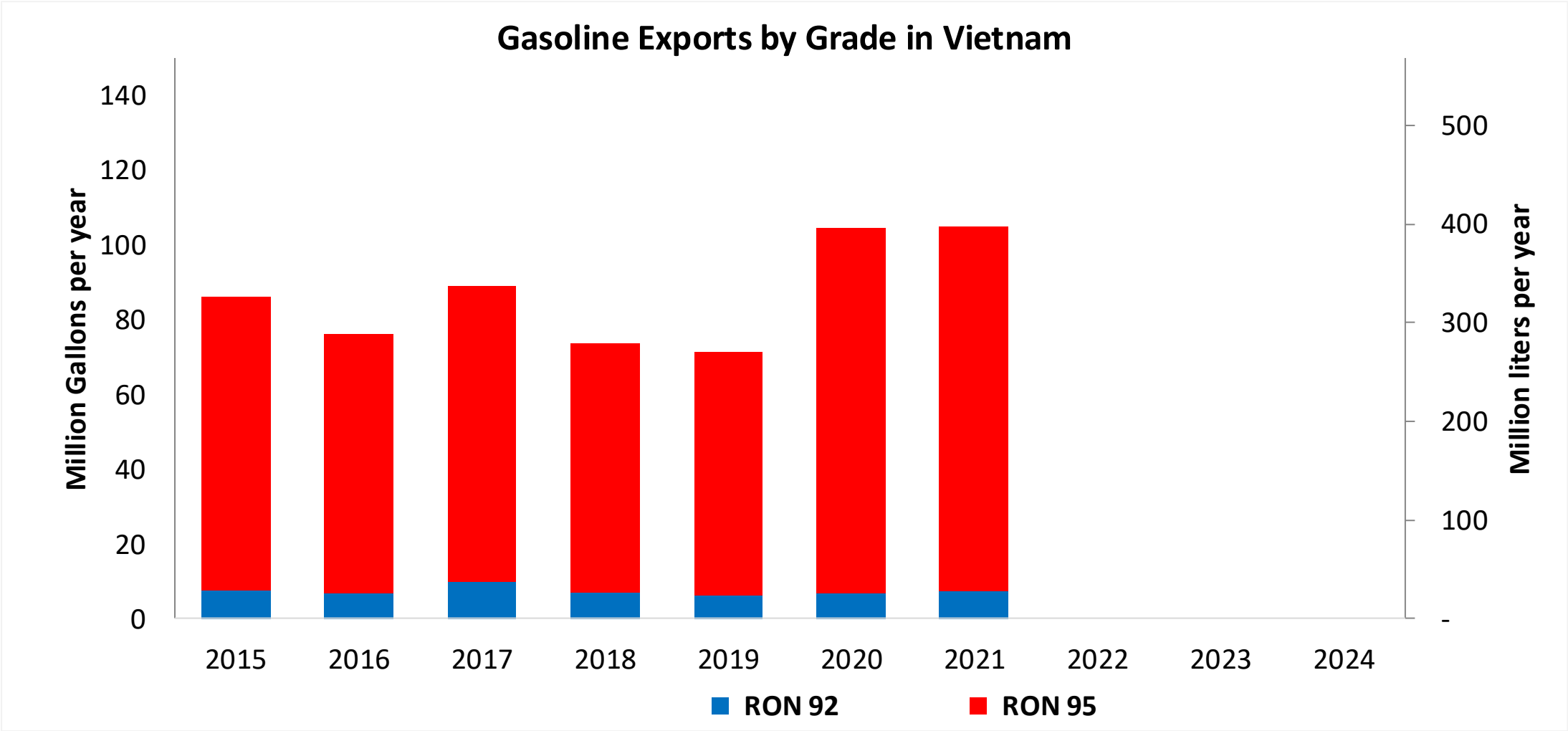
Source: Petro Vietnam, EIA,, EITdata

Gasoline Imports in Vietnam



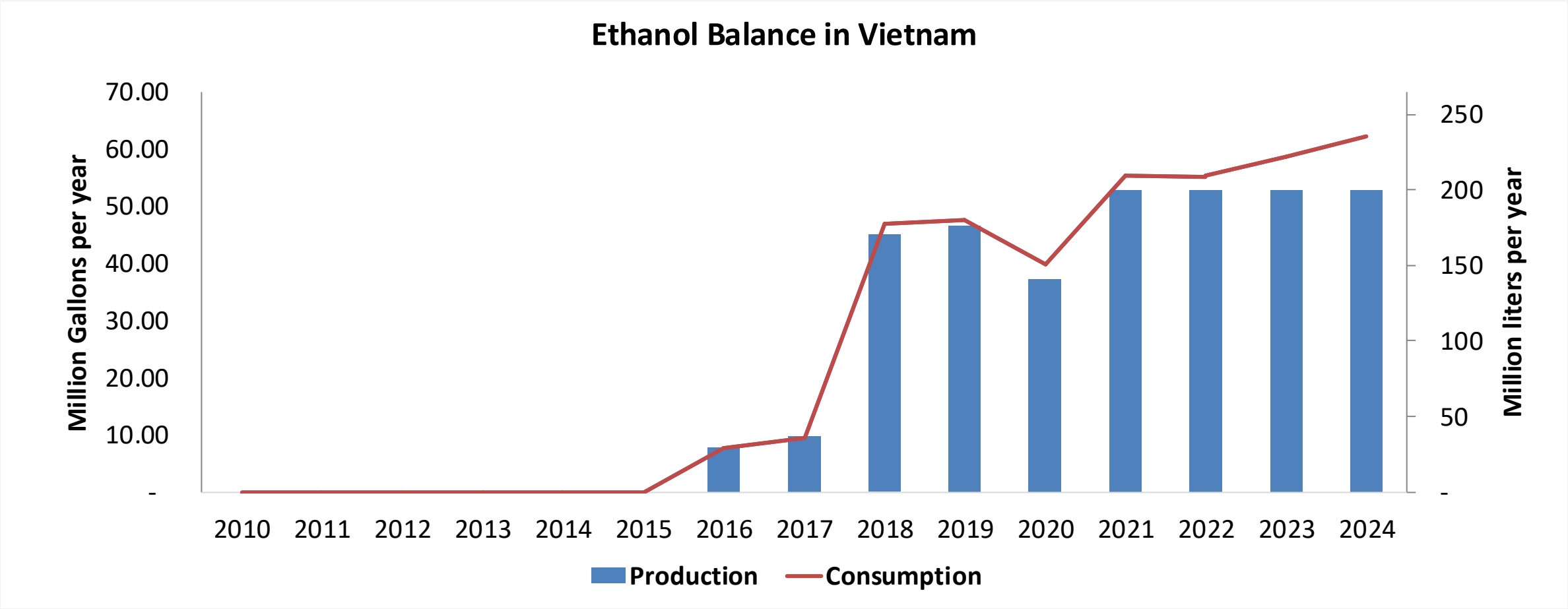
Source: Petro Vietnam, EIA,, EITdata

Gasoline Exports in Vietnam



Source: Petro Vietnam, EIA,, EITdata

Ethanol Balance in Vietnam



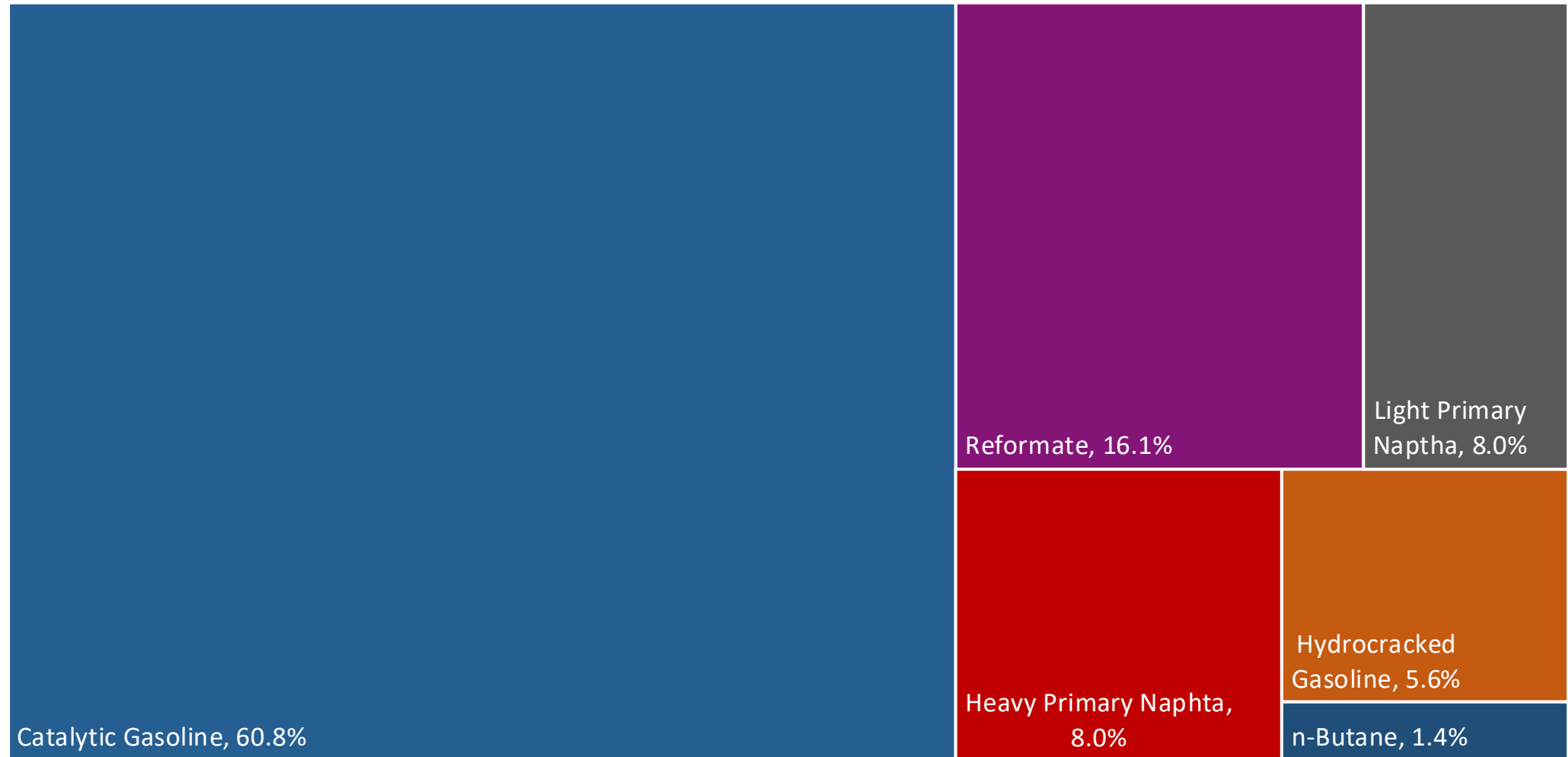
Source: Petro Vietnam, USDA

Gasoline Quality in Vietnam

Name	QCVN 01:2022/BKHCN					
Implementation Year	2023					
Applicability	All territory					
Grade	RON 92		RON 95		RON 97	
	Min	Max	Min	Max	Min	Max
RON	92.0		95.0		97.0	
MON	-		-		-	
AKI	-		-		-	
Benzene (%vol)		1.0		1.0		1.0
Aromatics (% vol)		35.0		35.0		35.0
Olefins (%vol)		18.0		18.0		18.0
Lead (g/L)		0.005		0.005		0.005
Manganese (mg/L)		5.0		5.0		5.0
Sulfur (ppm)		10		10		10
Oxygen (%w)		2.7		2.7		2.7
Ethanol (%vol)		-		-		-
MTBE (%vol)		-		-		-
RVP (psi)		8.7		8.7		8.7

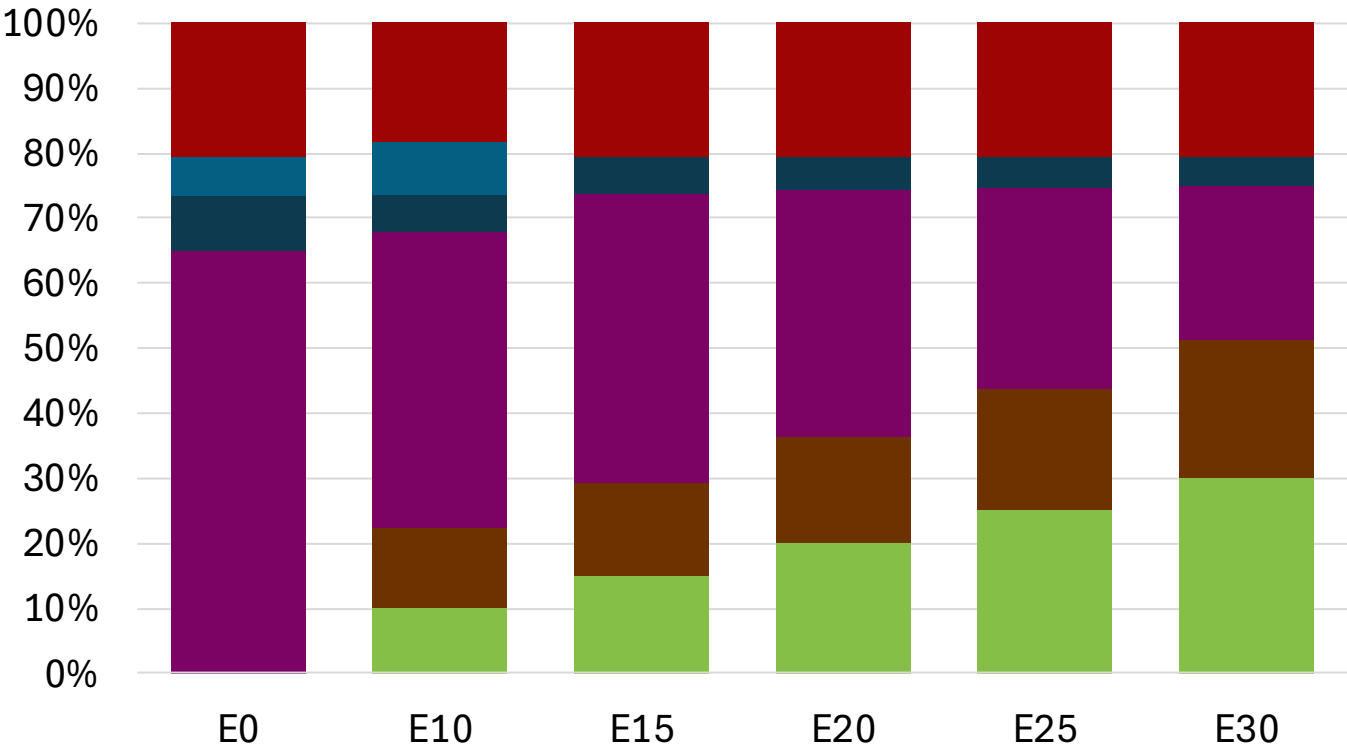
Source: Caselaw

Vietnam Available Blending Components



Source: Faro90

Vietnam – Gasoline 92 – Constant Octane

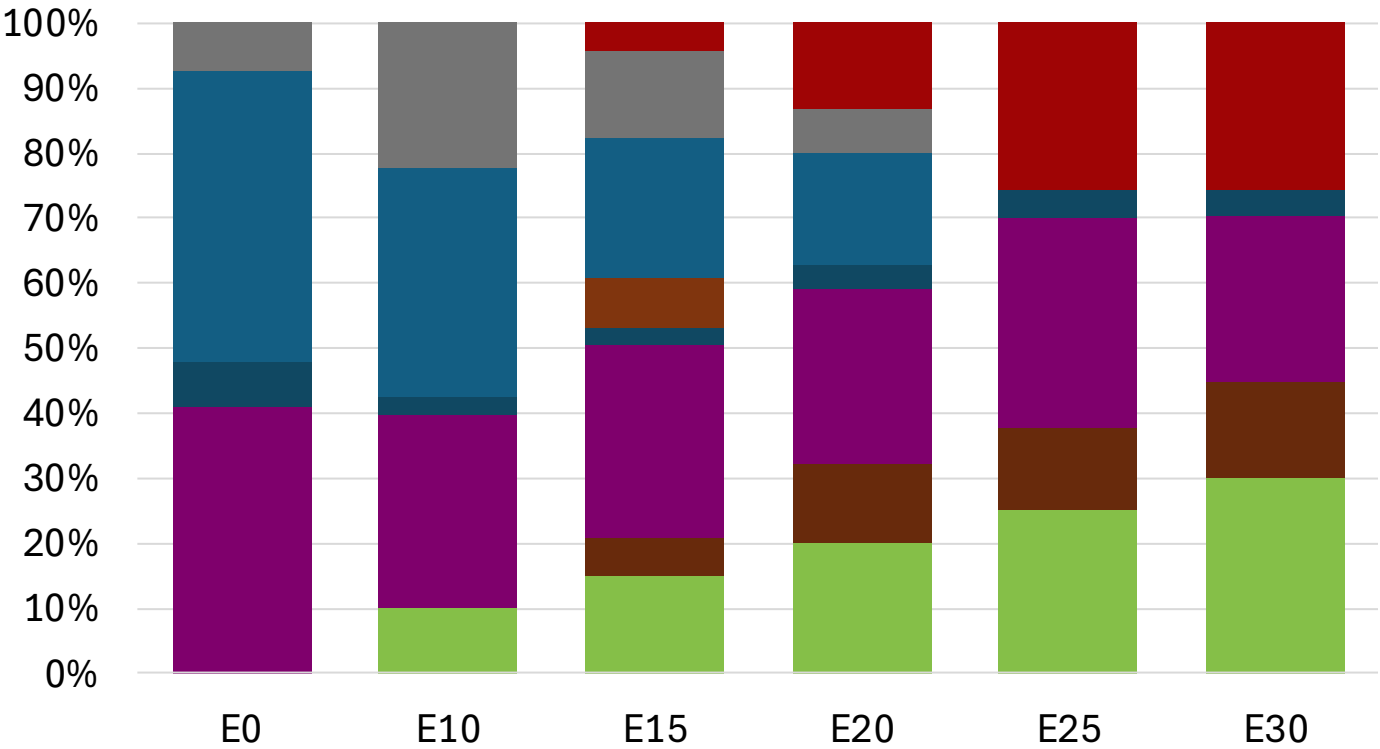


Average CIF 2024
Prices

Octane (RON)	92.0	92.0	92.9	94.0	95.0	96.0
Price (US\$/gal)	\$ 2.280	\$ 2.113	\$ 2.063	\$ 2.020	\$ 1.972	\$ 1.921

Source: Faro90

Vietnam – Gasoline 95 – Constant Octane

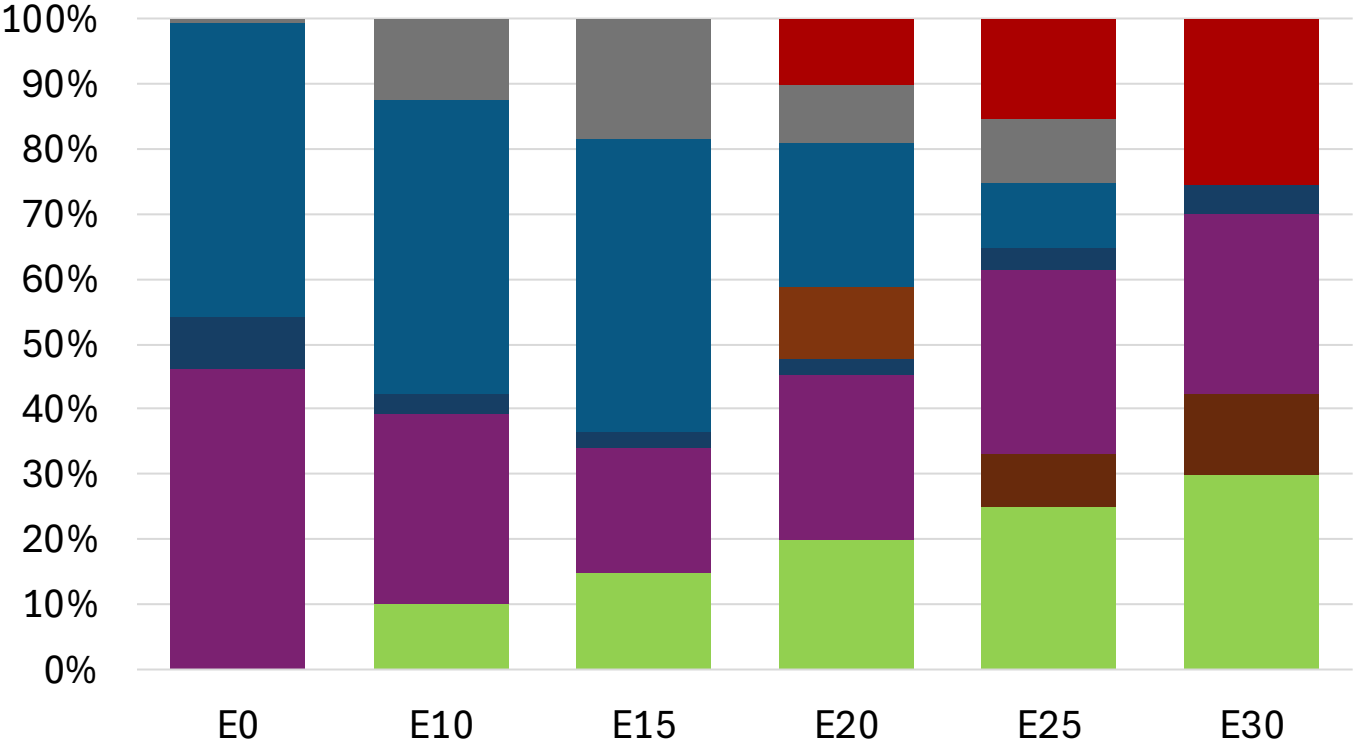


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.2	96.3
Price (US\$/gal)	\$ 2.436	\$ 2.305	\$ 2.214	\$ 2.090	\$ 1.981	\$ 1.933

Source: Faro90

Vietnam - Gasoline 97 - Constant Octane

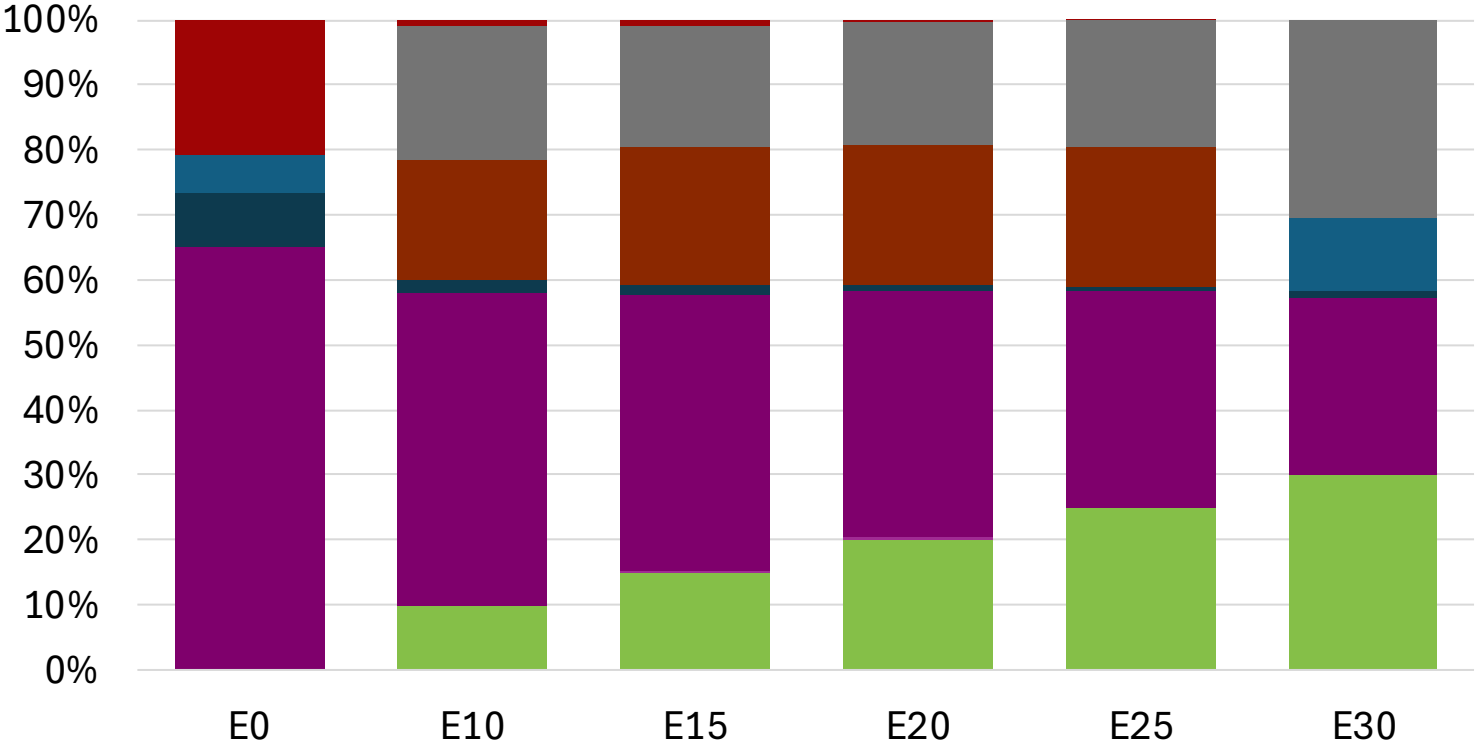


Average CIF 2024
Prices

Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.0
Price (US\$/gal)	\$ 2.542	\$ 2.416	\$ 2.322	\$ 2.228	\$ 2.105	\$ 1.983

Source: Faro90

Vietnam – Gasoline 92 – Increased Octane

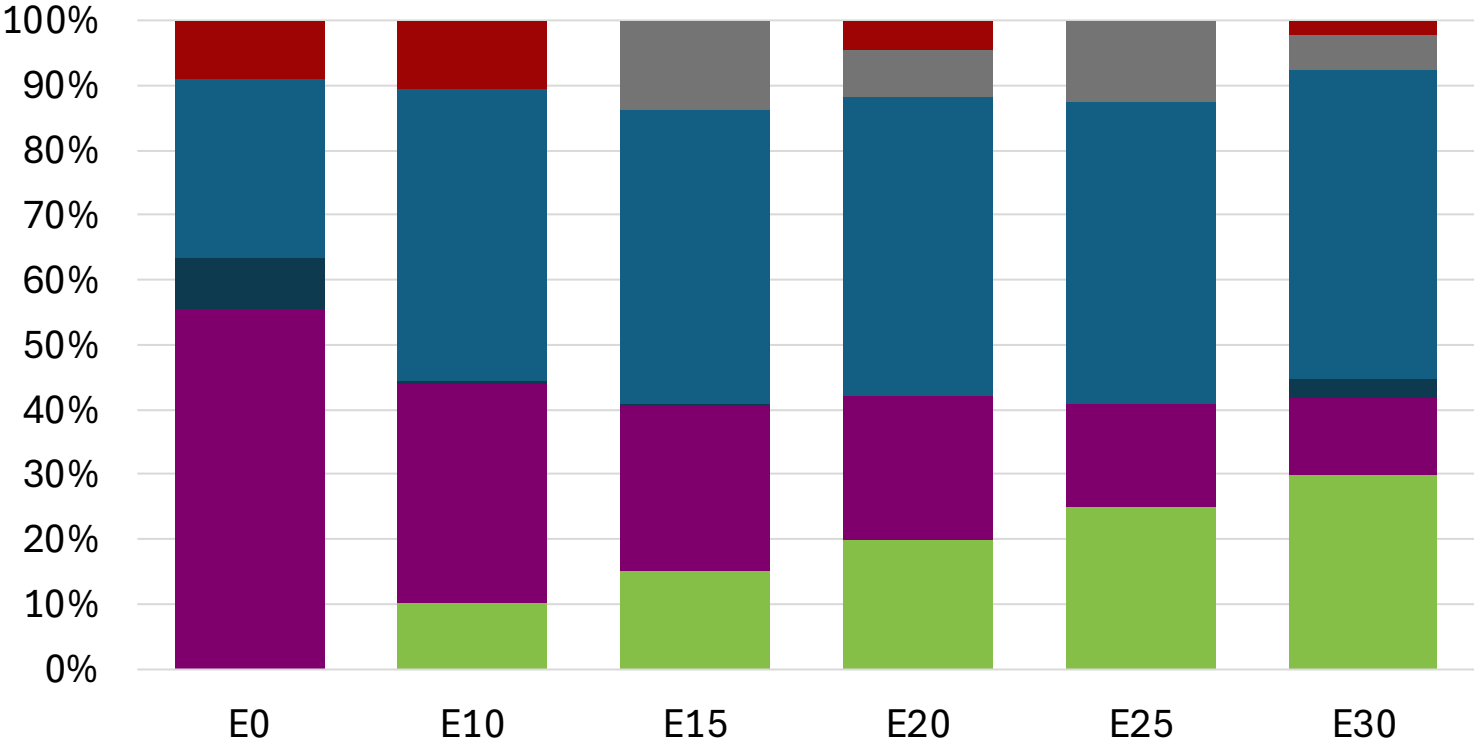


Average CIF 2024
Prices

Octane (RON)	92.0	94.1	95.7	97.4	99.0	101.0
Price (US\$/gal)	\$ 2.280	\$ 2.280	\$ 2.280	\$ 2.280	\$ 2.280	\$ 2.280

Source: Faro90

Vietnam – Gasoline 95 – Increased Octane

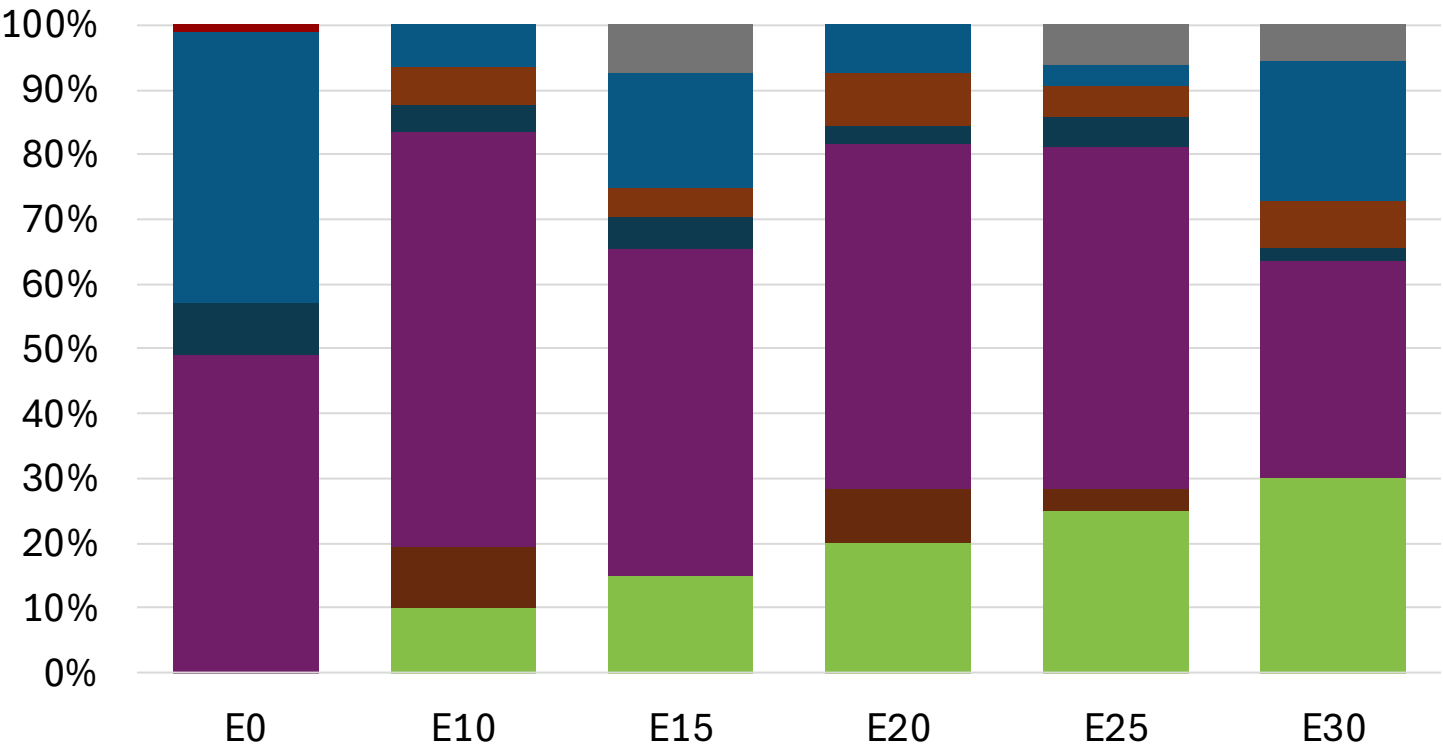


Average CIF 2024
Prices

Octane (RON)	95.0	96.8	98.4	100.0	101.6	103.7
Price (US\$/gal)	\$ 2.436	\$ 2.436	\$ 2.436	\$ 2.436	\$ 2.436	\$ 2.436

Source: Faro90

Vietnam – Gasoline 97 – Increased Octane

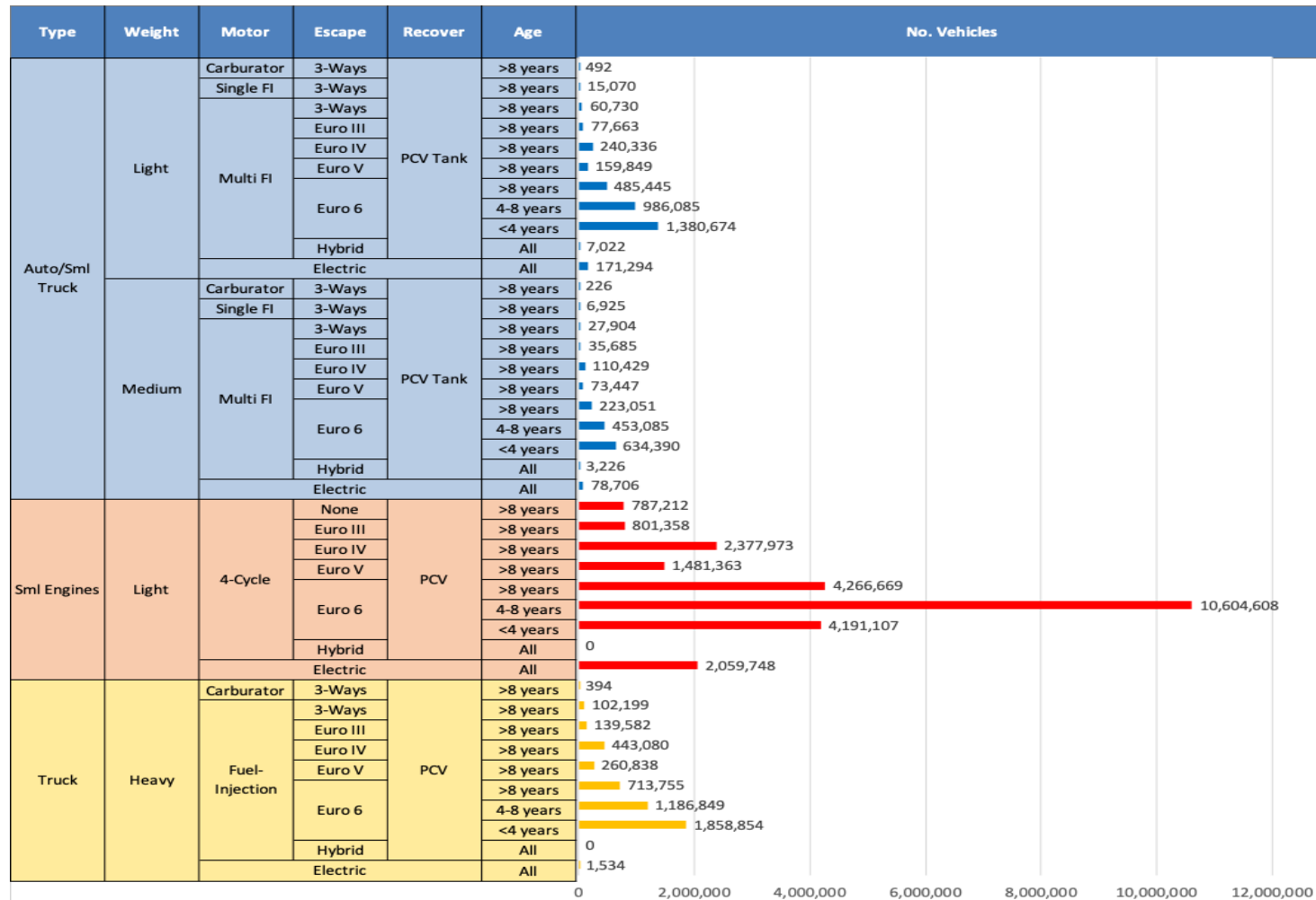


Average CIF 2024
Prices

Octane (RON)	97.0	99.1	101.1	101.9	103.9	104.9
Price (US\$/gal)	\$ 2.541	\$ 2.541	\$ 2.541	\$ 2.541	\$ 2.541	\$ 2.541

Source: Faro90

Gasoline Vehicle Fleet - Vietnam

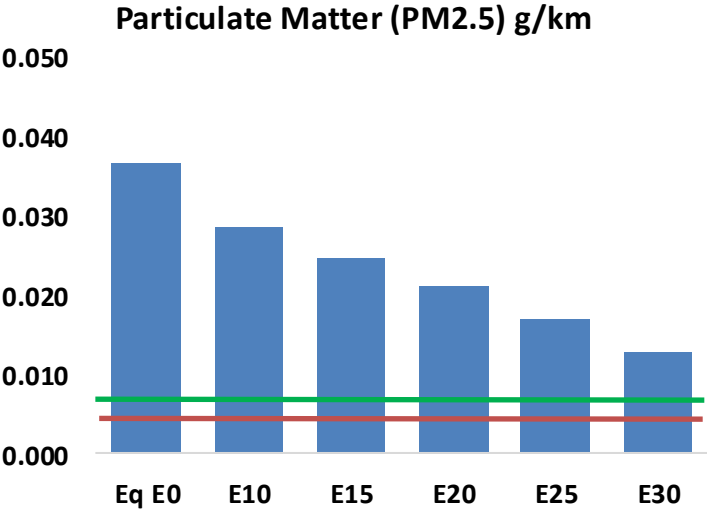
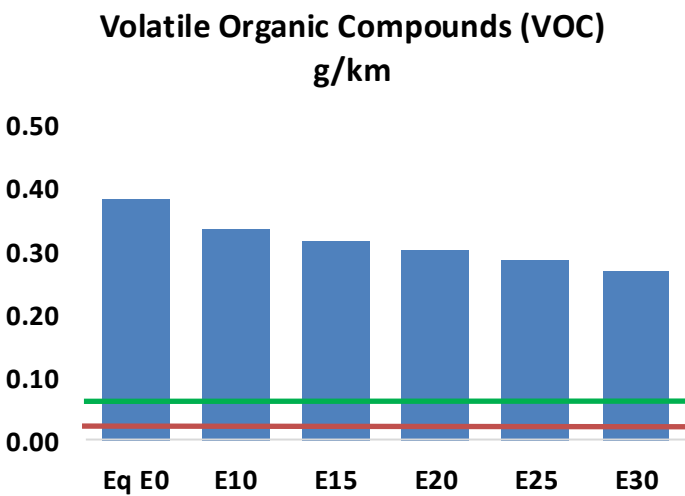
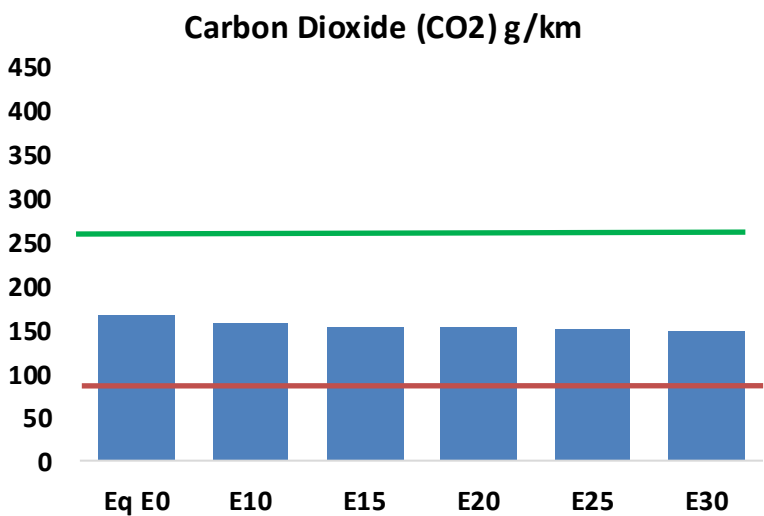
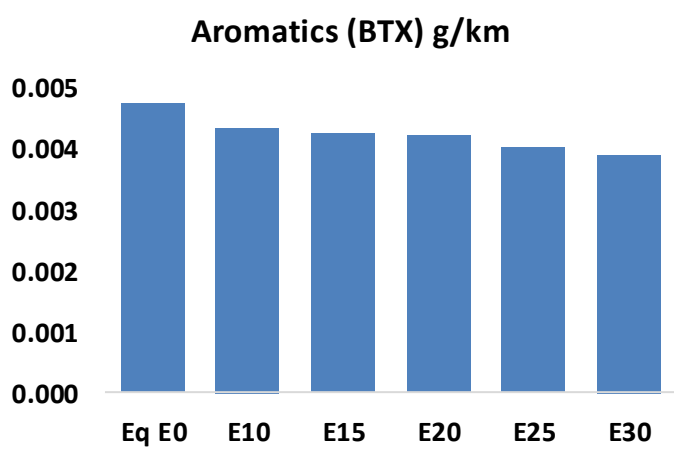
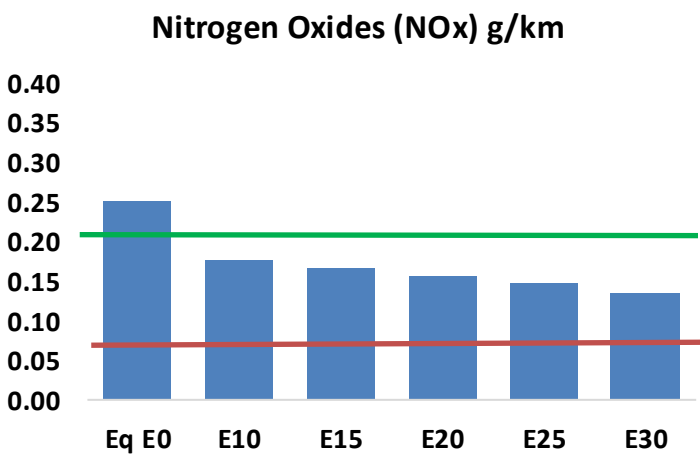
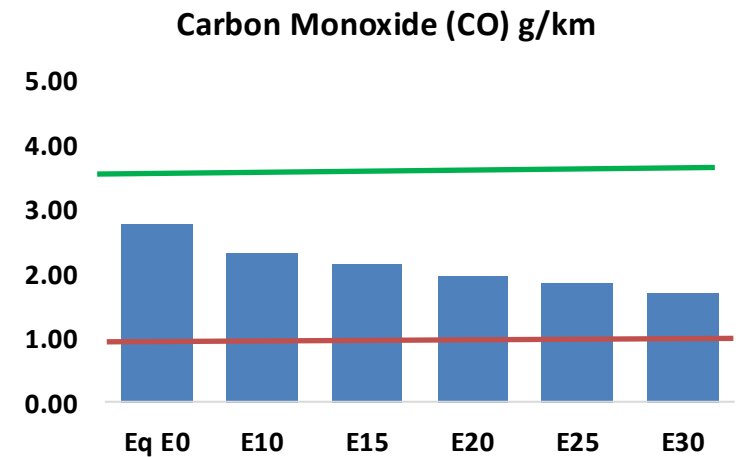
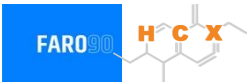


Vehicle Fleet: 35,508,859
Gasoline Vehicle Fleet: 25,550,917

Source: MoIT

Vietnam – Vehicular Emissions

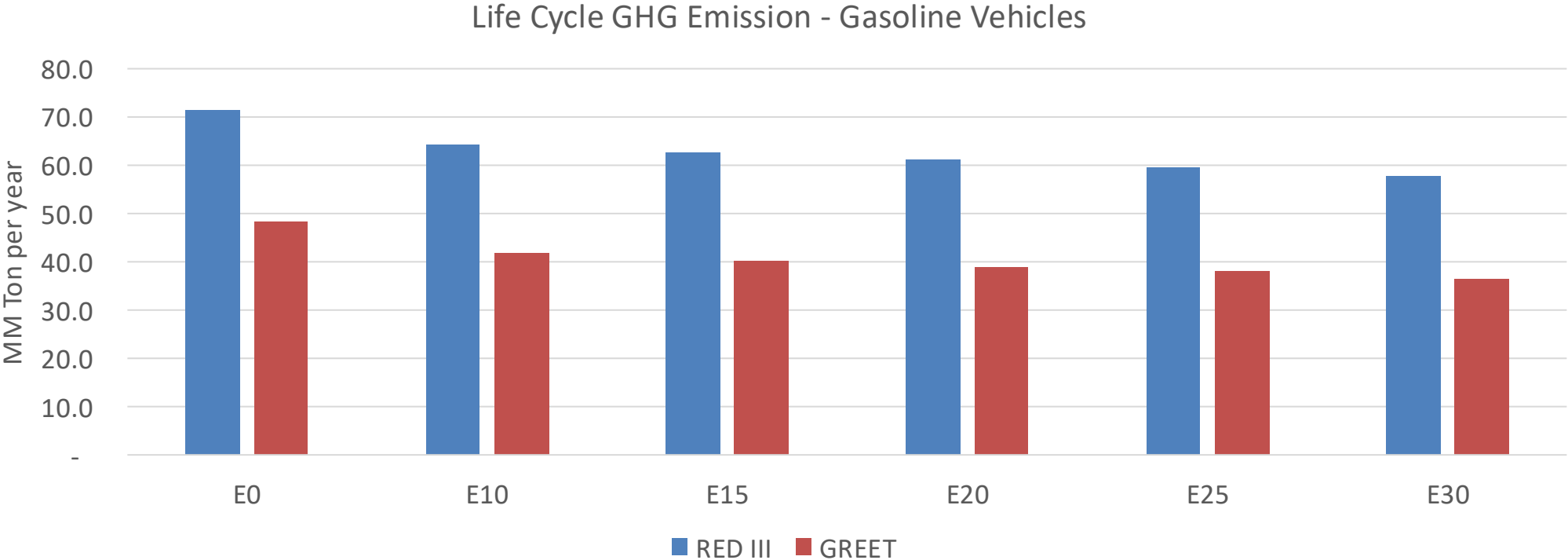
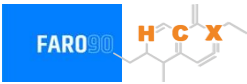
TIER III BIN 125 United States
Euro 6 Standard



Vietnam – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,359,820	1,252,428	1,145,036	1,056,476	967,099	898,296	825,989	-16%	-29%
CO2	80,379,150	78,312,257	76,360,192	74,824,950	74,065,253	73,754,180	72,394,679	-5%	-8%
VOC	186,910	175,504	164,098	155,329	146,771	140,271	131,313	-12%	-21%
NOx	122,884	92,163	86,019	81,073	76,459	71,371	65,989	-30%	-38%
SOx	872	806	740	685	631	573	512	-15%	-28%
NH3	33,331	33,001	32,671	32,826	33,195	33,455	33,671	-2%	0%
N2O	1,350	1,342	1,337	1,343	1,371	1,386	1,402	-1%	2%
CH4	47,919	47,919	47,919	48,637	49,692	50,602	51,465	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	1,465	1,391	1,318	1,265	1,216	1,179	1,126	-10%	-17%
Acetaldehyde	3,770	4,327	6,348	9,668	13,171	15,051	17,784	68%	249%
Formaldehyde	14,599	14,194	16,546	19,429	20,354	22,517	24,513	13%	39%
Benzene	2,328	2,237	2,120	2,086	2,069	1,979	1,915	-9%	-11%
PM2.5	5,773	5,126	4,479	3,889	3,303	2,680	2,031	-22%	-43%
PM10	12,169	10,805	9,442	8,199	6,963	5,650	4,281	-22%	-43%

Vietnam – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	489.9
Target @ 2035 (MMT/year)	330.6
Delta	159.3

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	13.2%	16.4%	19.0%	21.5%	24.8%
RED II/III	0.0%	9.8%	12.4%	14.5%	16.5%	19.1%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.0%	5.0%	5.8%	6.5%	7.5%
RED II/III	0.0%	4.4%	5.5%	6.5%	7.4%	8.5%

Source: Greet, RED II/III, climateactiontracker.org, F90